1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
			2010-07-23

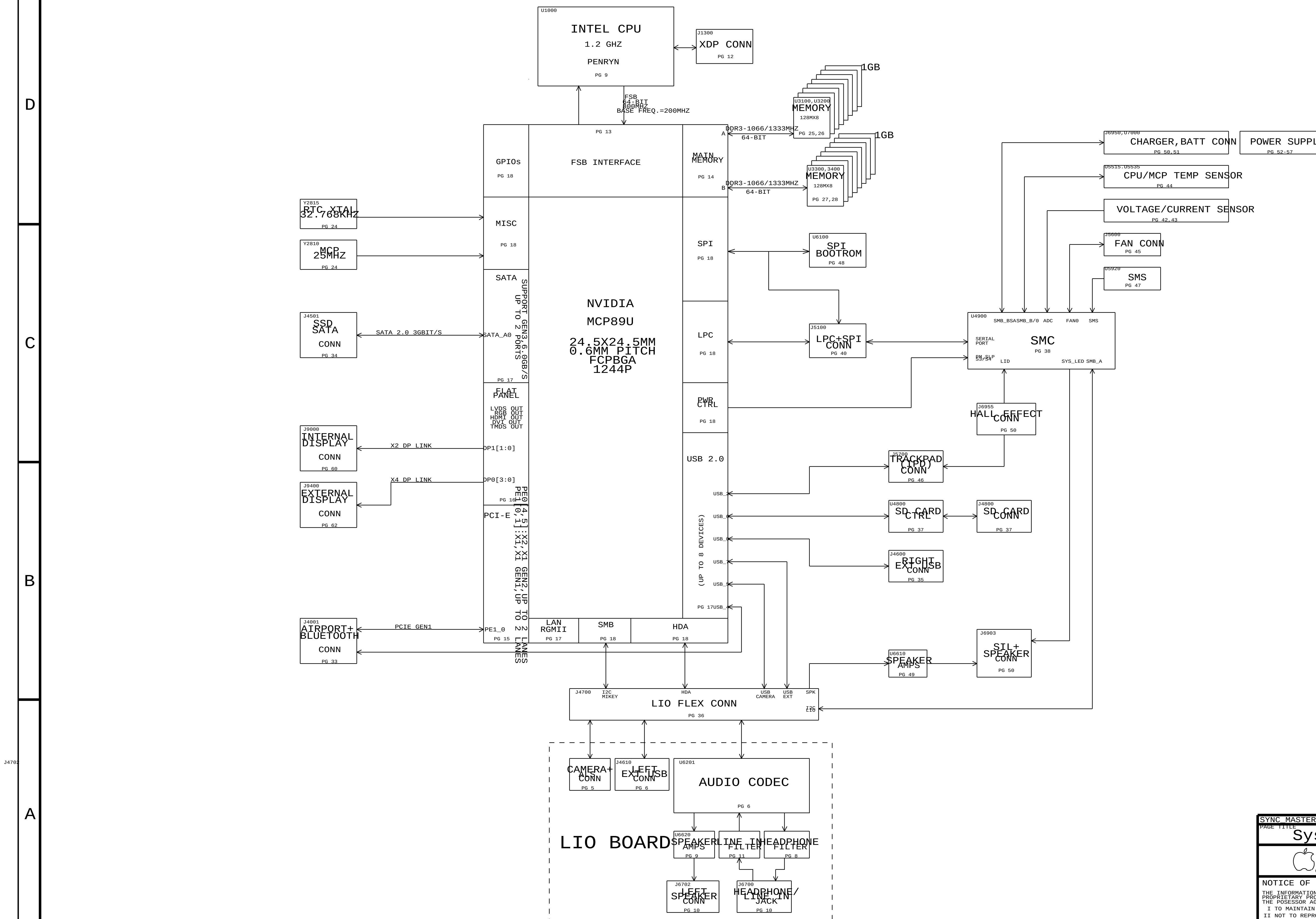
SCHEM,MLB,K16

07/23/2010

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8467	1	SCHEM,MLB,K16	SCH	CRITICAL	
820-2838	1	PCBF,MLB,K16	PCB	CRITICAL	

DRAWING TITLE: SCHEM,MLB,K16
DRAWING NUMBER: 051-8467
SIZE: D
Apple Inc.
REVISION: 3.3.0
BRANCH:
PAGE: 1 OF 110
SHEET: 1 OF 74



SYNC_MASTER=ALAN.DAI SYNC_DATE=02/26/2010

PAGE TITLE
System Block Diagram

DRAWING NUMBER 051-8467 SIZE D

Apple Inc.

REVISION 3.3.0

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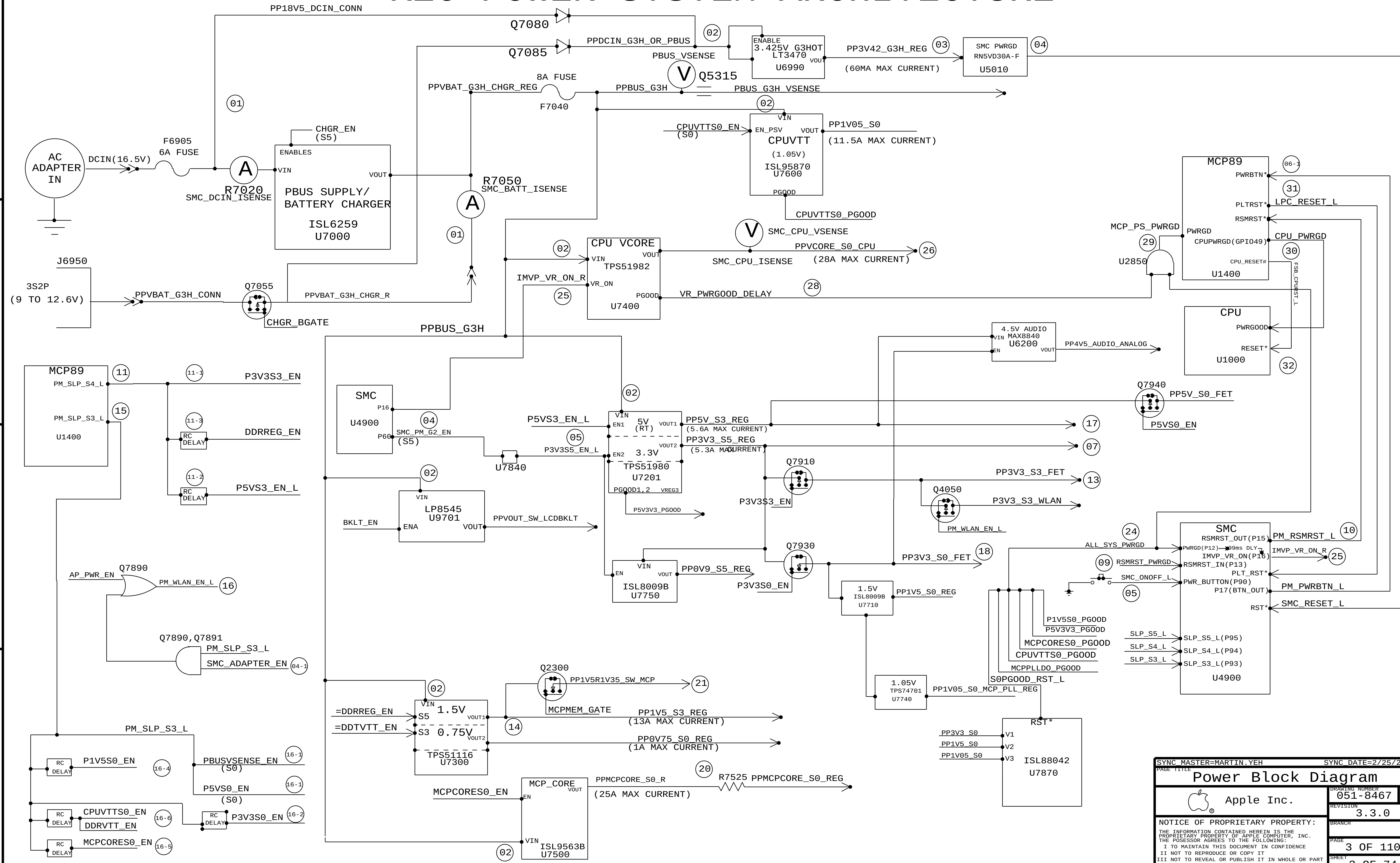
K16 POWER SYSTEM ARCHITECTURE



SYNC_MASTER=MARTIN.YEH SYNC_DATE=2/25/2010

PAGE TITLE: Power Block Diagram

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DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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DRAM CFG CHART

VENDOR	CFG 0	CFG 1
HYNIX	0	0
SAMSUNG	0	1
MICRON	1	0
ELPIDA	1	1

SIZE	CFG 2
2GB	0
4GB	1

DIE REV	CFG 3
A	0
B	1

K16 BOM Variants on following page

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3820	1	IC,MCP89U-A01,24.5MMX24.5MM,1244FCBGA	U1400	CRITICAL	MCP89U:A01
337S3868	1	IC,MCP89U-A02,24.5MMX24.5MM,1244FCBGA	U1400	CRITICAL	MCP89U:A02
337S3938	1	IC,MCP89U-A03,24.5MMX24.5MM,1244FCBGA	U1400	CRITICAL	MCP89U:A03
333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0552	4	HYNIX,LVDDR3,1GBIT,7.5X11.0	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:HYNIX_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0553	4	SAMSUNG,LVDDR3,1GBIT,7.5X11.0	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:SAMSUNG_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0554	4	MICRON,LVDDR3,1GBIT,8X11.5	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:MICRON_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0565	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:ELPIDA_2GB
333S0566	4	ELPIDA,LVDDR3,1GBIT,7.5X10.6	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0566	4	ELPIDA,LVDDR3,2GBIT,7.5X10.6	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0566	4	ELPIDA,LVDDR3,2GBIT,7.5X10.6	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0566	4	ELPIDA,LVDDR3,2GBIT,7.5X10.6	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0555	4	HYNIX,LVDDR3,2GBIT,9X11.1	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0556	4	SAMSUNG,LVDDR3,2GBIT,7.5X11.0	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3100,U3110,U3120,U3130	CRITICAL	DRAM_TYPE:MICRON_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3200,U3210,U3220,U3230	CRITICAL	DRAM_TYPE:MICRON_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3300,U3310,U3320,U3330	CRITICAL	DRAM_TYPE:MICRON_4GB
333S0557	4	MICRON,LVDDR3,2GBIT,9X11.5	U3400,U3410,U3420,U3430	CRITICAL	DRAM_TYPE:MICRON_4GB
353S2392	1	IC,ISL6259,BATCHARGER,4X4MM,QFN28	U7000	CRITICAL	ISL6259_SCREENED:NO
353S2929	1	IC,ISL6259,BATCHARGER,3%,4C4MM,QFN28	U7000	CRITICAL	ISL6259_SCREENED:YES

BOM Groups

BOM GROUP	BOM OPTIONS
K16_COMMON	COMMON,ALTERNATE,PROJ:K16,K16_MISC,MCP89U:A03,K16_DEBUG:ENG,K16_PROGPARTS,SPI:41MHZ,LVDDR3:YES,WLAN_PCTL:HW,IPD_5V:S5_INT,IPD_3V3:S5
K16_MISC	DP_ESD,DP_PWR:SMC,VFRQ:SLPS3,HVDDLDO:FIXED,MCPHVDD:P2V5,MCPLL_R:REG,S0PGOOD_BJT,ISL6259_SCREENED:YES,DPI2C:SMC
K16_PROGPARTS	BOOTROM:UNLOCKED,SMC:PROG
K16_DEVEL:ENG	BKLT:ENG,BMON:ENG,XDP_CONN,LPCPLUS,VREFMRGN:YES,EFI_DEBUG,S0PGOOD_ISL,MCPPLL_LDO,S3_S0_LED
K16_DEVEL:PVT	LPCPLUS
K16_DEBUG:ENG	DEVEL_BOM,SMC_DEBUG:YES,XDP
K16_DEBUG:PVT	DEVEL_BOM,BKLT:PROD,BMON:PROD,SMC_DEBUG:YES,XDP,VREFMRGN:NO
K16_DEBUG:PROD	BKLT:PROD,BMON:PROD,SMC_DEBUG:YES,XDP,VREFMRGN:NO
DDR3:HYNIX_2GB	DRAM_CFG0:L,DRAM_CFG1:L,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:HYNIX_2GB
DDR3:SAMSUNG_2GB	DRAM_CFG0:L,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:SAMSUNG_2GB
DDR3:MICRON_2GB	DRAM_CFG0:H,DRAM_CFG1:L,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:MICRON_2GB
DDR3:ELPIDA_2GB	DRAM_CFG0:H,DRAM_CFG2:L,DRAM_CFG3:L,DRAM_TYPE:ELPIDA_2GB
DDR3:ELPIDA_4GB	DRAM_CFG0:H,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:ELPIDA_4GB
DDR3:HYNIX_4GB	DRAM_CFG0:L,DRAM_CFG1:L,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:HYNIX_4GB
DDR3:SAMSUNG_4GB	DRAM_CFG0:L,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:SAMSUNG_4GB
DDR3:MICRON_4GB	DRAM_CFG0:H,DRAM_CFG1:L,DRAM_CFG2:H,DRAM_CFG3:L,DRAM_TYPE:MICRON_4GB
CAPS:SS	SS_CAP_2_2UF,SS_CAP_10UF,SS_CAP_1UF,SS_CAP_22UF
CAPS:MU	MU_CAP_2_2UF,MU_CAP_10UF,MU_CAP_1UF,MU_CAP_22UF
CAPS:TY	TY_CAP_2_2UF,TY_CAP_10UF,TY_CAP_1UF,TY_CAP_22UF

Programmable Parts

338S0563	1	IC,SMC,HS8/2117,9X9MM,TLP,HF	U4900	CRITICAL	SMC:BLANK
335S0610	1	IC,FLASH,SPI,32MBIT,3.3V,86MHZ,8-SOP	U6100	CRITICAL	BOOTROM:BLANK

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0681	138S0638		ALL	TAIYO YUDEN AS ALTERNATE
152S0874	152S0516		ALL	MAGLAYERS AS ALTERNATE
152S0847	152S0586		ALL	MAGLAYERS AS ALTERNATE
353S2987	353S2988	HVDDLDO:FIXED	ALL	TPS71725 ALTERNATE FOR U2590
104S0023	104S0018		ALL	CYNTEC/DALE AS ALTERNATES
107S0139	107S0075		ALL	CYNTEC AS ALTERNATE
138S0671	138S0673		ALL	TAIYO AS ALTERNATE
155S0578	155S0367		ALL	TAIYO AS ALTERNATE
376S0926	376S0610		ALL	FAIRCHILD AS ALTERNATE
155S0457	155S0329		ALL	MAGLAYERS AS ALTERNATE
377S0107	377S0066		ALL	ON SEMI AS ALTERNATE

SYNC_MASTER=K6_MLB SYNC_DATE=12/11/2009

PAGE TITLE

BOM Configuration

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-1070	PCBA,MLB,1.86GHZ HY 2GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWQ,CAPS:MU,DDR3:HYNIX_2GB
639-0837	PCBA,MLB,1.86GHZ,HY 2GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXW,CAPS:SS,DDR3:HYNIX_2GB
639-1096	PCBA,MLB,1.86GHZ HY 2GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXN,CAPS:TY,DDR3:HYNIX_2GB
639-1101	PCBA,MLB,1.86GHZ HY 4GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXV,CAPS:MU,DDR3:HYNIX_4GB
639-1098	PCBA,MLB,1.86GHZ HY 4GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXQ,CAPS:SS,DDR3:HYNIX_4GB
639-1068	PCBA,MLB,1.86GHZ HY 4GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWN,CAPS:TY,DDR3:HYNIX_4GB
639-1083	PCBA,MLB,1.86GHZ MI 2GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX6,CAPS:MU,DDR3:MICRON_2GB
639-1078	PCBA,MLB,1.86GHZ MI 2GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX1,CAPS:SS,DDR3:MICRON_2GB
639-1090	PCBA,MLB,1.86GHZ MI 2GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXG,CAPS:TY,DDR3:MICRON_2GB
639-1088	PCBA,MLB,1.86GHZ MI 4GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXD,CAPS:MU,DDR3:MICRON_4GB
639-1067	PCBA,MLB,1.86GHZ MI 4GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWM,CAPS:SS,DDR3:MICRON_4GB
639-1077	PCBA,MLB,1.86GHZ MI 4GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX0,CAPS:TY,DDR3:MICRON_4GB
639-1080	PCBA,MLB,1.86GHZ SA 2GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX3,CAPS:MU,DDR3:SAMSUNG_2GB
639-1095	PCBA,MLB,1.86GHZ SA 2GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXM,CAPS:SS,DDR3:SAMSUNG_2GB
639-1071	PCBA,MLB,1.86GHZ SA 2GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWR,CAPS:TY,DDR3:SAMSUNG_2GB
639-1097	PCBA,MLB,1.86GHZ SA 4GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXP,CAPS:MU,DDR3:SAMSUNG_4GB
639-1084	PCBA,MLB,1.86GHZ SA 4GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX7,CAPS:SS,DDR3:SAMSUNG_4GB
639-1091	PCBA,MLB,1.86GHZ SA 4GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXH,CAPS:TY,DDR3:SAMSUNG_4GB
639-1092	PCBA,MLB,2.13GHZ HY 2GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXJ,CAPS:MU,DDR3:HYNIX_2GB
639-1082	PCBA,MLB,2.13GHZ,HY 2GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX5,CAPS:SS,DDR3:HYNIX_2GB
639-1085	PCBA,MLB,2.13GHZ HY 2GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX8,CAPS:TY,DDR3:HYNIX_2GB
639-1089	PCBA,MLB,2.13GHZ HY 4GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXF,CAPS:MU,DDR3:HYNIX_4GB
639-1075	PCBA,MLB,2.13GHZ HY 4GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWX,CAPS:SS,DDR3:HYNIX_4GB
639-1079	PCBA,MLB,2.13GHZ HY 4GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX2,CAPS:TY,DDR3:HYNIX_4GB
639-1099	PCBA,MLB,2.13GHZ MI 2GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXR,CAPS:MU,DDR3:MICRON_2GB
639-1087	PCBA,MLB,2.13GHZ MI 2GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXC,CAPS:SS,DDR3:MICRON_2GB
639-1069	PCBA,MLB,2.13GHZ MI 2GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWP,CAPS:TY,DDR3:MICRON_2GB
639-1100	PCBA,MLB,2.13GHZ MI 4GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXT,CAPS:MU,DDR3:MICRON_4GB
639-1093	PCBA,MLB,2.13GHZ MI 4GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXK,CAPS:SS,DDR3:MICRON_4GB
639-1076	PCBA,MLB,2.13GHZ MI 4GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWY,CAPS:TY,DDR3:MICRON_4GB
639-1074	PCBA,MLB,2.13GHZ SA 2GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWW,CAPS:MU,DDR3:SAMSUNG_2GB
639-1072	PCBA,MLB,2.13GHZ SA 2GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWT,CAPS:SS,DDR3:SAMSUNG_2GB
639-1086	PCBA,MLB,2.13GHZ SA 2GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX9,CAPS:TY,DDR3:SAMSUNG_2GB
639-1073	PCBA,MLB,2.13GHZ SA 4GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWV,CAPS:MU,DDR3:SAMSUNG_4GB
639-1081	PCBA,MLB,2.13GHZ SA 4GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX4,CAPS:SS,DDR3:SAMSUNG_4GB
639-1094	PCBA,MLB,2.13GHZ SA 4GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXL,CAPS:TY,DDR3:SAMSUNG_4GB
639-1450	PCBA,MLB,1.86GHZ EL 2GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DG4W,CAPS:MU,DDR3:ELPIDA_2GB
639-1451	PCBA,MLB,1.86GHZ EL 2GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DG4Y,CAPS:SS,DDR3:ELPIDA_2GB
639-1455	PCBA,MLB,1.86GHZ EL 2GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DG53,CAPS:TY,DDR3:ELPIDA_2GB
639-1453	PCBA,MLB,2.13GHZ EL 2GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DG51,CAPS:MU,DDR3:ELPIDA_2GB
639-1454	PCBA,MLB,2.13GHZ EL 2GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DG52,CAPS:SS,DDR3:ELPIDA_2GB
639-1452	PCBA,MLB,2.13GHZ EL 2GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DG50,CAPS:TY,DDR3:ELPIDA_2GB
639-1458	PCBA,MLB,1.86GHZ EL 4GB,MU CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DG5P,CAPS:MU,DDR3:ELPIDA_4GB
639-1463	PCBA,MLB,1.86GHZ EL 4GB,SS CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DG5W,CAPS:SS,DDR3:ELPIDA_4GB
639-1460	PCBA,MLB,1.86GHZ EL 4GB,TY CAP,K16	K16_CMNPTS,CPU:1.86GHZ,EEEE:DG5T,CAPS:TY,DDR3:ELPIDA_4GB
639-1462	PCBA,MLB,2.13GHZ EL 4GB,MU CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DG5Q,CAPS:MU,DDR3:ELPIDA_4GB
639-1459	PCBA,MLB,2.13GHZ EL 4GB,SS CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DG5V,CAPS:SS,DDR3:ELPIDA_4GB
639-1461	PCBA,MLB,2.13GHZ EL 4GB,TY CAP,K16	K16_CMNPTS,CPU:2.13GHZ,EEEE:DG5R,CAPS:TY,DDR3:ELPIDA_4GB
607-6915	CMN PTS,PCBA,MLB,K16	K16_COMMON
085-1327	K16 MLB DEVELOPMENT BOM	K16_DEVEL:ENG

Bar Code Labels / EEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWM]	CRITICAL	EEEE:DCWM
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWN]	CRITICAL	EEEE:DCWN
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWP]	CRITICAL	EEEE:DCWP
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWQ]	CRITICAL	EEEE:DCWQ
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWR]	CRITICAL	EEEE:DCWR
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWT]	CRITICAL	EEEE:DCWT
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWV]	CRITICAL	EEEE:DCWV
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWW]	CRITICAL	EEEE:DCWW
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWX]	CRITICAL	EEEE:DCWX
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCWY]	CRITICAL	EEEE:DCWY
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX0]	CRITICAL	EEEE:DCX0
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX1]	CRITICAL	EEEE:DCX1
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX2]	CRITICAL	EEEE:DCX2
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX3]	CRITICAL	EEEE:DCX3
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX4]	CRITICAL	EEEE:DCX4
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX5]	CRITICAL	EEEE:DCX5
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX6]	CRITICAL	EEEE:DCX6
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX7]	CRITICAL	EEEE:DCX7
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX8]	CRITICAL	EEEE:DCX8
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCX9]	CRITICAL	EEEE:DCX9
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXC]	CRITICAL	EEEE:DCXC
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXD]	CRITICAL	EEEE:DCXD
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXF]	CRITICAL	EEEE:DCXF
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXG]	CRITICAL	EEEE:DCXG
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXH]	CRITICAL	EEEE:DCXH
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXJ]	CRITICAL	EEEE:DCXJ
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXK]	CRITICAL	EEEE:DCXK
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXL]	CRITICAL	EEEE:DCXL
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXM]	CRITICAL	EEEE:DCXM
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXN]	CRITICAL	EEEE:DCXN
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXP]	CRITICAL	EEEE:DCXP
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXQ]	CRITICAL	EEEE:DCXQ
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXR]	CRITICAL	EEEE:DCXR
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXT]	CRITICAL	EEEE:DCXT
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXV]	CRITICAL	EEEE:DCXV
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DCXW]	CRITICAL	EEEE:DCXW
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG4W]	CRITICAL	EEEE:DG4W
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG4Y]	CRITICAL	EEEE:DG4Y
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG53]	CRITICAL	EEEE:DG53
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG51]	CRITICAL	EEEE:DG51
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG52]	CRITICAL	EEEE:DG52
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG50]	CRITICAL	EEEE:DG50
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG5P]	CRITICAL	EEEE:DG5P
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG5W]	CRITICAL	EEEE:DG5W
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG5T]	CRITICAL	EEEE:DG5T
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG5Q]	CRITICAL	EEEE:DG5Q
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG5V]	CRITICAL	EEEE:DG5V
825-7557	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_DG5R]	CRITICAL	EEEE:DG5R

SAMSUNG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0632	1	CAP, 2.2UF, 6.3V, 20%, 0402	C4807	CRITICAL	SS_CAP_2_2UF

MURATA

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0633	1	CAP, 2.2UF, 6.3V, 20%, 0402	C4807	CRITICAL	MU_CAP_2_2UF

TAIYO YUDEN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0634	1	CAP, 2.2UF, 6.3V, 20%, 0402	C4807	CRITICAL	TY_CAP_2_2UF

K16-Specific BOM Tables

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3751	1	PDC,SLGAB,PRQ,1.86,17W,1066,ED,6M,BGA	U1000	CRITICAL	CPU:1.86GHZ
337S3758	1	PDC,SLGEQ,PRQ,2.13,17W,1066,ED,6M,BGA	U1000	CRITICAL	CPU:2.13GHZ
341T0276	1	IC ASSY,SMC EXTERNAL,K16	U4900	CRITICAL	SMC:PROG
341T0275	1	IC ASSY,EFI UNLOCKED,K16	U6100	CRITICAL	BOOTROM:UNLOCKED
341S2785	1	IC EFI ROM,PVT,LOCKED,K16	U6100	CRITICAL	BOOTROM:LOCKED

Sub-BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-1327	1	K16 MLB DEVELOPMENT BOM	DEVEL	CRITICAL	DEVEL_BOM
607-6915	1	CMN PTS,PCBA,MLB,K16	CMNPTS	CRITICAL	K16_CMNPTS

SYNC_MASTER=N/A SYNC_DATE=N/A

PAGE TITLE
K16 BOM Variants

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

BRANCH

8 | 7 | 6 | 5 | 4 | 3 | 2 | 1

Revision History

Proto 0 (ECO #0000876215, v1.0.0, P4 change #210266, 03/16/2010)

v1.1.0 (P4 change #211399, 03/24/2010)

MCP:
7742015 - Added RC to DDC pass FETs to avoid glitch (pp. 7, 93).
7788138 - Added feedback divider and BOM tables for more HVDD LDOs (pp. 4, 25).
SMC:
7761747 - Added resistors to connect TCON to SMC or MCP SMBus (pp. 4, 52, 90).
7787883 - Added support for DP HPD wake / S4 state (pp. 4, 7, 8, 19, 49, 50, 78, 94).
SMS:
7765466 - Added S3 pull-up to SMS_INT_L to prevent leakage path (pp. 50, 59).
7769139 - Unstuffed SMS circuit (pg. 4).
General:
7787897 - Property/page fixes to reduce CheckPlus warnings/errors (pp. 7, 8, 12, 17, 74, 93, 108).

v1.2.0 (P4 change #211839, 03/26/2010)

USB:
7796626 - Changed port switch from TPS2052B to TPS2069 (pg. 46).
SMC:
7787883 - Added PLACE_NEAR property on R5022 to avoid stub (pg. 50).
SMBus:
7761747 - Added TCON I2C nets to FUNC_TEST list for J9000 (pg. 7).
Power:
7796648 - Changed DP and LCD power from PP3V3_S3 to PP3V3_S5 (pp. 8, 90).
7796658 - Changed backlight driver to E00 version (pg. 97).
BOM:
7796661 - Set up primary & alternate for power supply FET (pp. 4, 72).
7796654 - Consolidated SSM6N15FE to SSM6N37FE (pg. 48).
7796658 - Changed RCs on some SMC analog inputs (pg. 54).
7796683 - Stuffed RC on backlight driver PWM input (pg. 97).
General:
7796631 - Sorted BOM variants for easier verification (pg. 5).
7796631 - Cosmetic clean-up (pg. 76).

v1.3.0 (P4 change #212050, 03/26/2010)

SMBus:
7761747 - Added isolation FET and unstuffed series R's on TCON I2C for now (pp. 4, 90, 108).
Power Supply:
7796661 - Removed alternate FET, made some FETs primary to other APN (pp. 4, 72, 73, 76).
7798425 - R/C value changes for 3.42V G3Hot power supply (pg. 69).
7798399 - R/C value changes for 5V/3.3V power supply (pg. 72).
7800179 - R value changes for CPU VCore power supply (pg. 74).
7798445 - R value changes for 0.9V S5 power supply (pg. 77).
7796658 - Changed backlight driver back to non-E00 version (pp. 4, 97).
BOM:
7796658 - Added alternates for two caps per GSM and removed unused alternates (pg. 4).
7798399 - Consolidated 100pF caps (pp. 74, 75).

v1.4.0 (P4 change #212757, 03/31/2010)

MCP SPI:
7809733 - Changed strapping to select 62.5MHz SPI bus frequency (pg. 4).
SMBus:
7796631 - Added XDP connection to SMBus aliases page (pp. 13, 52).
7808530 - Changed SMC 'MGMT' SMBus pull-ups from 4.7K to 2K (pg. 52).
7761747 - Documented SMBus addresses for panel (pg. 52).
SD Card:
7800415 - Changed SD Card discharge R to more standard value (pg. 48).
Power Supplies:
7803283 - Changed 5V S3 regulator output from 5.02V to 5.12V nominal (pg. 72).
7809760 - Stuffed C9799 and clarified tables/BOMOPTIONs around these parts (pg. 97).

Proto 1 (ECO #0000884508, v2.0.0, P4 change #212783, 03/31/2010)

v2.1.0 (P4 change #??????, ??/??/2010)

BOM:
7796658 - Changed OMITs to OMIT_TABLEs (pp. 10-11, 14-20, 26, 31-36, 49, 61).

NOTE: All page numbers are .csa, not PDF. See page 1 for .csa -> PDF mapping.

SYNC_MASTER=N/A SYNC_DATE=N/A
PAGE TITLE: Revision History
Apple Inc.
DRAWING NUMBER: 051-8467 SIZE: D
REVISION: 3.3.0
BRANCH

8 | 7 | 6 | 5 | 4 | 3 | 2 | 1

Functional Test Points

J4001: AirPort / BT Connector

FUNC_TEST	Net	Pages
TRUE	PP3V3_WLAN_F	7 34 40
TRUE	WIFI_EVENT_L	34 39 40
TRUE	PCIE_AP_R2D_N	34 68
TRUE	PCIE_AP_R2D_P	34 68
TRUE	PCIE_CLK100M_AP_N	16 34 68
TRUE	PCIE_CLK100M_AP_P	16 34 68
TRUE	USB_BT_P	18 34 69
TRUE	USB_BT_N	18 34 69
TRUE	PCIE_AP_D2R_P	16 34 68
TRUE	PCIE_AP_D2R_N	16 34 68
TRUE	PCIE_WAKE_L	16 34
TRUE	AP_RESET_CONN_L	34
TRUE	AP_CLKREQ_Q_L	34
TRUE	=PP3V3_S3_BT	8 34

(Need 5 TPs)
(Need to add 6 GND TPs)

J4501: SATA SSD Connector

FUNC_TEST	Net	Pages
TRUE	PP3V3_S0_HDD_R	7 35
TRUE	SATA_HDD_D2R_C_P	35 68
TRUE	SATA_HDD_D2R_C_N	35 68
TRUE	SATA_HDD_R2D_N	35 68
TRUE	SATA_HDD_R2D_P	35 68
TRUE	SMC_HDD_OOB_TEMP	35 39
TRUE	SMC_HDD_TEMP_CTL	35 39

(Need 5 TPs)
(Need to add 6 GND TPs)

J4700: LIO Connector

FUNC_TEST	Net	Pages
TRUE	=PP3V42_G3H_ONEWIRE	8 37
TRUE	=PP3V3_S0_AUDIO	8 37
TRUE	=PP1V8R1V5_S0_AUDIO	8 37
TRUE	SYS_ONEWIRE	37 39
TRUE	SMC_BC_ACOK	9 37 39 40
TRUE	=USB_PWR_EN	36 37 58
TRUE	SMC_LID	7 37 39 40 47
TRUE	=I2C_LIO_SDA	37 42
TRUE	=I2C_LIO_SCL	37 42
TRUE	=I2C_MIKEY_SCL	37 42
TRUE	=I2C_MIKEY_SDA	37 42
TRUE	AUD_IPHS_SWITCH_EN	19 37
TRUE	AUD_IP_PERIPHERAL_DET	17 37
TRUE	AUD_I2C_INT_L	19 37
TRUE	AUD_GPIO_3	37 49
TRUE	SPKRAMP_INR_N	37 49 72
TRUE	SPKRAMP_INR_P	37 49 72
TRUE	USB_EXTD_N	18 37 69
TRUE	USB_EXTD_P	18 37 69
TRUE	USB_CAMERA_N	18 37 69
TRUE	USB_CAMERA_P	18 37 69
TRUE	HDA_SDOUT	19 37 69
TRUE	HDA_BIT_CLK	19 37 69
TRUE	HDA_SDIN0	19 37 69
TRUE	USB_EXTD_OC_L	18 37
TRUE	HDA_RST_L	19 37 69
TRUE	HDA_SYNC	19 37 69

(Need 2 TPs)
(Need to add 5 GND TPs)

J4800: SD Card Connector

FUNC_TEST	Net	Pages
TRUE	PP3V3_SW_SD_PWR	38
TRUE	SD_CLK	38 70
TRUE	SD_CMD	38 70
TRUE	SD_D<7..0>	38 70
TRUE	SD_CD_L	38
TRUE	SD_WP	38

(Need to add 2 GND TPs)

J5100: LPC+SPI Connector

FUNC_TEST	Net	Pages
TRUE	=PP3V3_S5_LPCPLUS	8 41
TRUE	=PP5V_S0_LPCPLUS	8 41
TRUE	LPC_AD<3..0>	19 39 41 69
TRUE	SPI_ALT_MOSI	41 69
TRUE	SPI_ALT_MISO	41 69
TRUE	LPC_FRAME_L	19 39 41 69
TRUE	PM_CLKRUN_L	19 39 41
TRUE	SMC_TMS	39 40 41
TRUE	LPCPLUS_RESET_L	25 41
TRUE	SMC_TDO	39 40 41
TRUE	SMC_TRST_L	39 41
TRUE	SMC_MD1	39 41
TRUE	SMC_TX_L	36 39 40 41
TRUE	LPC_CLK33M_LPCPLUS	25 41 69
TRUE	SPIROM_USE_MLB	19 41 48
TRUE	SPI_ALT_CLK	41 69
TRUE	SPI_ALT_CS_L	41 69
TRUE	LPC_SERIRQ	19 39 41
TRUE	LPC_PWRDWN_L	19 39 41
TRUE	SMC_TDI	39 40 41
TRUE	SMC_TCK	39 40 41
TRUE	SMC_RESET_L	39 40 41 51
TRUE	SMC_NMI	39 41
TRUE	SMC_RX_L	36 39 40 41
TRUE	LPCPLUS_GPIO	19 41

(Need to add 6 GND TPs)

J5600: Fan Connector

FUNC_TEST	Net	Pages
TRUE	PP5V_S0	7 8 58
TRUE	FAN_RT_TACH	46
TRUE	FAN_RT_PWM	46

(Need to add 1 GND TP)

J5700: IPD Flex Connector

FUNC_TEST	Net	Pages
TRUE	=PP5V_S3_TPAD	8 57
TRUE	=PP3V42_G3H_TPAD	8 47
TRUE	=PP3V3_S3_TPAD	8 47
TRUE	USB_TPAD_CONN_P	47 72
TRUE	USB_TPAD_CONN_N	47 72
TRUE	=I2C_TPAD_SDA	42 47
TRUE	=I2C_TPAD_SCL	42 47
TRUE	SMC_ONOFF_L	39 40 47
TRUE	SMC_LID	7 37 39 40 47
TRUE	SMC_TPAD_RST_L	40 47

(Need to add 5 GND TPs)

J6900: DC-In Connector

FUNC_TEST	Net	Pages
TRUE	=PP18V5_DCIN_CONN	8 59
TRUE	=PP5V_S3_LIO_CONN	8 59

(Need 6 TPs)
(Need to add 6 GND TPs)

J6903: Speaker Connector

FUNC_TEST	Net	Pages
TRUE	SPKRAMP_R_P_OUT	49 50
TRUE	SPKRAMP_R_N_OUT	49 50

J6950: Battery Connector

FUNC_TEST	Net	Pages
TRUE	PPVBAT_G3H_CONN	50 51
TRUE	SMBUS_SMC_BSA_SCL	42 71
TRUE	SMBUS_SMC_BSA_SDA	42 71
TRUE	SYS_DETECT_L	50

(Need 4 TPs)
(Need to add 4 GND TPs near J6950 and 1 for shield)

J9000: Internal DP Connector

FUNC_TEST	Net	Pages
TRUE	PPVOUT_SW_LCDBKLT	7 43 60 63
TRUE	PP3V3_SW_LCD	60
TRUE	=I2C_TCON_SDA	42 60
TRUE	LED_RETURN_6	60 63
TRUE	LED_RETURN_5	60 63
TRUE	LED_RETURN_4	60 63
TRUE	LED_RETURN_3	60 63
TRUE	LED_RETURN_2	60 63
TRUE	LED_RETURN_1	60 63
TRUE	DP_INT_HPD_CONN	60
TRUE	DP_INT_AUX_CH_C_N	60 72
TRUE	DP_INT_AUX_CH_C_P	60 72
TRUE	DP_INT_ML_F_P<0>	60 72
TRUE	DP_INT_ML_F_N<0>	60 72
TRUE	DP_INT_ML_F_P<1>	60 72
TRUE	DP_INT_ML_F_N<1>	60 72
TRUE	=I2C_TCON_SCL	42 60

(Need 2 TPs)
(Need 2 TPs)
(Need to add 5 GND TPs)

Misc Voltages & Control Signals

FUNC_TEST	Net	Pages
TRUE	PPVOUT_SW_LCDBKLT	7 43 60 63
TRUE	PPDCIN_S5_S5	8
TRUE	PPBUS_G3H	8 43 50
TRUE	PPBUS_G3H_ISNS	8
TRUE	PP5V_S3	8
TRUE	PP5V_S3_RTUSB_A_F	36
TRUE	PP5V_S0	7 8 58
TRUE	PP3V42_G3H	8
TRUE	PP3V3_S5	8 58 72
TRUE	PP3V3_SW_DPPWR	62
TRUE	PP3V3_S3	8
TRUE	PP3V3_WLAN_F	7 34 40
TRUE	PP3V3_S0	8 58 72
TRUE	PP3V3_S0_HDD_R	7 35
TRUE	PP3V3_ENET	8
TRUE	PP1V5R1V35_S3	8 72
TRUE	PP1V5_S0	8 58 72
TRUE	PP1V05_S0	8 58
TRUE	PP1V05_S0_MCP_PLL_UF	8
TRUE	PP0V9_S5	8
TRUE	PP0V9_ENET	8
TRUE	PPVCORE_S0_CPU	8 43
TRUE	PPVCORE_S0_MCP	8 43

(Need to add 27 GND TPs)

FUNC_TEST	Net	Pages
TRUE	SMC_PM_G2_EN	39 58
TRUE	PM_SLP_S4_L	19 39 58
TRUE	PM_SLP_S3_L	19 39 40 58

NO_TEST Nets

FSB Signals (Covered via CPU/MCP JTAG)

NO_TEST	Net	Pages
TRUE	FSB_A_L<35..3>	10 14 66
TRUE	FSB_ADS_L	10 14 66
TRUE	FSB_ADSTB_L<1..0>	10 14 66
TRUE	FSB_D_L<63..0>	10 14 66
TRUE	FSB_DINV_L<3..0>	10 14 66
TRUE	FSB_DSTB_L_N<3..0>	10 14 66
TRUE	FSB_DSTB_L_P<3..0>	10 14 66
TRUE	FSB_HIT_L	10 14 66
TRUE	FSB_HITM_L	10 14 66
TRUE	FSB_LOCK_L	10 14 66
TRUE	FSB_REQ_L<4..0>	10 14 66

SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/16/2010)

PAGE TITLE: Functional Test / No Test

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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PAGE: 7 OF 110

SHEET: 7 OF 74

"G3Hot" (Always-Present) Rails

=PP18V5_DCIN_CONN → PPDCIN_S5_S5
MIN_LINE_WIDTH=1mm
MIN_NECK_WIDTH=0.20mm
VOLTAGE=18.5V
MAKE_BASE=TRUE
- =PPDCIN_S5_CHGR

=PPBUS_G3H → PPBUS_G3H
MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.25 mm
VOLTAGE=8.4V
MAKE_BASE=TRUE
- =PPBUS_G3H_R_IN
- =PPBUS_S0_LCDBKLT
- =PPVIN_S5_P5VP3V3
- =PPBUS_5V_S5

=PPBUS_G3H_R_OUT → PPBUS_G3H_ISNS
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.25 MM
VOLTAGE=8.4V
MAKE_BASE=TRUE
- =PPVIN_S5_CPU_IMVP
- =PPVIN_S3_DDRREG
- =PPVIN_S0_MCPCORE
- =PPVIN_S0_CPUVTTS0

=PP3V42_G3H_REG (0.064) → PP3V42_G3H
MIN_LINE_WIDTH=0.6 MM
MIN_NECK_WIDTH=0.2 MM
VOLTAGE=3.42V
MAKE_BASE=TRUE
- PP3V3_G3_RTC
- =PP3V42_G3H_SMCUSBMUX
- =PP3V42_G3H_ONEWIRE
- =PP3V42_G3H_SMBUS_SMC_BSA
- =PP3V42_G3H_BMON_ISNS
- =PP3V42_G3H_TPAD
- =PP3V42_G3H_CHGR
- =PP3V42_G3H_PWRCTL
- =PP3V3_S5_SMC
- =PP3V3_S5_LPCPLUS
- =PPVIN_S5_SMCVREF

"S5" Rails

=PP3V3_S5_REG (1.07 A + S3 + S0) → PP3V3_S5
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=3.3V
MAKE_BASE=TRUE
- =PP3V3_S5_MCP_GPIO
- =PP3V3_S5_MCP
- =PP3V3_S5_MCPPWRGD
- =PP3V3_S5_SMBUS_SMC_MGMT
- =PP3V3_S5_ROM
- =PP3V3_S5_P0V9S5
- =PP3V3_S5_VMON
- =PP3V3_S5_P3V3S3FET
- =PP3V3_S5_P3V3ENETFET
- =PP3V3_S5_P3V3S0FET
- =PP3V3_S5_P0V9ENETFET
- =PP3V3_S5_DP_PORT_PWR
- =PP3V3_S0_LCD
- =PP3V3_S5_TPAD
- =PP3V3_SMC_PME

=PP0V9_S5_REG (0.290 A) → PP0V9_S5
MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=0.9V
MAKE_BASE=TRUE
- 105 mA/241 mA =PP0V9_S5_MCP_VDD_AUXC
- =PP0V9_ENET_P0V9ENETFET

"S3" Rails

=PP3V3_S3_FET (1.274 A) → PP3V3_S3
MIN_LINE_WIDTH=0.5 mm
MIN_NECK_WIDTH=0.25 mm
VOLTAGE=3.3V
MAKE_BASE=TRUE
- =PP3V3_S3_MCP_GPIO
- =PP3V3_S3_VREFMRGN
- =PP3V3_S3_BT
- =PP3V3_S3_WLAN
- =PP3V3_S3_CARDREADER
- =PP3V3_S3_SMBUS_SMC_A_S3
- =PP3V3_S3_1V5S3ISNS
- =PP3V3_S3_WLANISNS
- =PP3V3_S3_TPAD
- =PP3V3_S3_SMS
- =PP3V3_S3_DBGLEDS
- =PP3V3_S3_PDCISENS

=PP5V_S3_REG (5.40 A) → PP5V_S3
MIN_LINE_WIDTH=0.5 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=5V
MAKE_BASE=TRUE
- =PP5V_S3_MCPDDRFET
- =PP5V_S3_RTUSB
- =PP5V_S3_SYSLED
- =PP5V_S3_TPAD
- =PP5V_S3_AUDIO_AMP
- =PP5V_S3_LIO_CONN
- =PP5V_S3_DDRREG
- =PP5V_S3_P5VS0FET

=PPDDR_S3_REG (11.30 A) → PP1V5R1V35_S3
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=1.5V
MAKE_BASE=TRUE (OR 1.35V)
- =PP1V5R1V35_S3_MCP_MEM
- =PP1V5R1V35_S0_MCPDDRFET
- 0 mA =PPLVDDR_S3_MEM_A
- 4250 mA =PPLVDDR_S3_MEM_B
- =PPVIN_S0_DDRREG_LDO

=PPVTT_S3_DDR_BUF → PPDDRVREF_S3
MIN_LINE_WIDTH=0.3 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=0.75V
MAKE_BASE=TRUE (OR 0.675V)

"ENET" Rails

=PP3V3_ENET_FET_R (300mA) → PP3V3_ENET
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=3.3V
MAKE_BASE=TRUE
- 300mA =PP3V3_ENET_MCP_RMGT
- =PP3V3_ENET_MCP_PLL_MAC

=PP0V9_ENET_FET → PP0V9_ENET
MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=0.9V
MAKE_BASE=TRUE
- =PP0V9_ENET_MCP_RMGT

"S0" Rails

=PP5V_S0_FET (0.100 A) → PP5V_S0
MIN_LINE_WIDTH=0.6 MM
MIN_NECK_WIDTH=0.2 MM
VOLTAGE=5V
MAKE_BASE=TRUE
- =PP5V_S0_MCPFSBFET
- =PP5V_S0_LPCPLUS
- =PP5V_S0_FAN
- =PP5V_S0_CPU_IMVP
- =PP5V_S0_MCPREG
- =PP5V_S0_CPUVTTS0
- =PP5V_S0_BKL
- =PP5VR3V3_S0_DPCADET

=PP3V3_S0_FET (3.30 A) → PP3V3_S0
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=3.3V
MAKE_BASE=TRUE
- =PP3V3_S0_XDP
- =PP3V3_S0_MCP_GPIO
- =PP3V3_S0_MCP_HVDD
- =PP3V3_S0_MCP
- =PP3V3_S0_MCP_PLL_UF
- =PP3V3_S0_HDD
- =PP3V3_S0_AUDIO
- =PP3V3_S0_SMC
- =PP3V3_S0_DEBUGROM
- =PP3V3_S0_SMBUS_MCP_0
- =PP3V3_S0_SMBUS_MCP_1
- =PP3V3_S0_SMBUS_SMC_B_S0
- =PP3V3_S0_SMBUS_SMC_0_S0
- =PP3V3_S0_BKLTISNS
- =PP3V3_S0_HDDISNS
- =PP3V3_S0_CSREGISNS
- =PP3V3_S0_MCPCOREISNS
- =PP3V3_S0_MCPDDRISNS
- =PP3V3_S0_CPUTHMSNS
- =PP3V3_S0_MCPTHMSNS
- =PP3V3_S0_FAN
- =PP3V3_S0_IMVP
- =PP3V3_S0_MCP_PLL_VLDO
- =PP3V3_S0_P1V5S0
- =PP3V3_S0_PWRCTL
- =PP3V3_S0_DPCONN
- =PP3V3_S0_BKL_VDDIO

=PP1V5_S0_REG (0.210 A) → PP1V5_S0
MIN_LINE_WIDTH=0.6 MM
MIN_NECK_WIDTH=0.2 MM
VOLTAGE=1.5V
MAKE_BASE=TRUE
- =PP1V5_S0_CPU
- =PP1V5_S0_MCP_PLL_VLDO
- =PP3V3R1V5_S0_MCP_HDA
- =PP1V8R1V5_S0_AUDIO

=PPCPUVTT_S0_REG (9.40 A) → PP1V05_S0
MIN_LINE_WIDTH=0.6 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=1.05V
MAKE_BASE=TRUE
- =PP1V05_S0_CPU
- =PP1V05_S0_XDP
- =PP1V05_S0_MCP_FSB
- =PP1V05_S0_MCP_M2CLK_DLL
- =PP1V05_S0_MCP_DP0_VDD
- =PP1V05_S0_MCP_PE_DVDD
- =PP1V05_S0_MCP_SATA_DVDD
- =PP1V05_S0_MCP_AVDD_UF
- =PP1V05_S0_MCP_PLL_UF_R
- =PP1V05_SW_MCP_FSB

=PP1V05_S0_MCP_PLL_OR → PP1V05_S0_MCP_PLL_UF
MIN_LINE_WIDTH=0.6 MM
MIN_NECK_WIDTH=0.2 MM
VOLTAGE=1.05V
MAKE_BASE=TRUE
- =PP1V05_S0_MCP_PLL_UF

=PPVCORE_S0_CPU_REG (18 A) → PPVCORE_S0_CPU
MIN_LINE_WIDTH=0.6 MM
MIN_NECK_WIDTH=0.25 MM
VOLTAGE=1.25V
MAKE_BASE=TRUE
- =PPVCORE_S0_CPU

=PPMCPCORE_S0_REG (23.8 A) → PPVCORE_S0_MCP
MIN_LINE_WIDTH=0.6 MM
MIN_NECK_WIDTH=0.2 MM
VOLTAGE=1.05V
MAKE_BASE=TRUE
- =PPVCORE_S0_MCP
- =PPVCORE_S0_MCPGFXFET

=PPVTT_S0_DDR_LDO (1.20 A) → PPDDRVTT_S0
MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=0.75V
MAKE_BASE=TRUE (OR 0.675V)
- =PPDDRVTT_S0_MEM_A
- =PPDDRVTT_S0_MEM_B

SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/11/2010)

PAGE TITLE: Power Aliases

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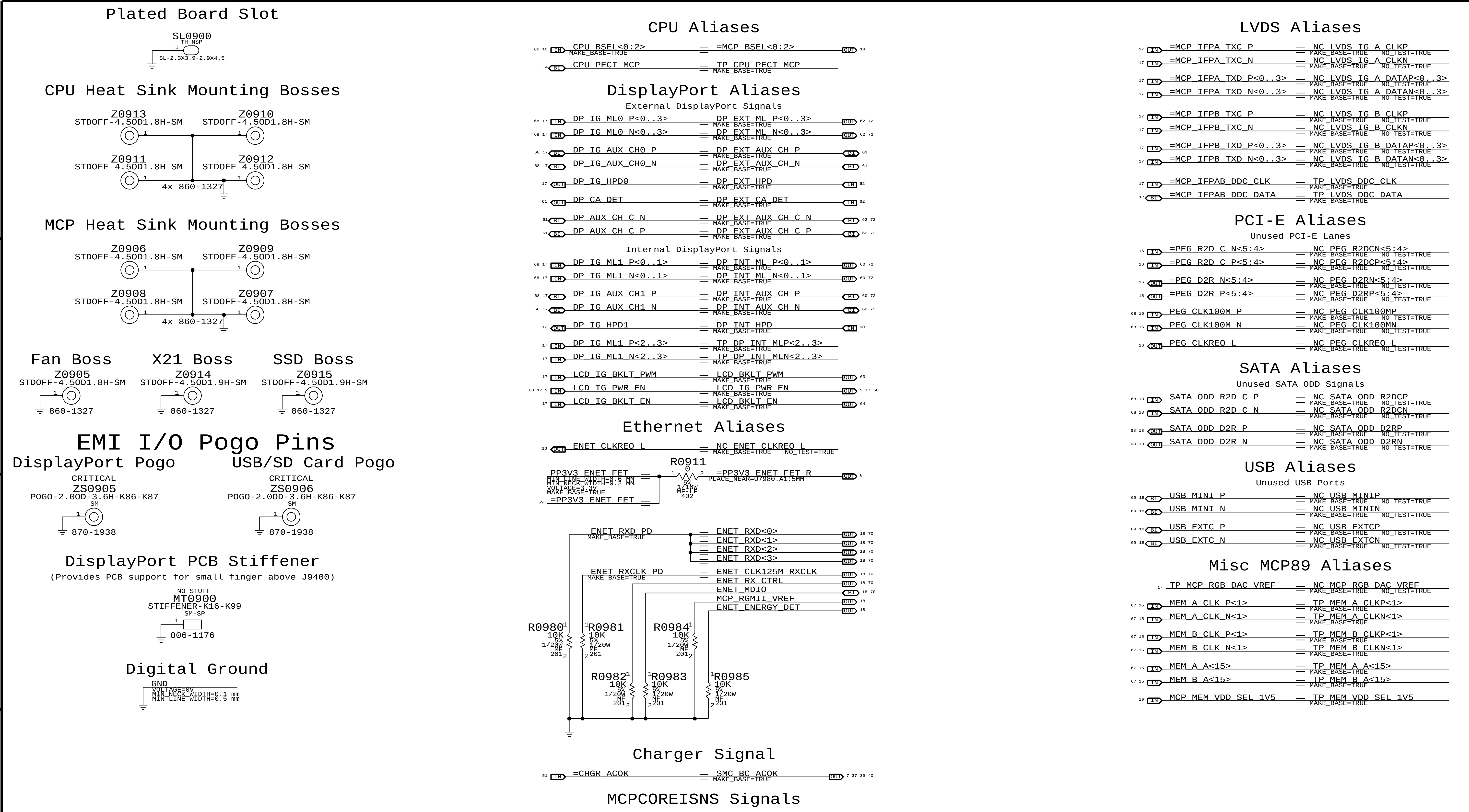

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PAGE TITLE: Signal Aliases

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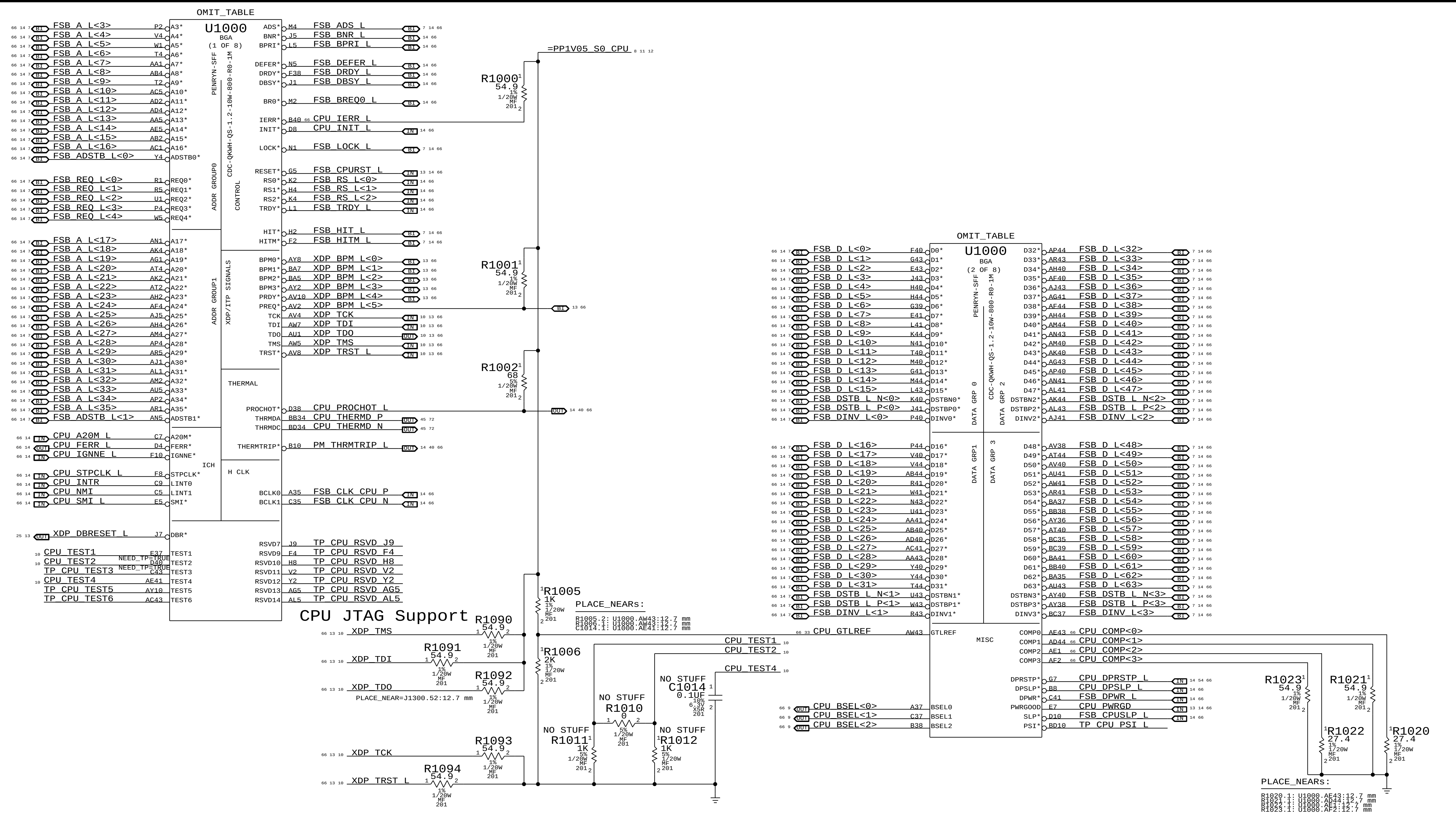

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

CPU FSB

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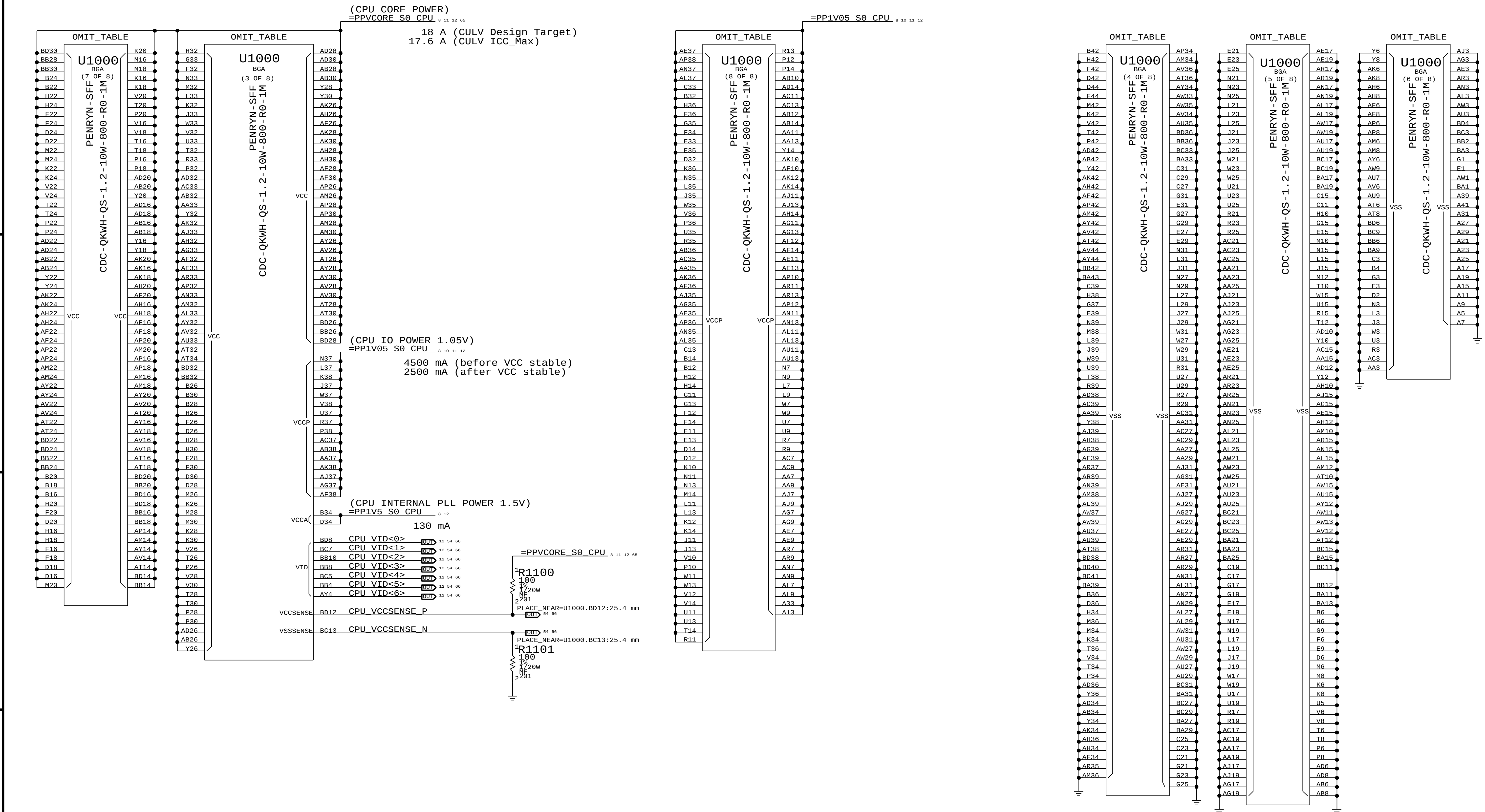

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

CPU Power & Ground

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CPU VCORE HF AND BULK DECOUPLING

4x 270uF. 32x 10uF 0603, 28x 2.2uF 0402 + 40x 2.2uF 0402

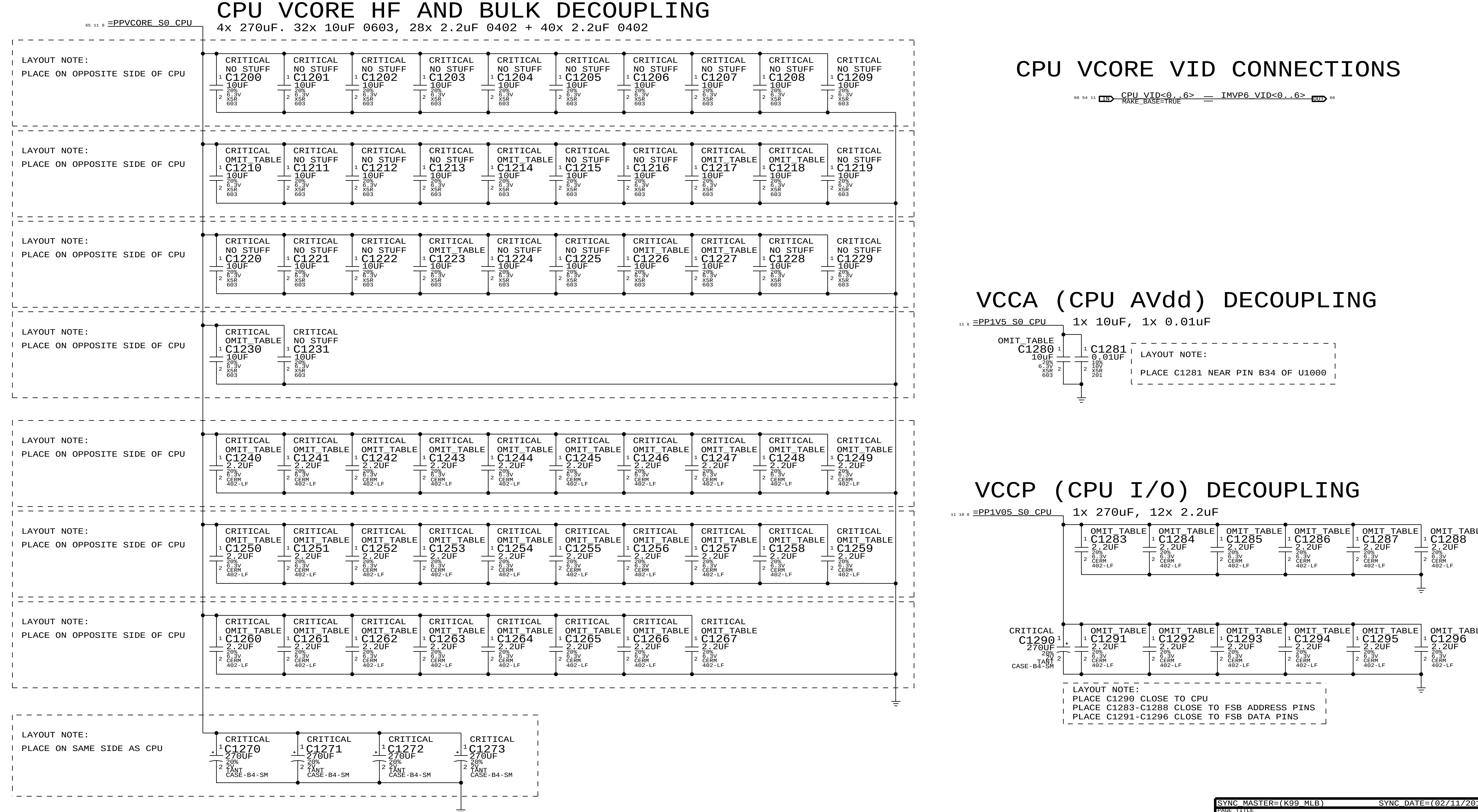
CPU VCORE VID CONNECTIONS

VCCA (CPU AVdd) DECOUPLING

1x 10uF, 1x 0.01uF

VCCP (CPU I/O) DECOUPLING

1x 270uF, 12x 2.2uF



SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/11/2010)

PAGE TITLE: CPU Decoupling & VID

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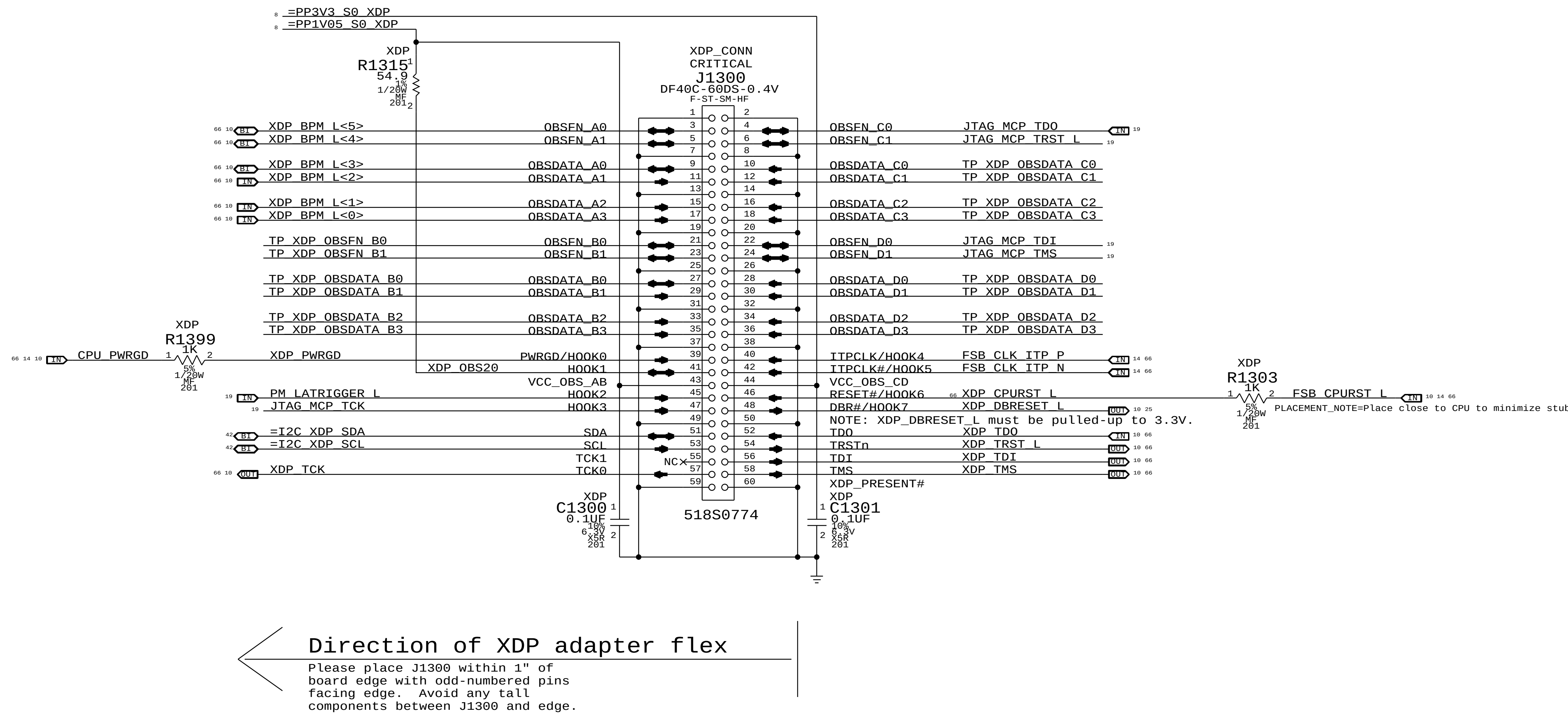
Micro2-XDP Connector

NOTE: This is not the standard XDP pinout.

Use with 920-0782 Adapter Flex to support chipset debug.



SYNC_MASTER=K99_MLB SYNC_DATE=03/01/2010

PAGE TITLE
eXtended Debug Port (Micro-XDP)

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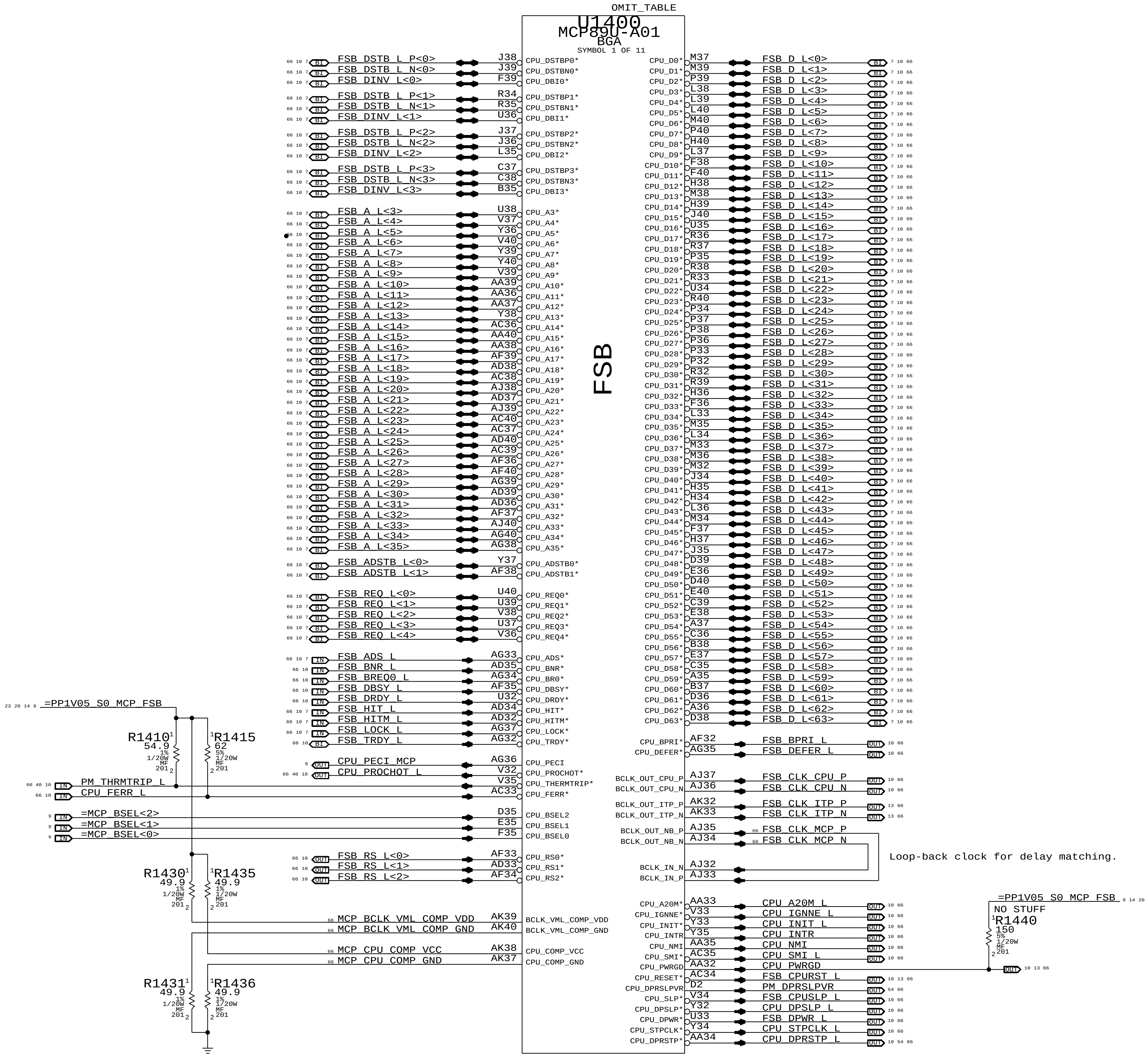

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PAGE TITLE

MCP CPU Interface

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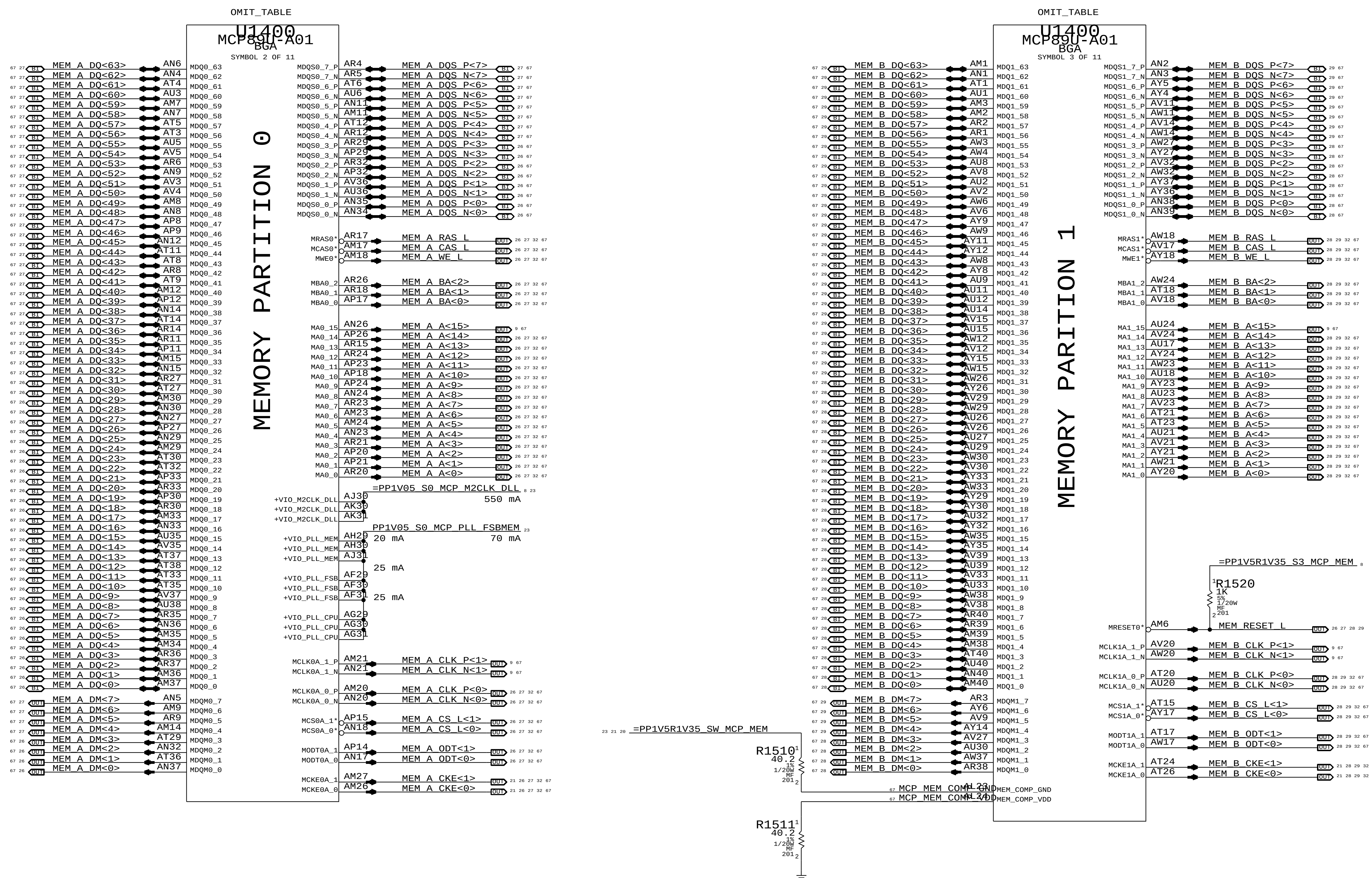

SYNC_MASTER=K99_MLB	SYNC_DATE=04/08/2010
PAGE TITLE: MCP Memory Interface	
Apple Inc.	DRAWING NUMBER 051-8467 SIZE D
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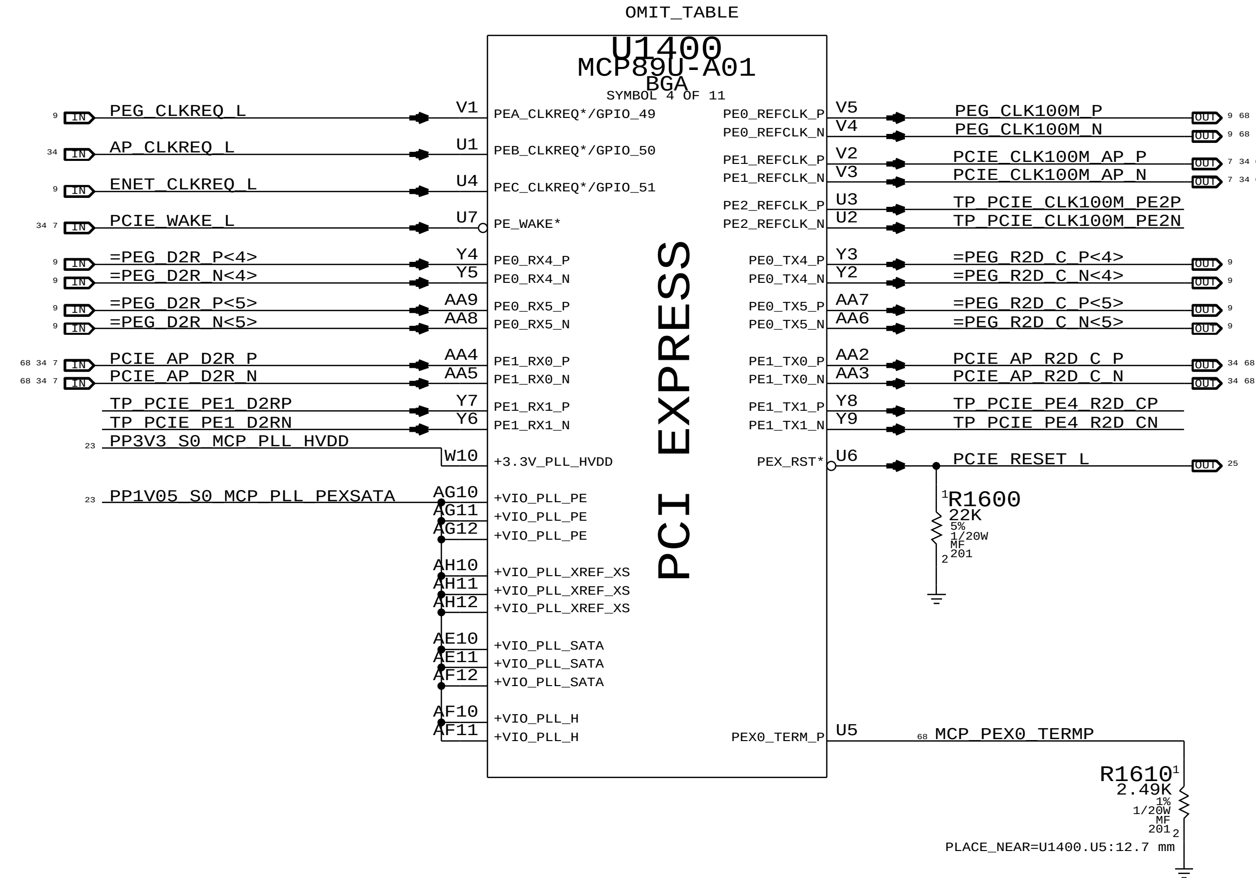
	SHEET 15 OF 74

Current numbers from MCP89 A01 Bring-Up Support document (MCP80_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.



PE0 ports are Gen2-capable. 4 RCs: 4x, x2, x1, x1
PE1 ports are Gen1-only. 2 RCs: x1, x1

If PE0[3:0] are not used,
+VIO_PE_AVDD0 and +VIO_PE_DVDD0 can be GND

If PE0[4:5] and PE1[0:1] are not used,
+VIO_PE_AVDD1 and +VIO_PE_DVDD1 can be GND

Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

MCP PCIe Interfaces

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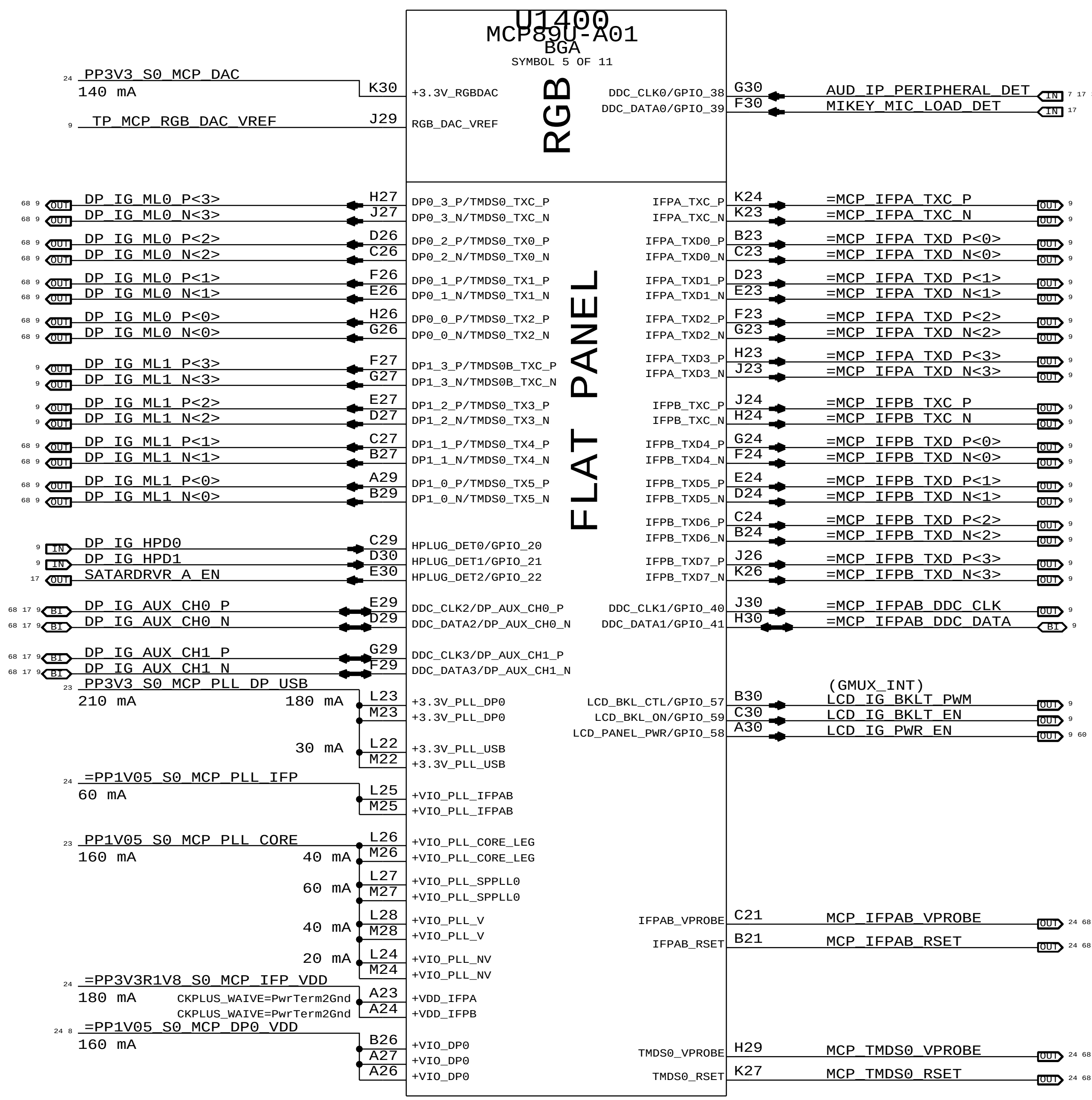
OMIT_TABLE



NOTE: 100K pull-downs required if HPLUG_DET0/HPLUG_DET1 are not used.

RGB DAC Disable:

Okay to float all RGB_DAC signals.
DDC_CLK0/DDC_DATA0 pull-ups still required (or use as GPIOs).
Connect +3.3V_RGBDAC pin to GND.

NOTE: No Composite/S-Video/Component Video support on MCP89

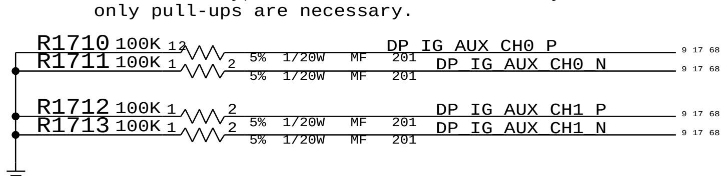
Interface Mode

MCP Signal	TMDS/HDMI	LVDS
=MCP_IFPA_TXC_P/N	TMDS_IG_TXC_P/N	LVDS_IG_A_CLK_P/N
=MCP_IFPA_TXD_P/N<0>	TMDS_IG_TXD_P/N<0>	LVDS_IG_A_DATA_P/N<0>
=MCP_IFPA_TXD_P/N<1>	TMDS_IG_TXD_P/N<1>	LVDS_IG_A_DATA_P/N<1>
=MCP_IFPA_TXD_P/N<2>	TMDS_IG_TXD_P/N<2>	LVDS_IG_A_DATA_P/N<2>
=MCP_IFPA_TXD_P/N<3>	(UNUSED)	LVDS_IG_A_DATA_P/N<3>
=MCP_IFPB_TXC_P/N	(UNUSED)	LVDS_IG_B_CLK_P/N
=MCP_IFPB_TXD_P/N<0>	TMDS_IG_TXD_P/N<3>	LVDS_IG_B_DATA_P/N<0>
=MCP_IFPB_TXD_P/N<1>	TMDS_IG_TXD_P/N<4>	LVDS_IG_B_DATA_P/N<1>
=MCP_IFPB_TXD_P/N<2>	TMDS_IG_TXD_P/N<5>	LVDS_IG_B_DATA_P/N<2>
=MCP_IFPB_TXD_P/N<3>	(UNUSED)	LVDS_IG_B_DATA_P/N<3>
=MCP_IFPAB_DDC_CLK	TMDS_IG_DDC_CLK	LVDS_IG_DDC_CLK
=MCP_IFPAB_DDC_DATA	TMDS_IG_DDC_DATA	LVDS_IG_DDC_DATA

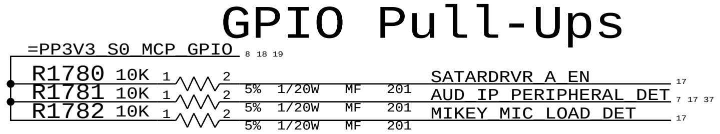
LVDS: Power +VDD_IFPx at 1.8V
TMDS: Power +VDD_IFPx at 3.3V

DDC Mode Pull-downs

NOTE: DP_AUX_CH1 also requires pull-downs if used for dual-mode DisplayPort (DP++). If unused no pulls are necessary, if used for TMDS/HDMI only then only pull-ups are necessary.



GPIO Pull-Ups



Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE
MCP Graphics

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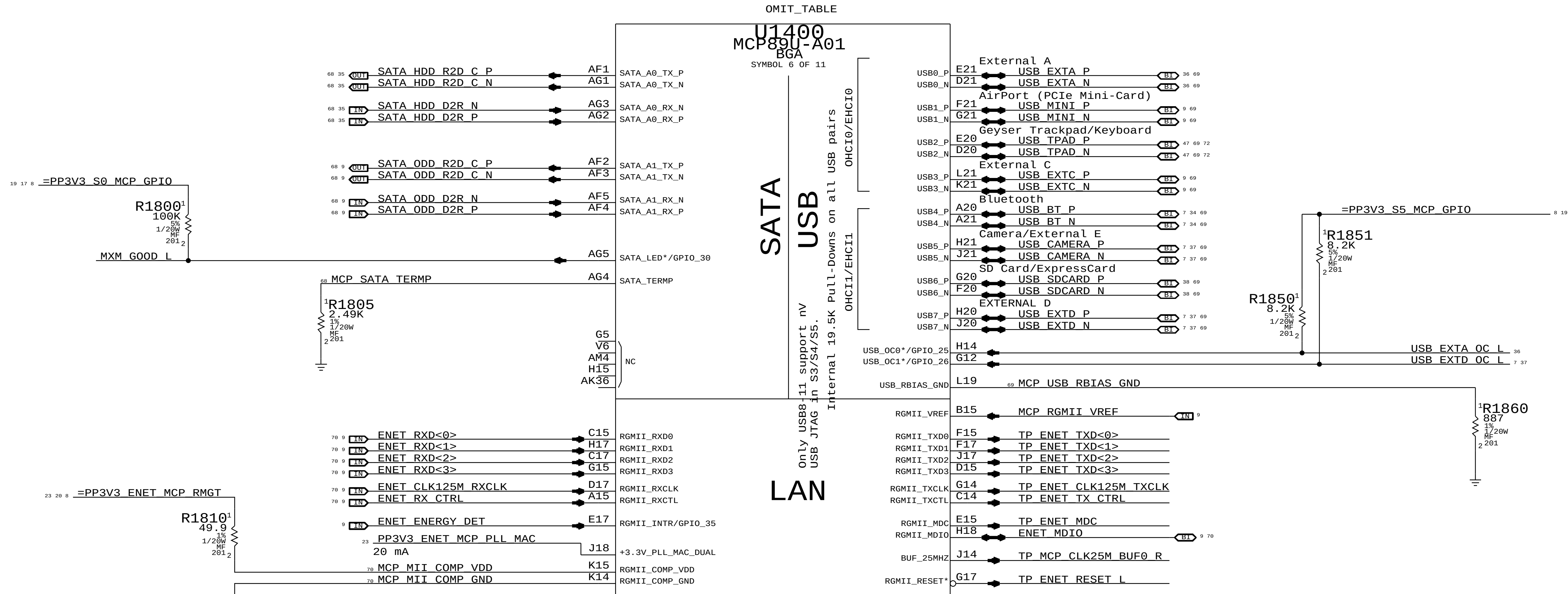

Internal MAC Disable:

Connect RGMII_RXD<0:3> together to 10K pull-down.
Connect RGMII_RXCLK to 10K pull-down.
Connect RGMII_RXCTL to 10K pull-down.
Connect RGMII_INTR to 10K pull-down (if not used as GPIO).
+3.3V_PLL_MAC_DUAL must remain connected to 3.3V RMGT rail.
RGMII_COMP_VDD/_GND must remain connected as shown.
Connect RGMII_VREF to 10K pull-down.
Connect RGMII_MDIO to 10K pull-down.
All other pins can be left TP or NC.

Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE
MCP SATA, USB & Ethernet

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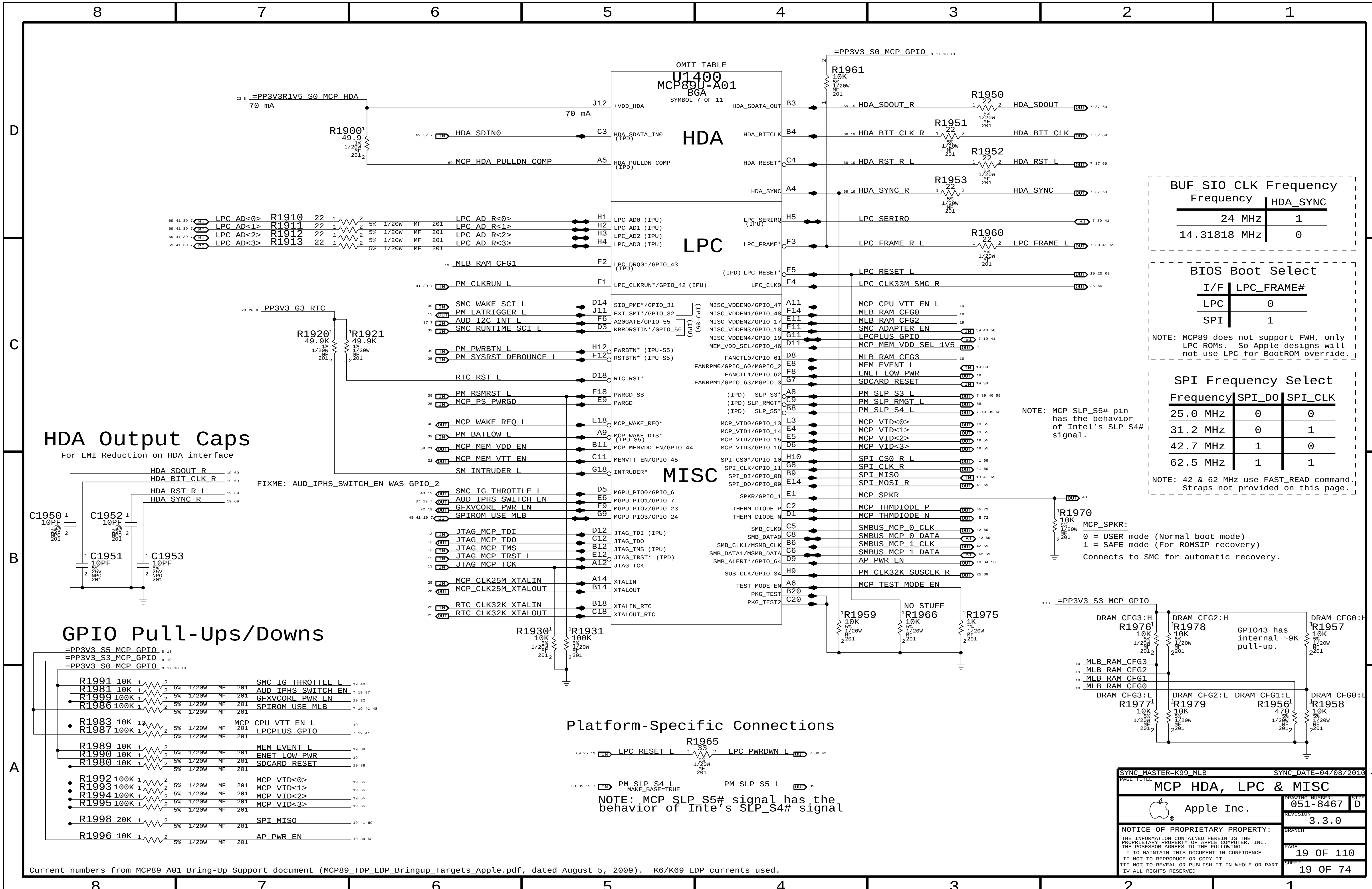

BUF_SIO_CLK Frequency

Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

BIOS Boot Select

I/F	LPC_FRAME#
LPC	0
SPI	1

NOTE: MCP89 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.

SPI Frequency Select

Frequency	SPI_DO	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
42.7 MHz	1	0
62.5 MHz	1	1

NOTE: 42 & 62 MHz use FAST_READ command. Straps not provided on this page.

Current numbers from MCP89 A01 Bring-Up Support document (MCP89_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: MCP HDA, LPC & MISC

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REVISION: 3.3.0

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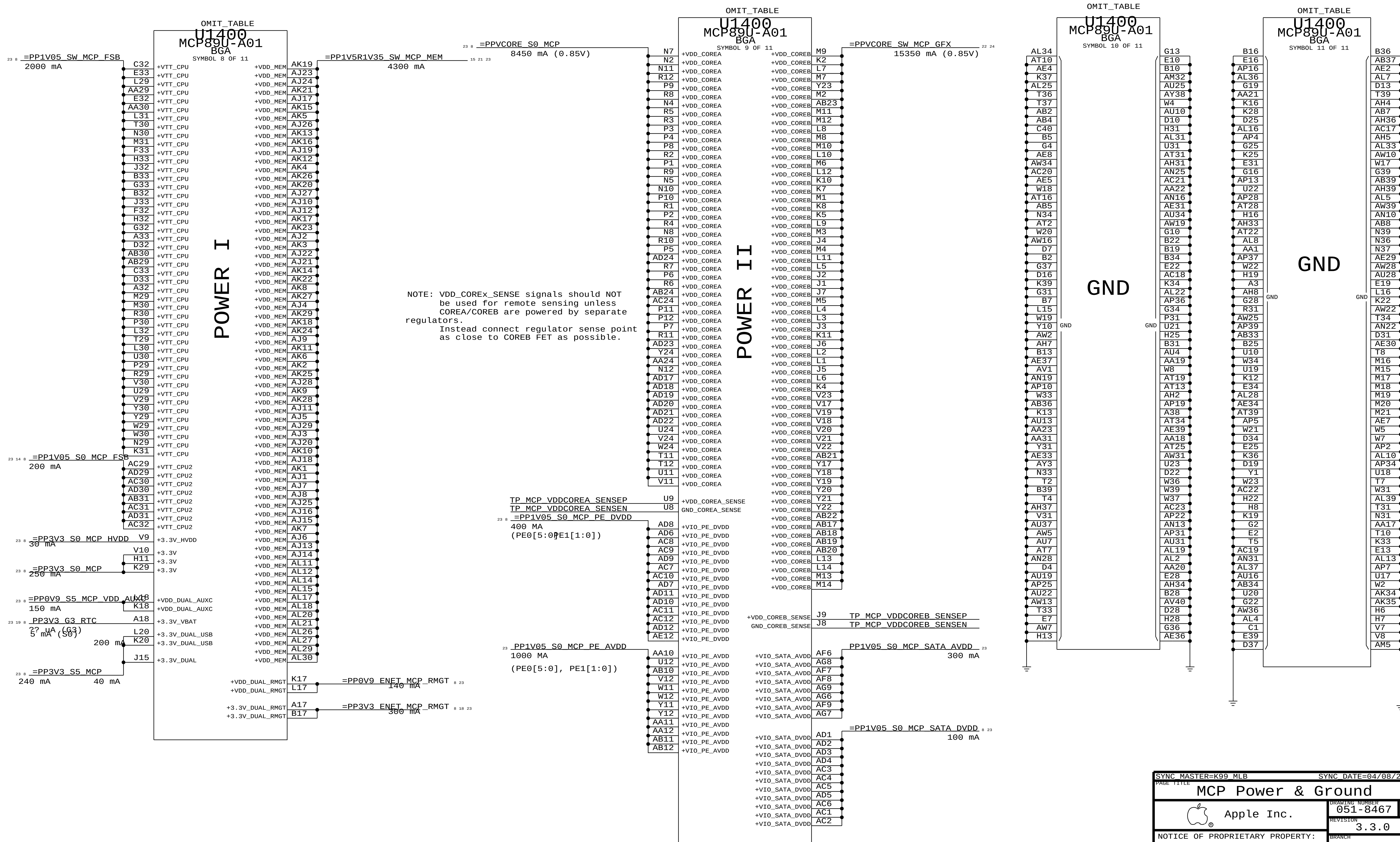
PAGE: 19 OF 110

SHEET: 19 OF 74

NOTE: "SW" rails are dynamically switched in the S0 state as needed, controlled by MCP89 GPIOs.



Current numbers from MCP89 A01 Bring-Up Support document (MCP80_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

MCP Power & Ground

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REVISION 3.3.0

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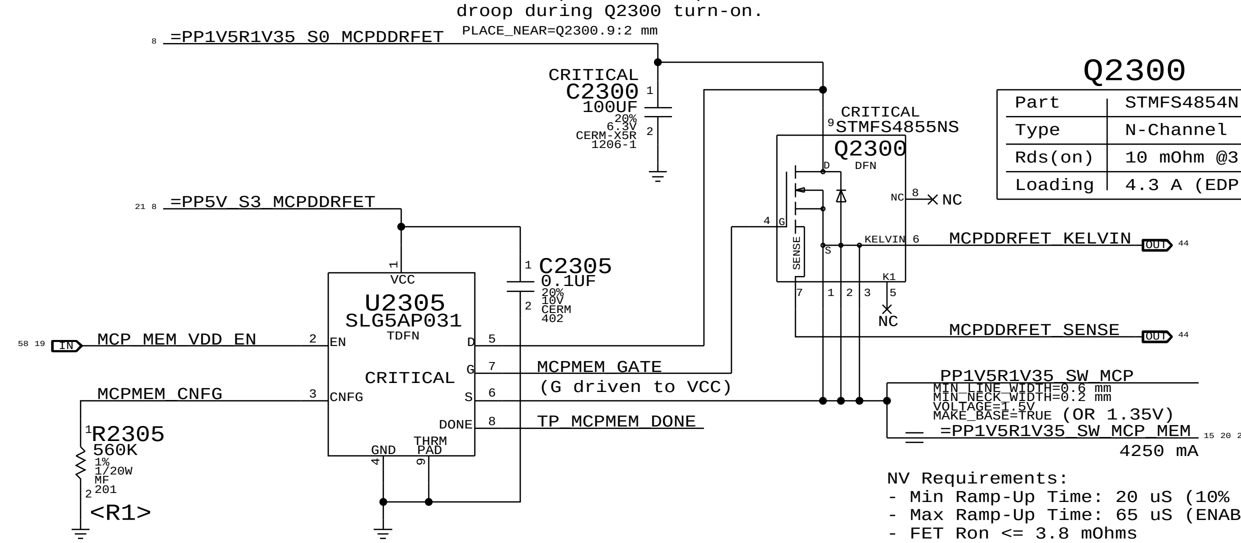


C2300 helps reduce input rail droop during Q2300 turn-on.

Q2300

Part	STMFS4854N
Type	N-Channel
Rds(on)	10 mOhm @3.2V
Loading	4.3 A (EDP)

Approx. Ramp Time (EN to 1.35V, uS): 7.91 + 0.0678 * R1(Kohms)

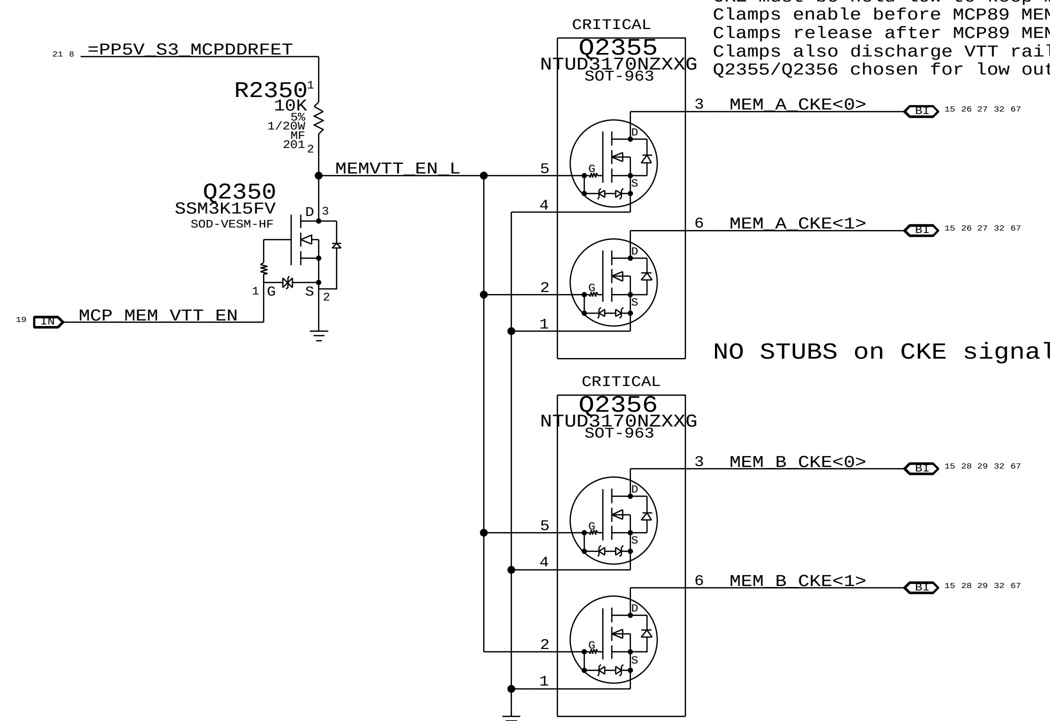
NV Requirements:
- Min Ramp-Up Time: 20 uS (10% to 90%)
- Max Ramp-Up Time: 65 uS (ENABLE to 90%)
- FET Ron <= 3.8 mOhms

NOTE: nVidia recommends Infineon BSC030N03MS for Q2300.

Gated Rail Savings: 120mW

DIMM CKE Clamps

CKE must be held low to keep memory in self-refresh.
Clamps enable before MCP89 MEMVDD rail switched off.
Clamps release after MCP89 MEMVDD is up and CKEs are driven by MCP89.
Clamps also discharge VTT rail via termination resistor on each CKE signal on DIMM.
Q2355/Q2356 chosen for low output capacitance.



NO STUBS on CKE signals!

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: MCP89 Memory Rail Gating

DRAWING NUMBER: 051-8467 SIZE: D

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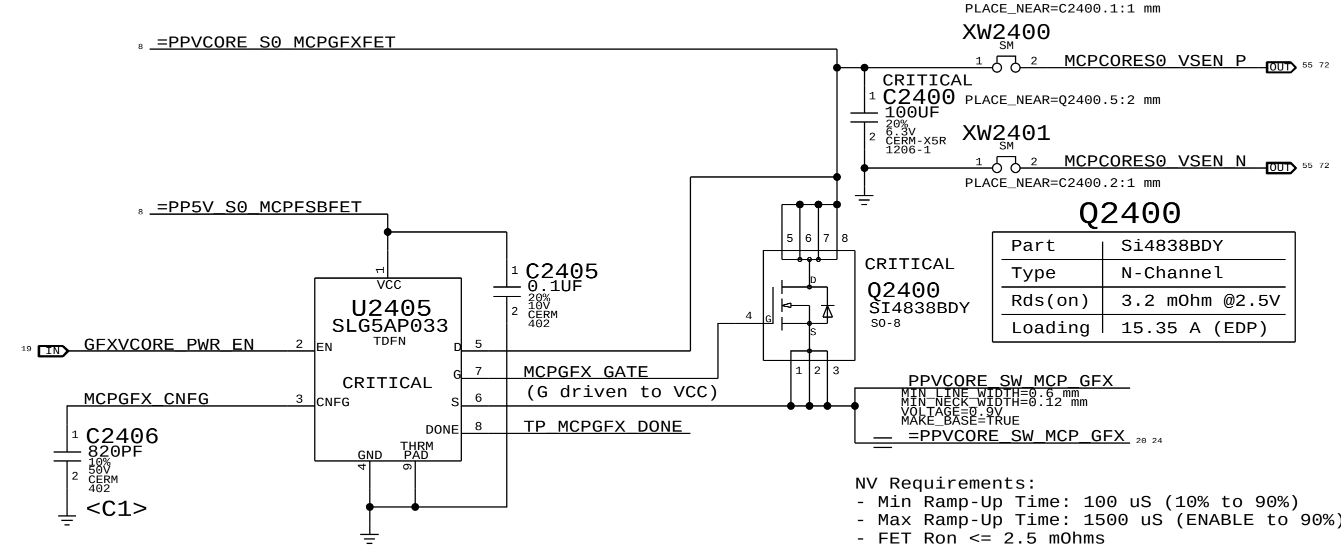
C2400 helps reduce input rail
droop during Q2400 turn-on.



Approx. Ramp Time (EN to 1V, uS): 43.9 + 0.6943 * C1(pF)

Q2400

Part	Si4838BDY
Type	N-Channel
Rds(on)	3.2 mOhm @2.5V
Loading	15.35 A (EDP)

NV Requirements:
- Min Ramp-Up Time: 100 uS (10% to 90%)
- Max Ramp-Up Time: 1500 uS (ENABLE to 90%)
- FET Ron <= 2.5 mOhms

NOTE: nVidia recommends Infineon BSC020N03MS for Q2400.

Gated Rail Savings: 860mW

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE
MCP89 GFX Core Rail Gating

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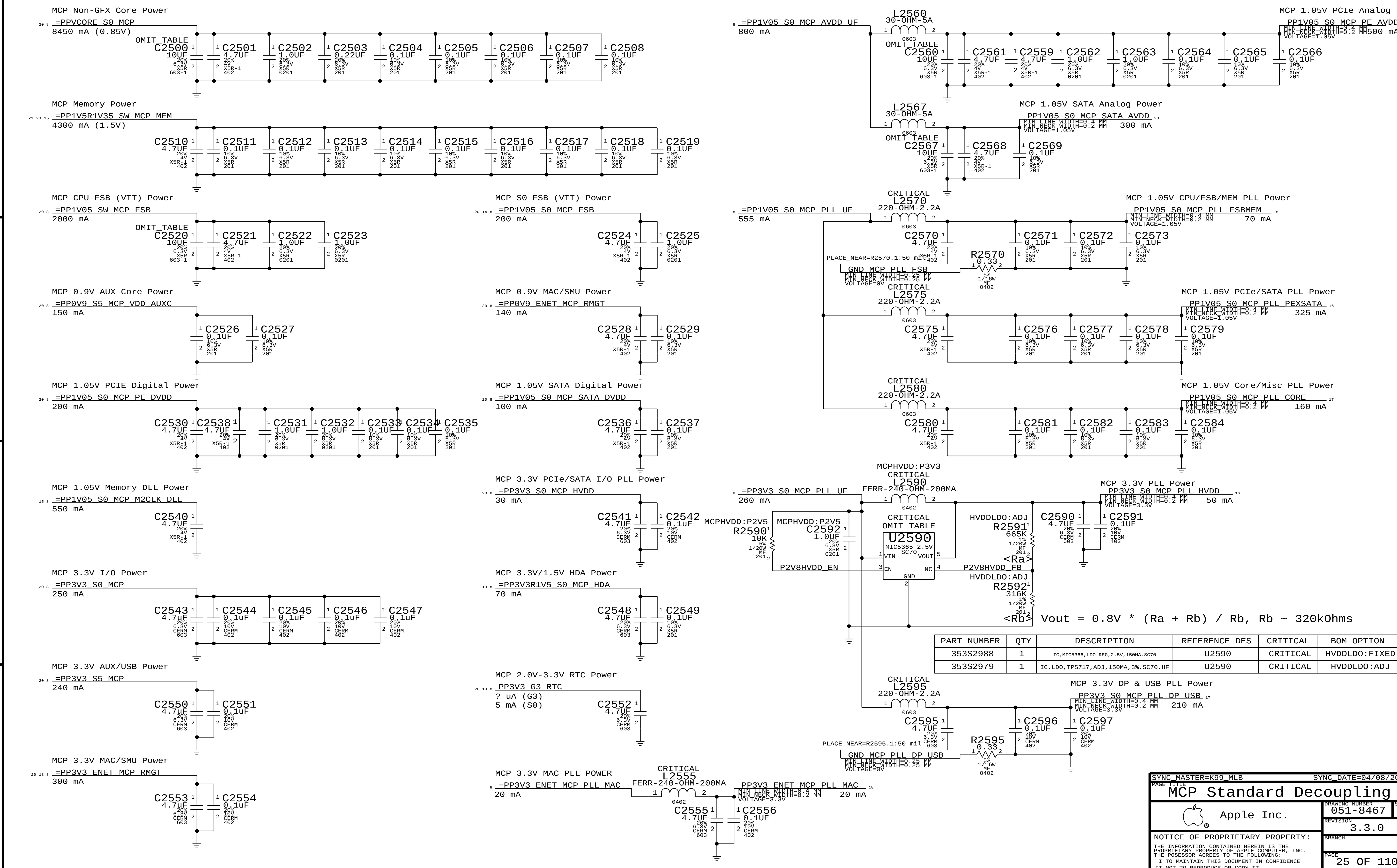
PAGE 24 OF 110
SHEET 22 OF 74



Vout = 0.8V * (Ra + Rb) / Rb, Rb ~ 320kOhms

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S2988	1	IC,MIC5366,LDO REG,2.5V,150MA,SC70	U2590	CRITICAL	HVDDLDO:FIXED
353S2979	1	IC,LDO,TPS717,ADJ,150MA,3%,SC70,HF	U2590	CRITICAL	HVDDLDO:ADJ

Current numbers from MCP89 A01 Bring-Up Support document (MCP80_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE
MCP Standard Decoupling

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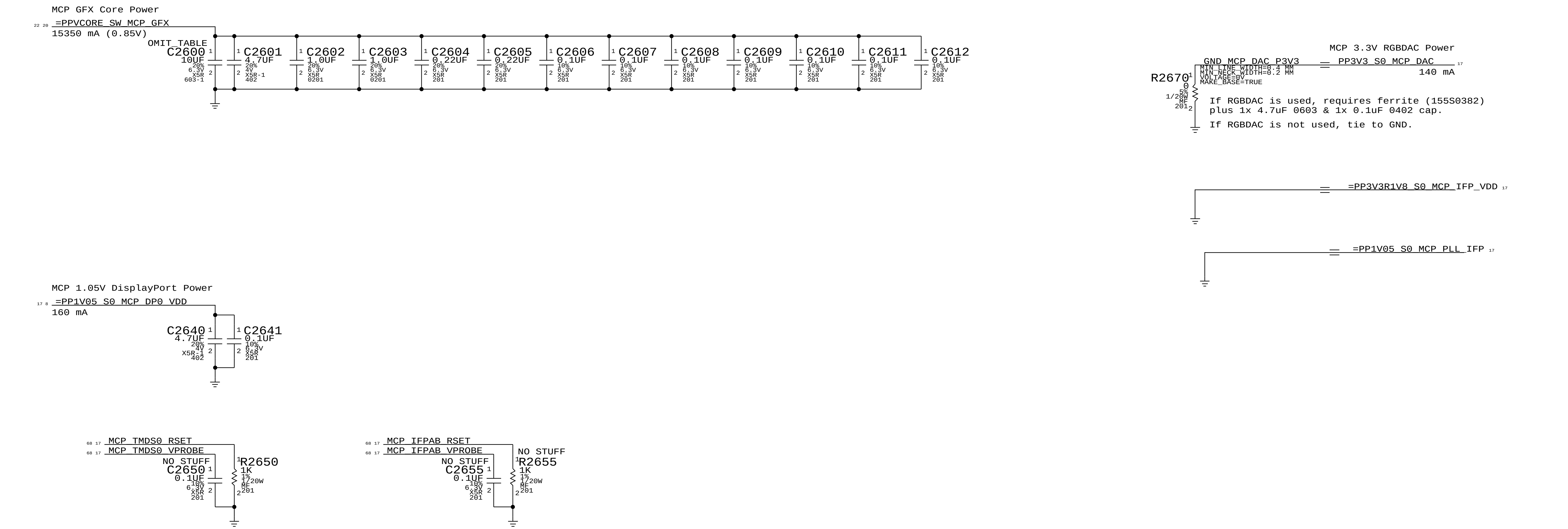

Current numbers from MCP89 A01 Bring-Up Support document (MCP80_TDP_EDP_Bringup_Targets_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE
MCP Graphics Support

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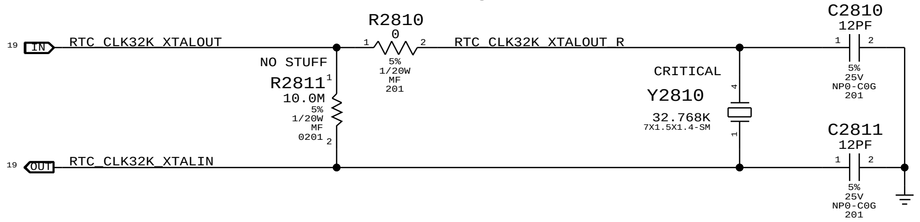
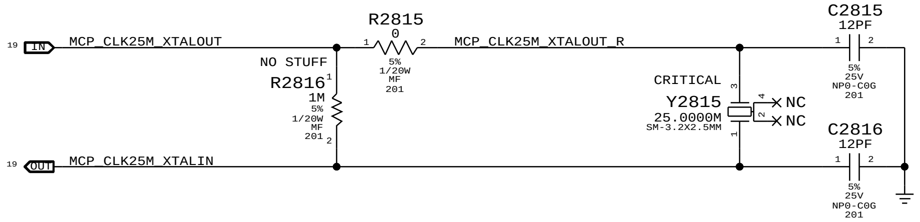
RTC Crystal

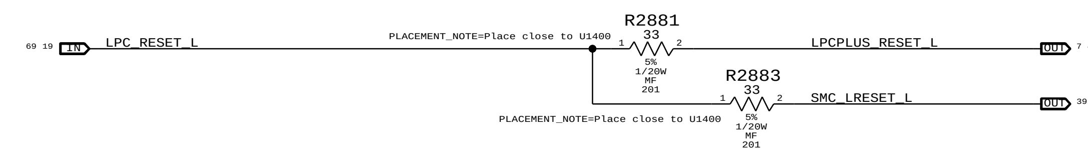


MCP 25MHz Crystal

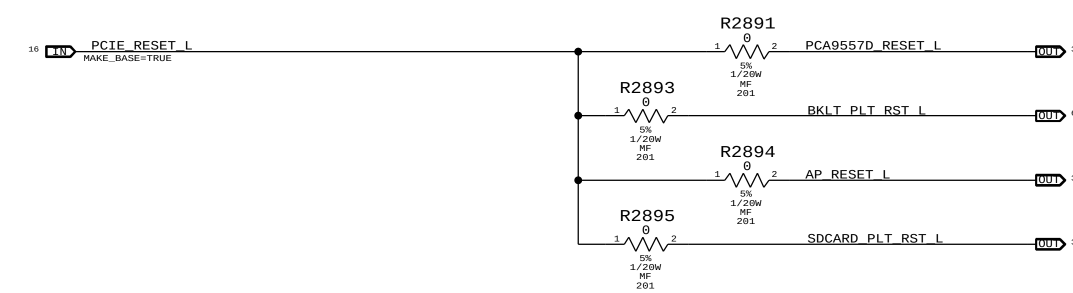


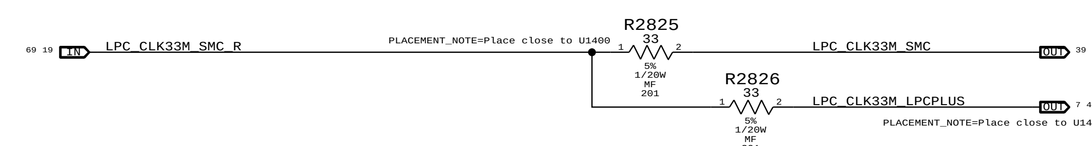
Platform Reset Connections

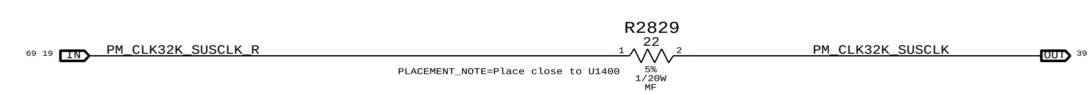
LPC Reset (Unbuffered)



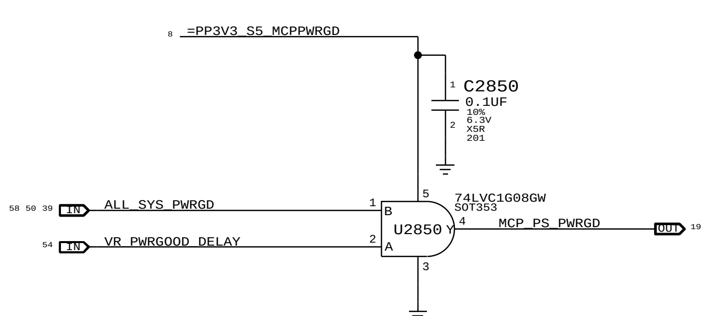
PCIE Reset (Unbuffered)



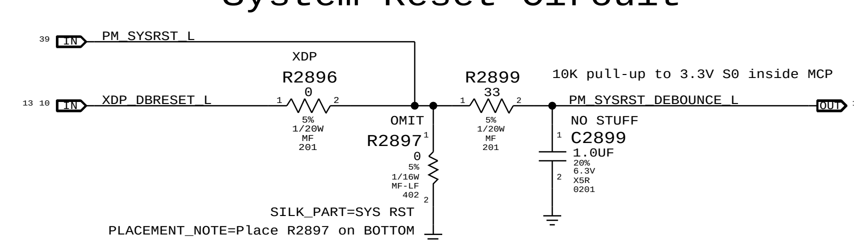




MCP S0 PWRGD & CPU_VLD



System Reset Circuit



SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/11/2010)

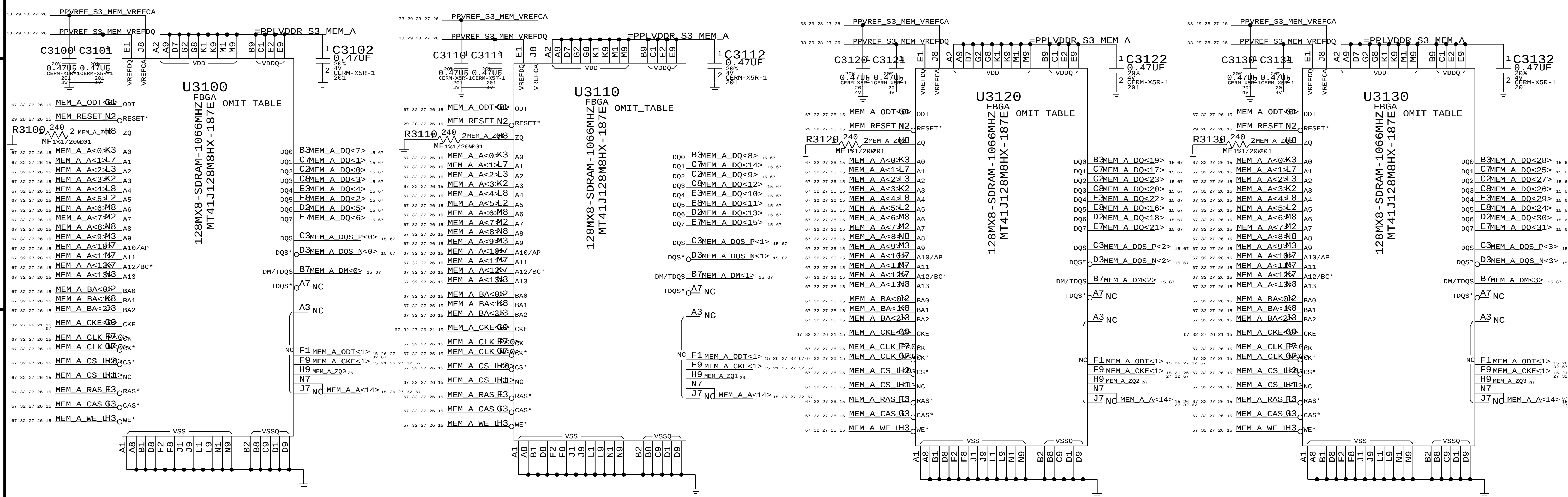
PAGE TITLE: SB Misc

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

BRANCH



A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

DDR3 DRAM Channel A (0-31)

Apple Inc.

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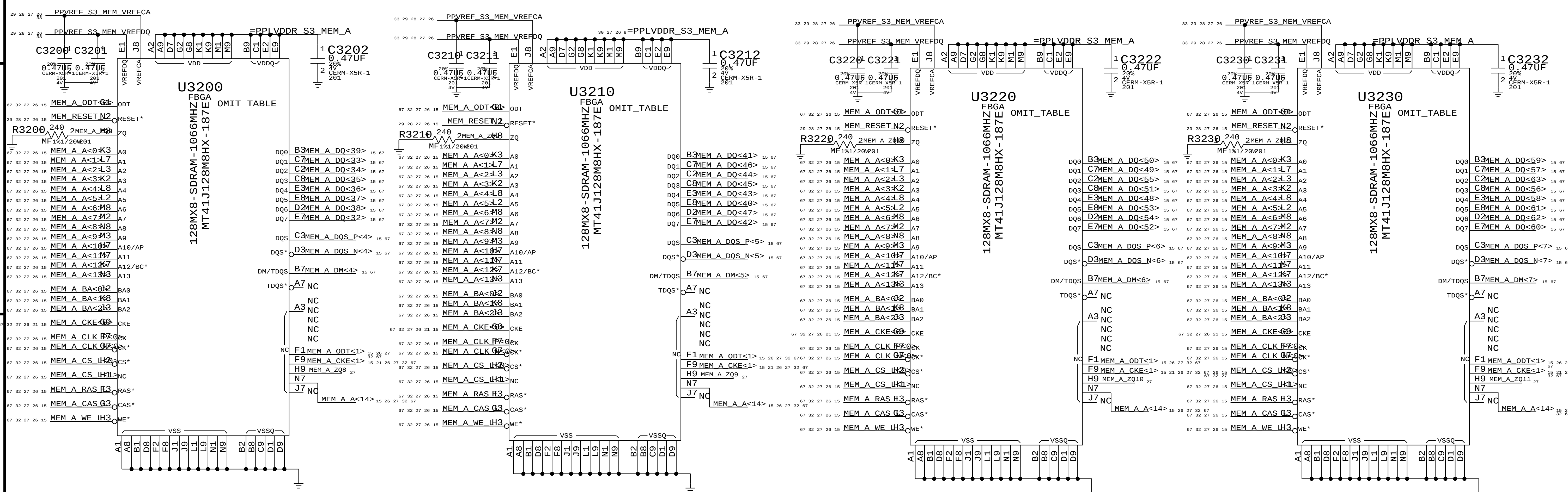
REVISION 3.3.0

BRANCH



A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

DDR3 DRAM Channel A (32-63)

Apple Inc.

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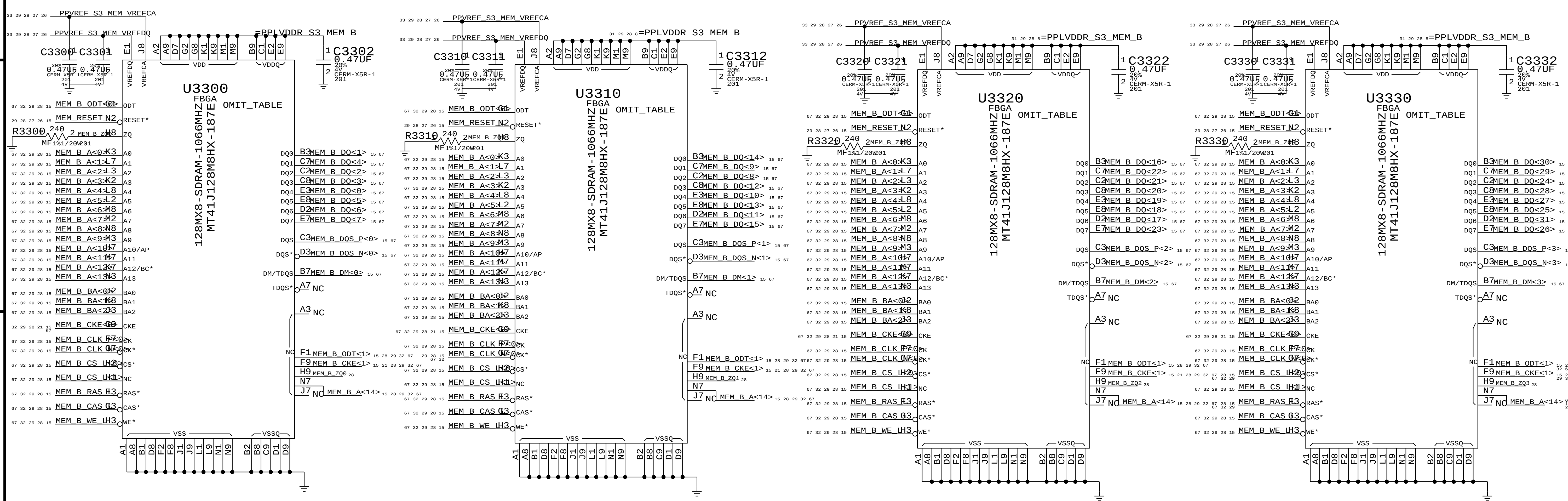
REVISION 3.3.0

BRANCH

PAGE 32 OF 110

SHEET 27 OF 74



A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

DDR3 DRAM Channel B (0-31)

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REVISION 3.3.0

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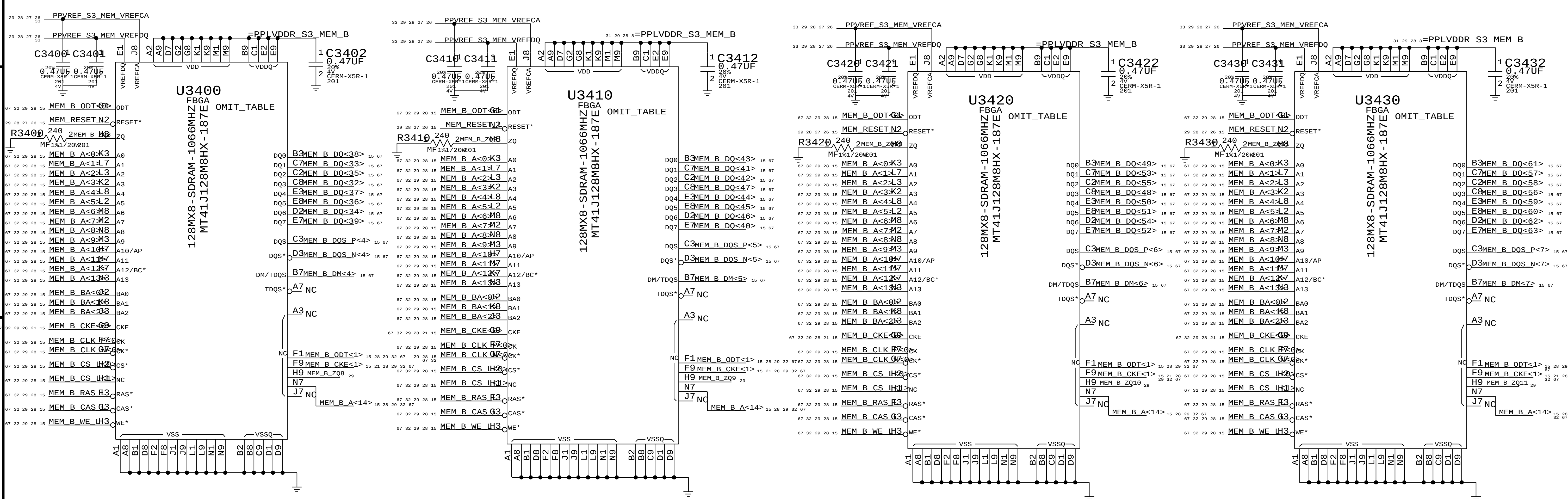

A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

DDR3 DRAM Channel B (32-63)

Apple Inc.

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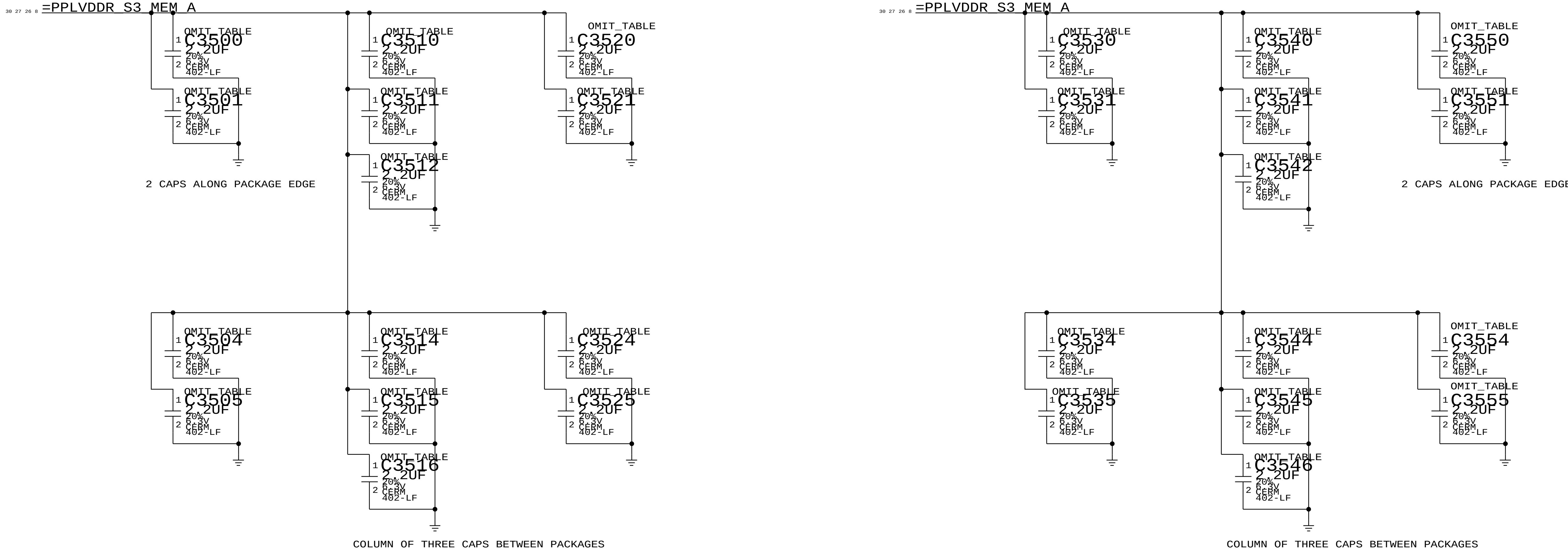
PAGE 34 OF 110

SHEET 29 OF 74



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

DDR BYPASSING 1

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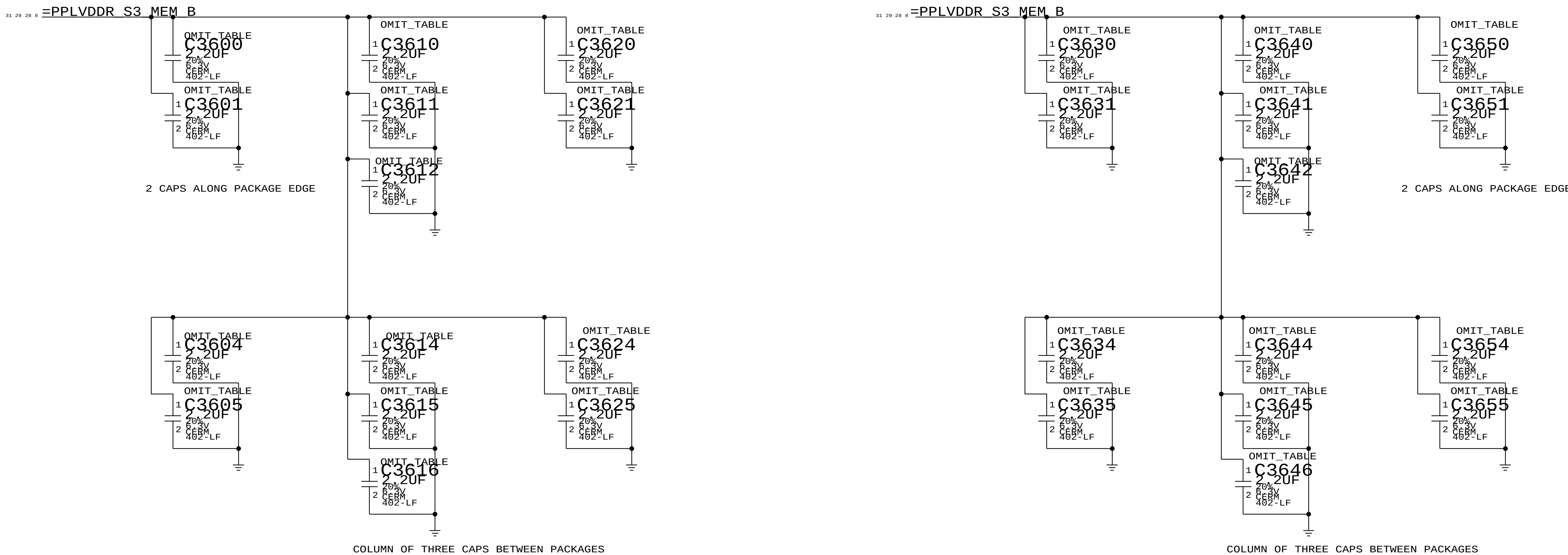

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

DDR BYPASSING 2

Apple Inc.

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REVISION 3.3.0

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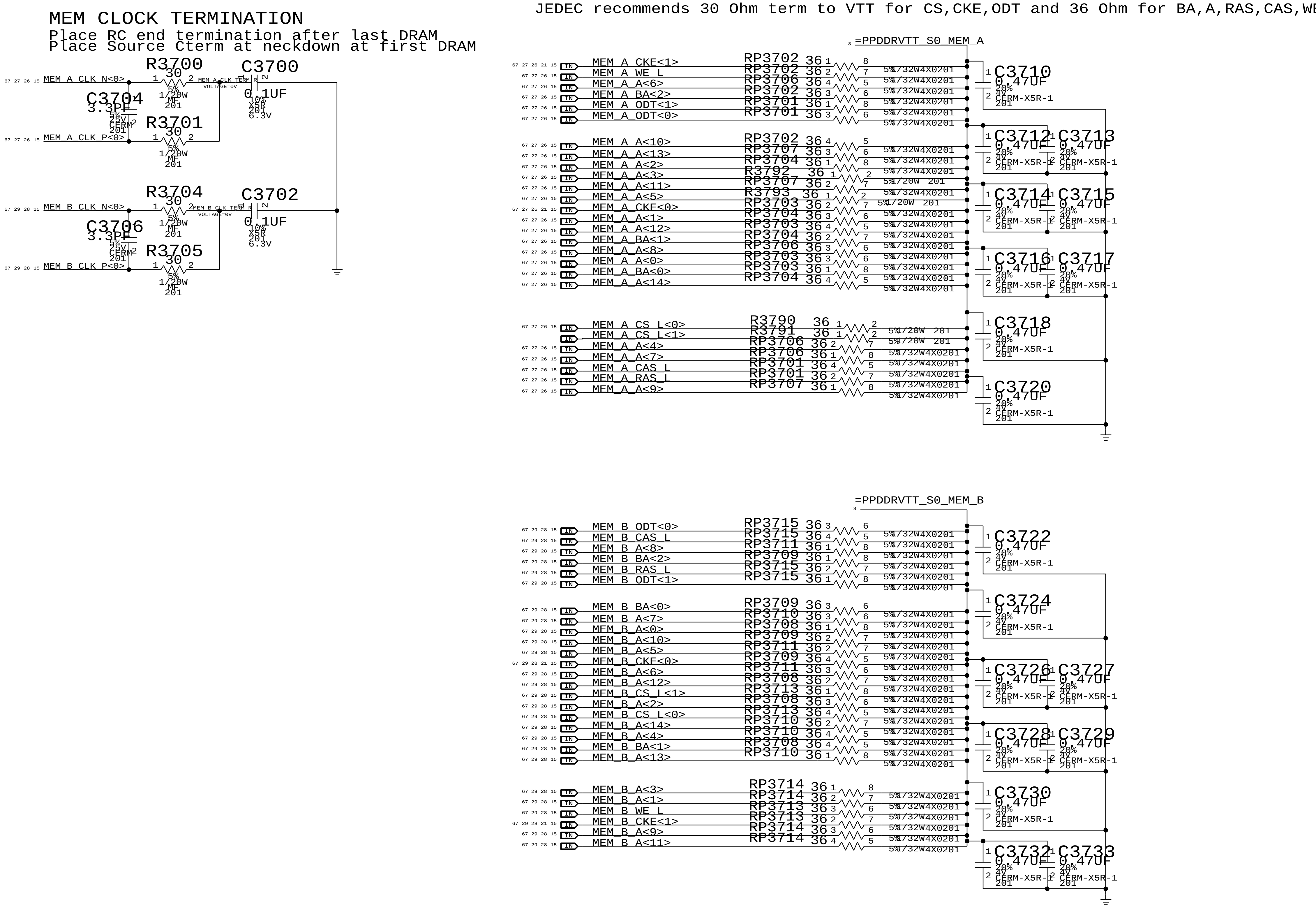


SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

Memory Active Termination

Apple Inc.

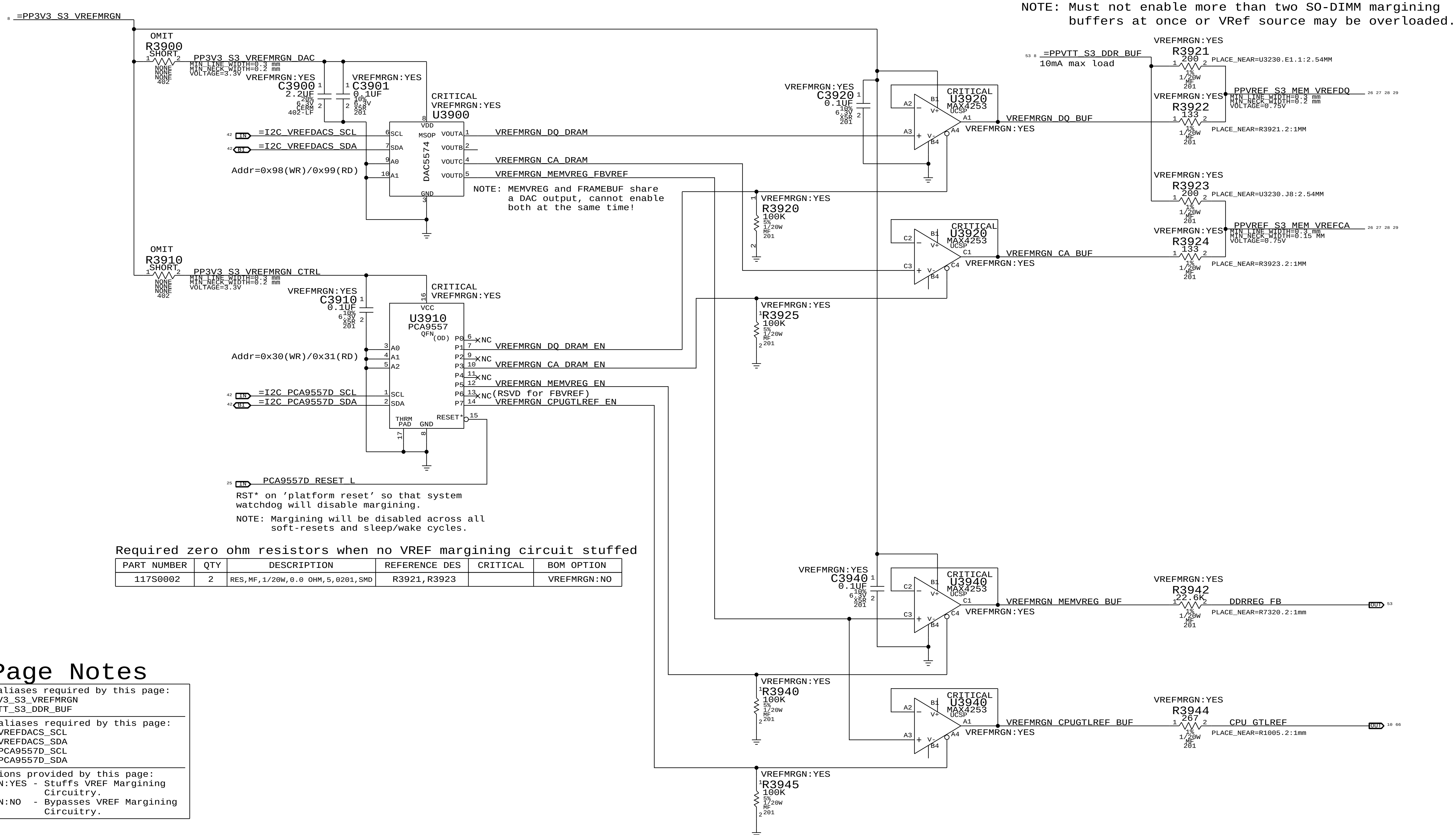
DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

BRANCH

PAGE 37 OF 110

SHEET 32 OF 74



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0002	2	RES,MF,1/20W,0.0 OHM,5,0201,SMD	R3921,R3923		VREFMRGN:NO

Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN:YES - Stuffs VREF Margining Circuitry.
VREFMRGN:NO - Bypasses VREF Margining Circuitry.

	MEM VREF DQ	MEM VREF CA	MEM VREG	CPU GTLREF (FSB)
DAC Channel:	A	C	D	D
PCA9557D Pin:	1	3	5	7
Nominal value	0.75V (DAC: 0x3A)		1.5V (DAC: 0x3A)	0.7V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)		1.998V - 1.002V (+/- 498mV)	0.200V - 1.050V (+/- 500mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)		0.000V - 1.501V (0x00 - 0x74)	0.000V - 1.191V (0x00 - 0x5C)
VRef current:	+3.4mA - -3.4mA (- = sourced)		+33uA - -33uA (- = sourced)	+750uA - -528uA (- = sourced)
DAC step size:	7.69mV / step @ output		8.59mV / step @ output	9.24mV / step @ output

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: FSB/DDR3 Vref Margining

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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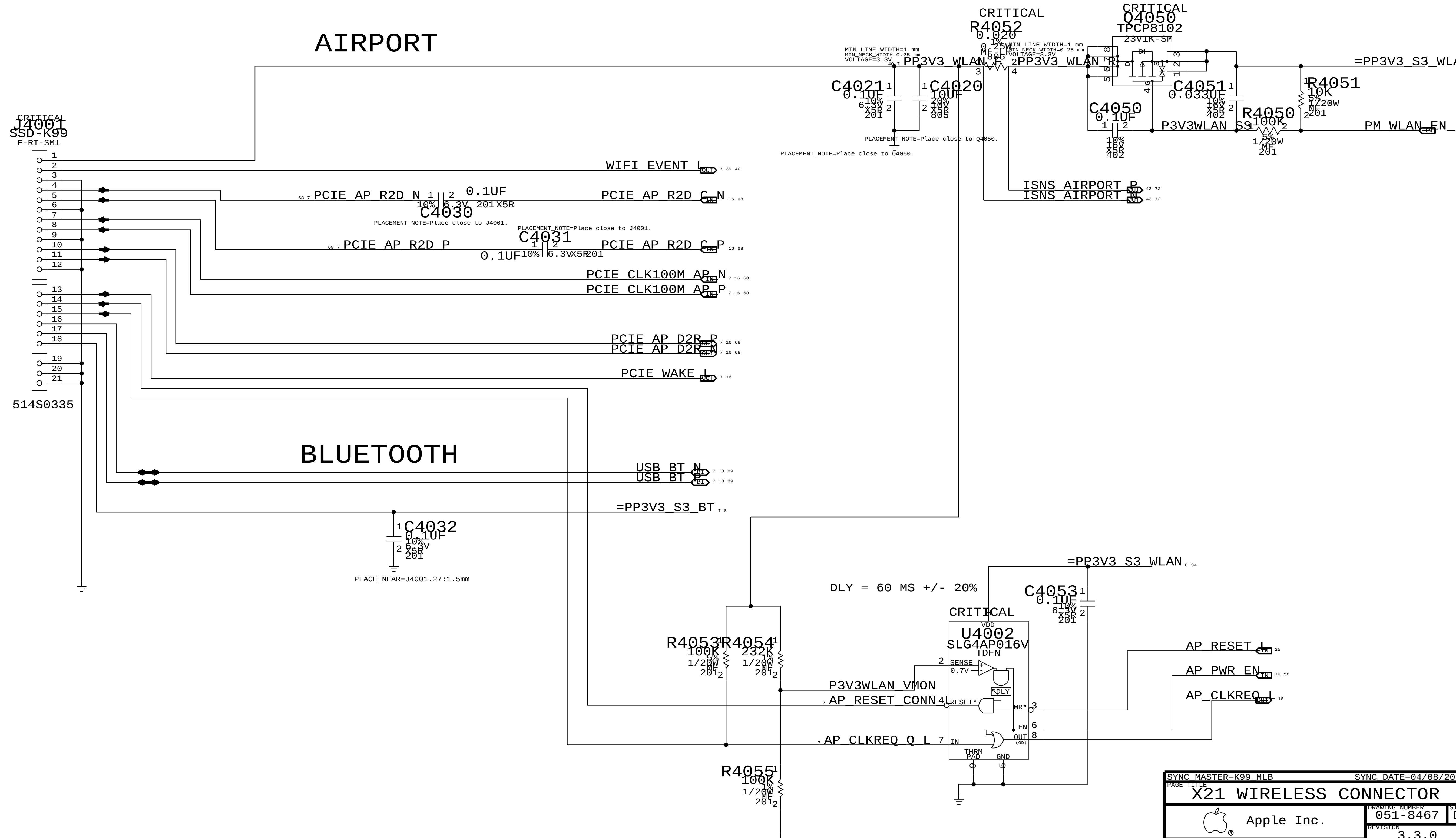
PAGE 39 OF 110

SHEET 33 OF 74

3V S3 WLAN FET

MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	20-30 MOHM @2.5V
LOADING	0.750 A (EDP)



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

X21 WIRELESS CONNECTOR

DRAWING NUMBER 051-8467 SIZE D

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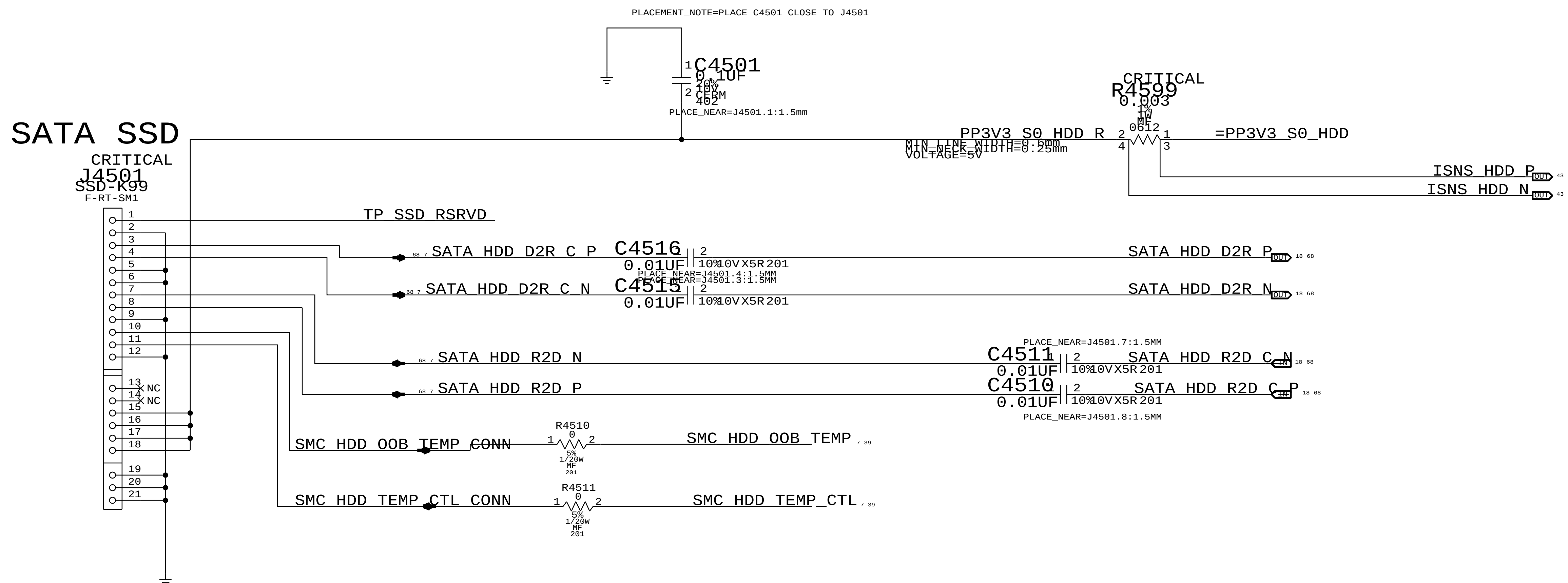
PAGE 40 OF 110

SHEET 34 OF 74



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

SATA CONNECTOR

Apple Inc.

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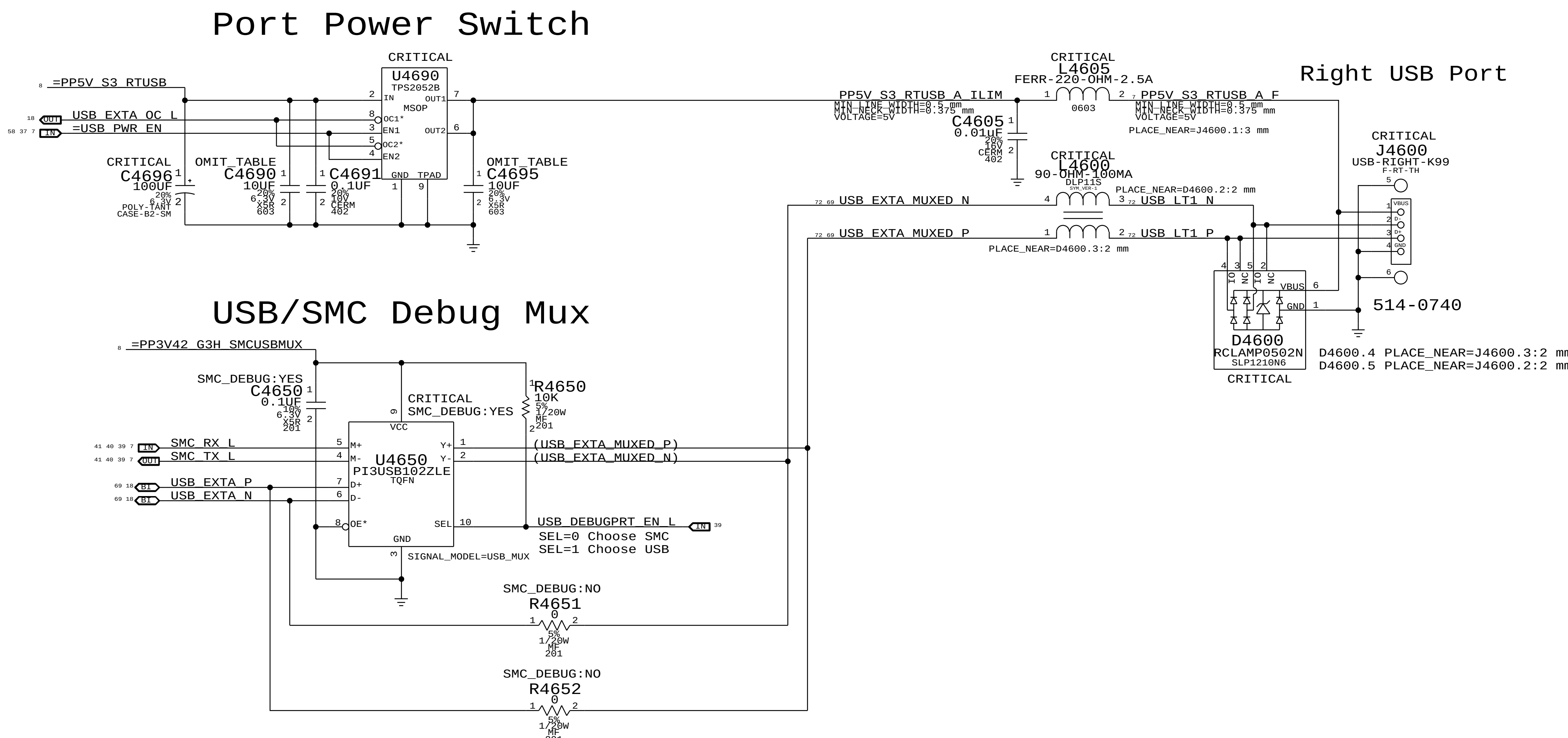

SYNC_MASTER=K99_MLB SYNC_DATE=03/01/2010

PAGE TITLE

External USB Connectors

Apple Inc.

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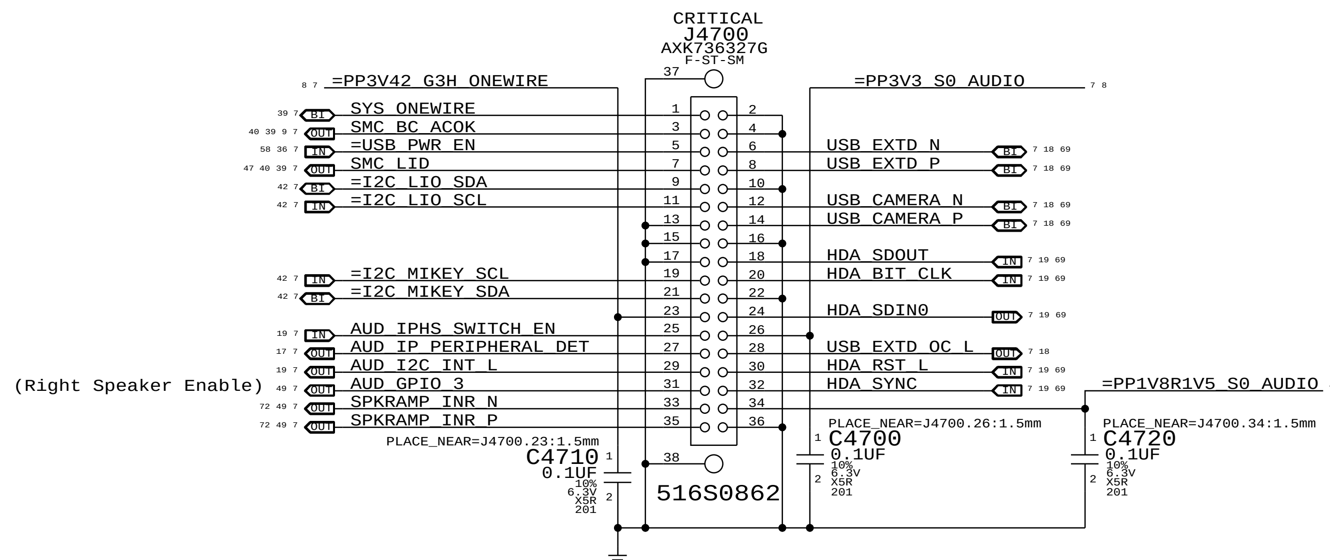

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE

Left I/O (LIO) Connector

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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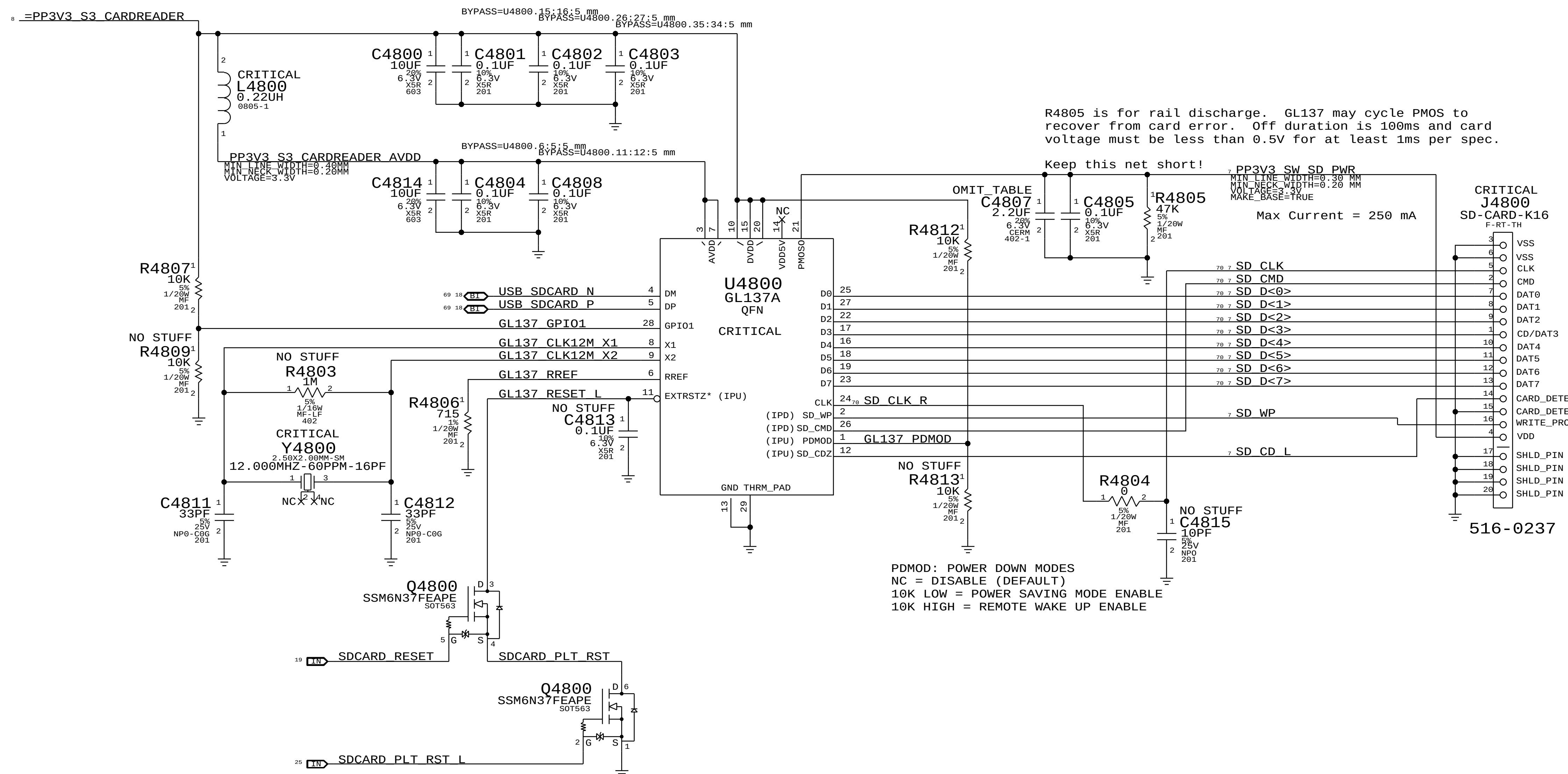

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE
SecureDigital Card Reader

Apple Inc.

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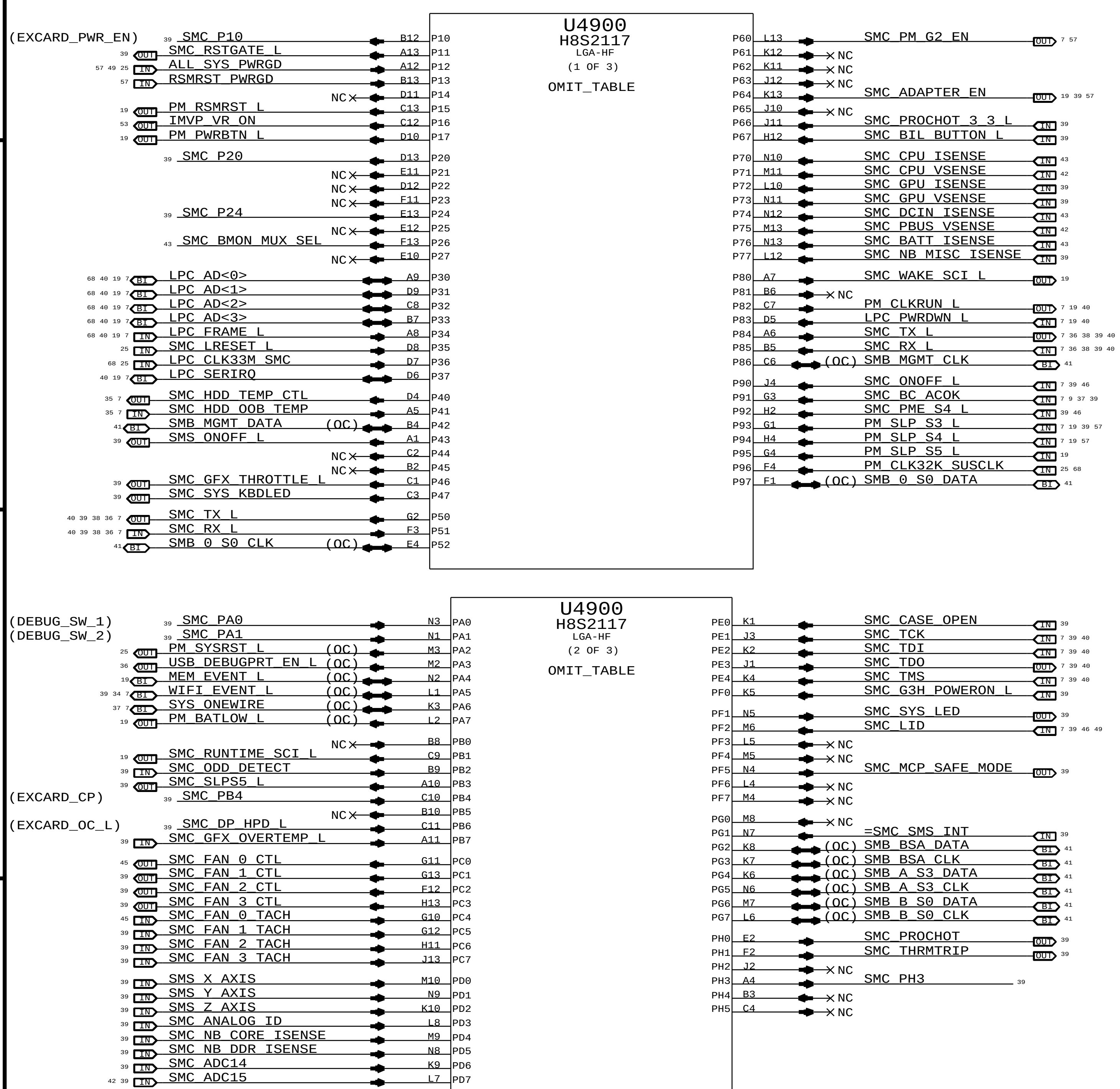
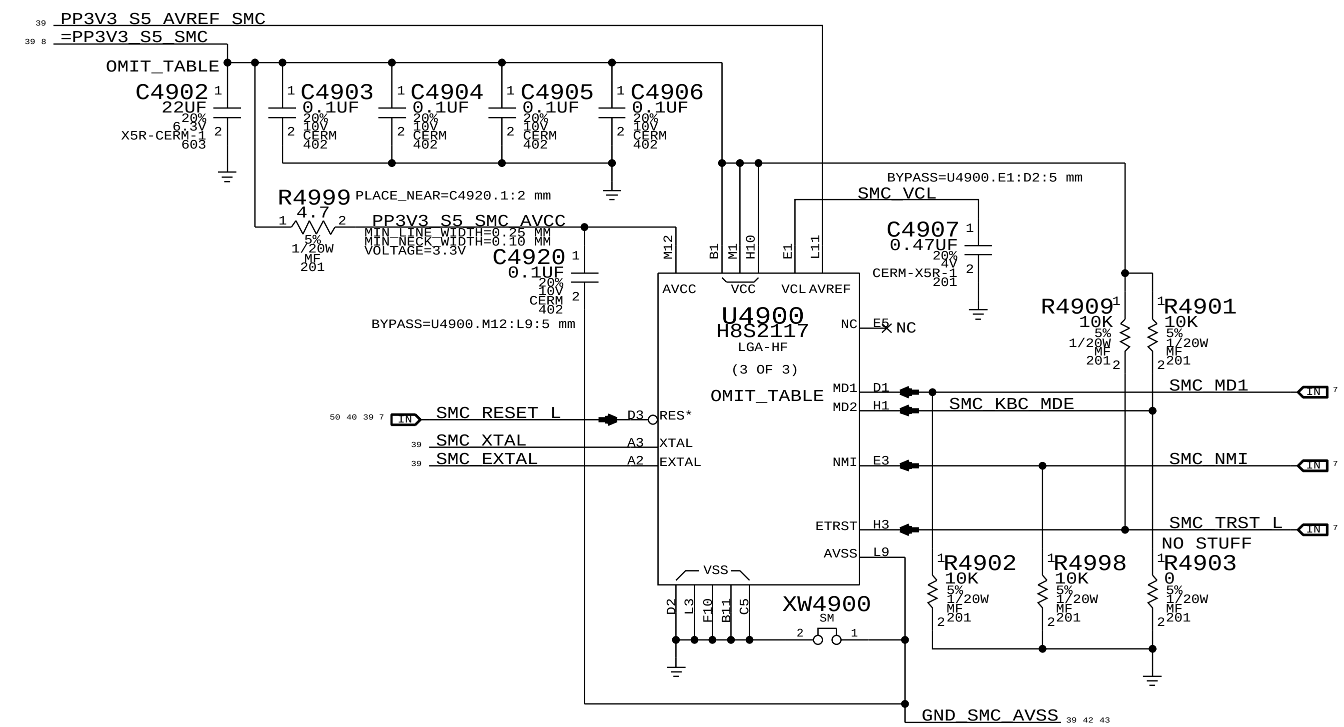
NOTE: Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.





NOTE: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

H8S2117-R:
(SMC_PECI)
(SMC_PECI_VREF)
(SMC_PECI_VSTP)

SYNC_MASTER=K16_MLB SYNC_DATE=06/01/2010

PAGE TITLE: SMC

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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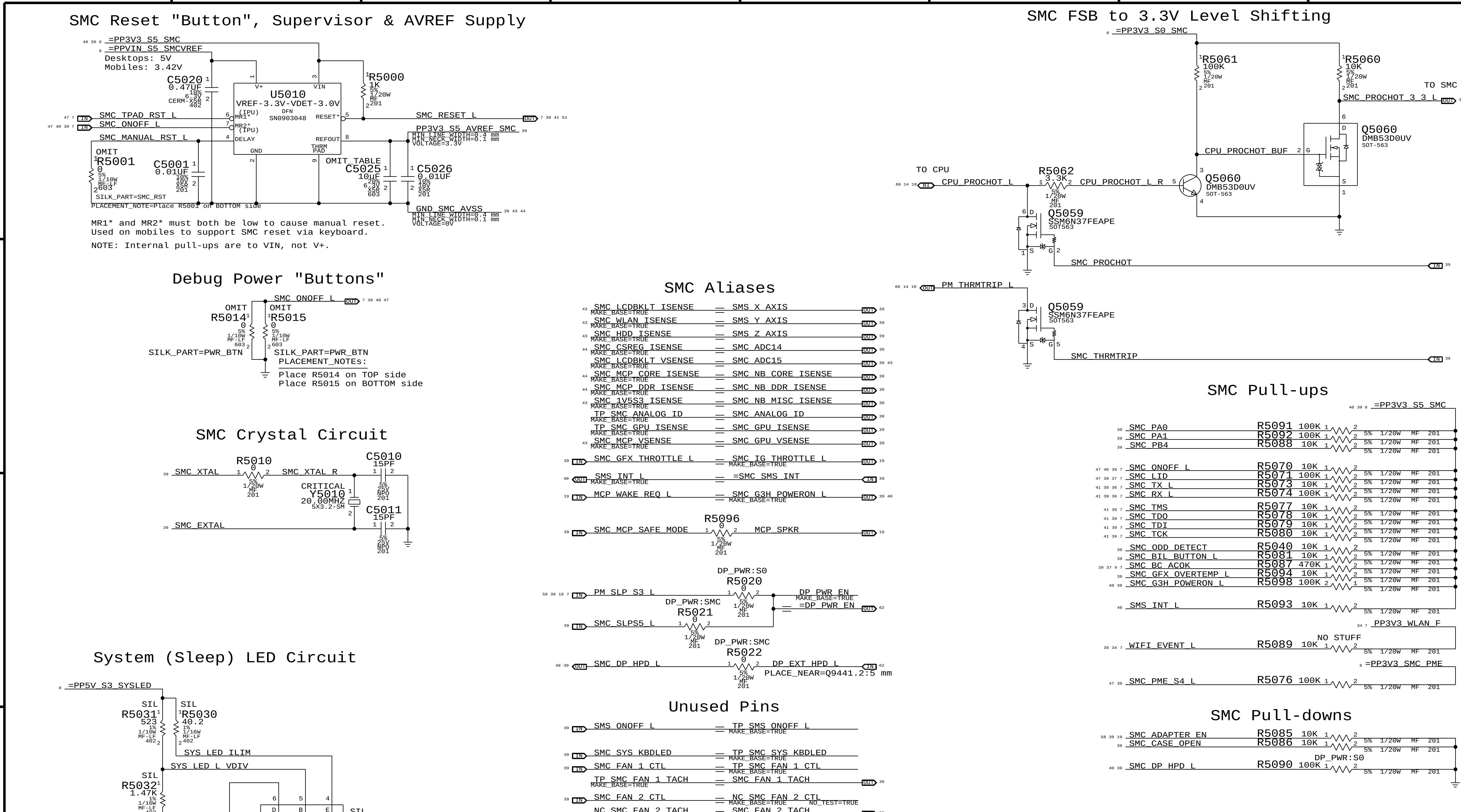

SYNC_MASTER=(K99_MLB) SYNC_DATE=(03/01/2010)

PAGE TITLE

SMC Support

Apple Inc.

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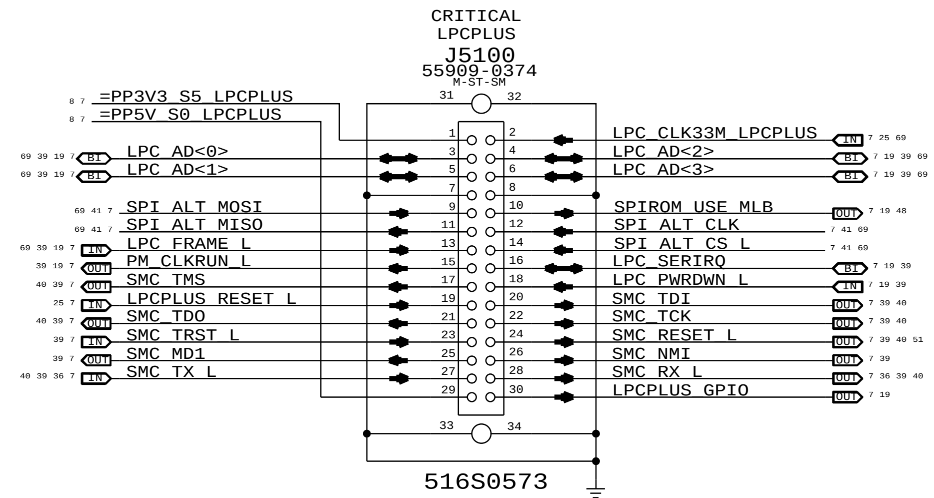
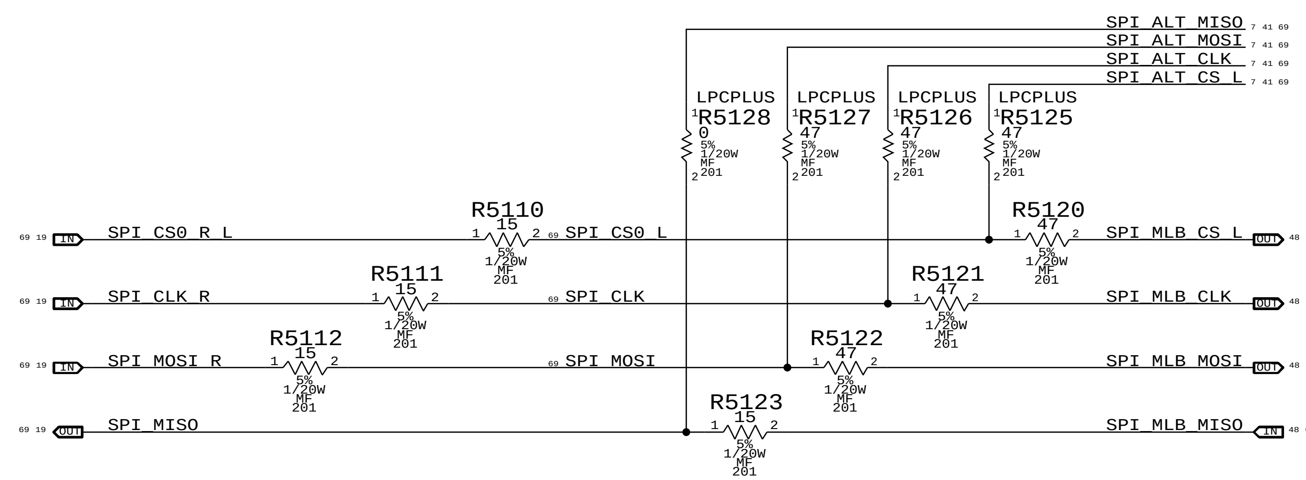
LPC+SPI Connector

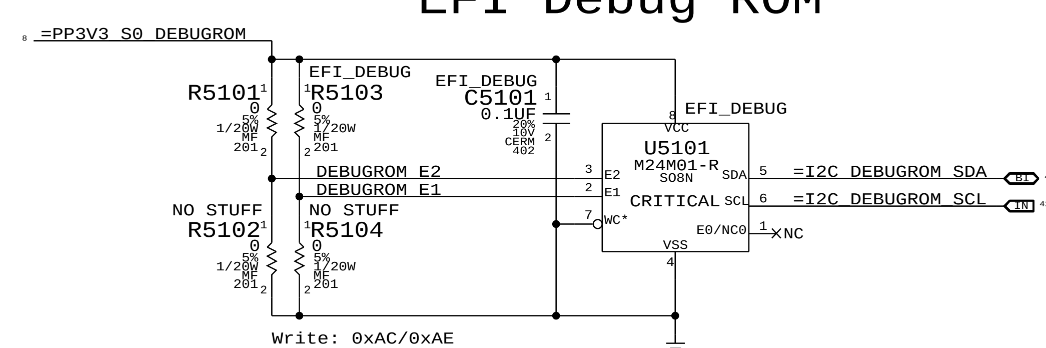


SPI Bus Series Termination



EFI_DEBUG

EFI Debug ROM



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

LPC+SPI Debug Connector

Apple Inc.

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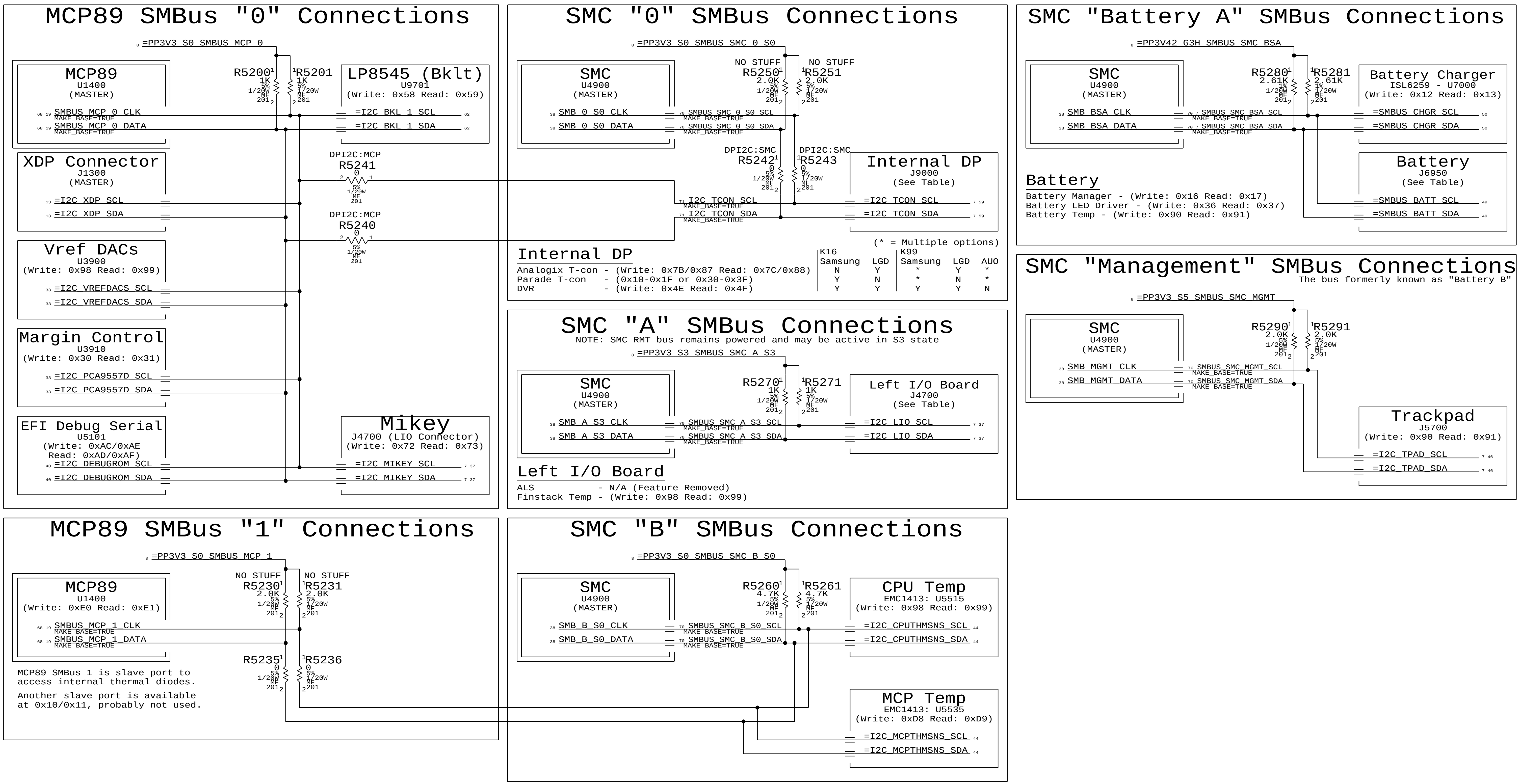

SYNC_MASTER=K99_MLB SYNC_DATE=07/23/2010

PAGE TITLE

K16/K99 SMus Connections

Apple Inc.

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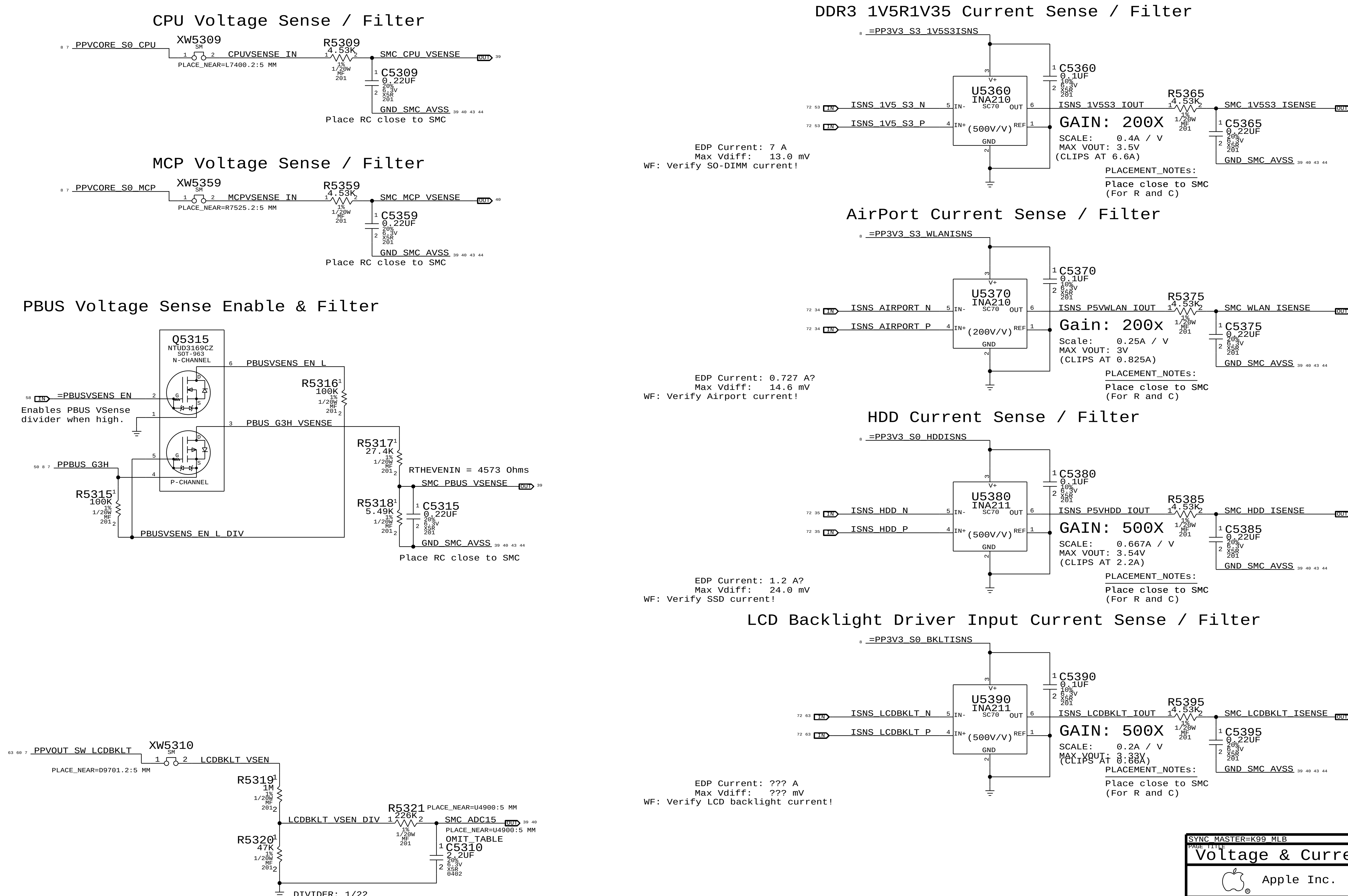
PAGE 52 OF 110

SHEET 42 OF 74



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE
Voltage & Current Sensing

Apple Inc.

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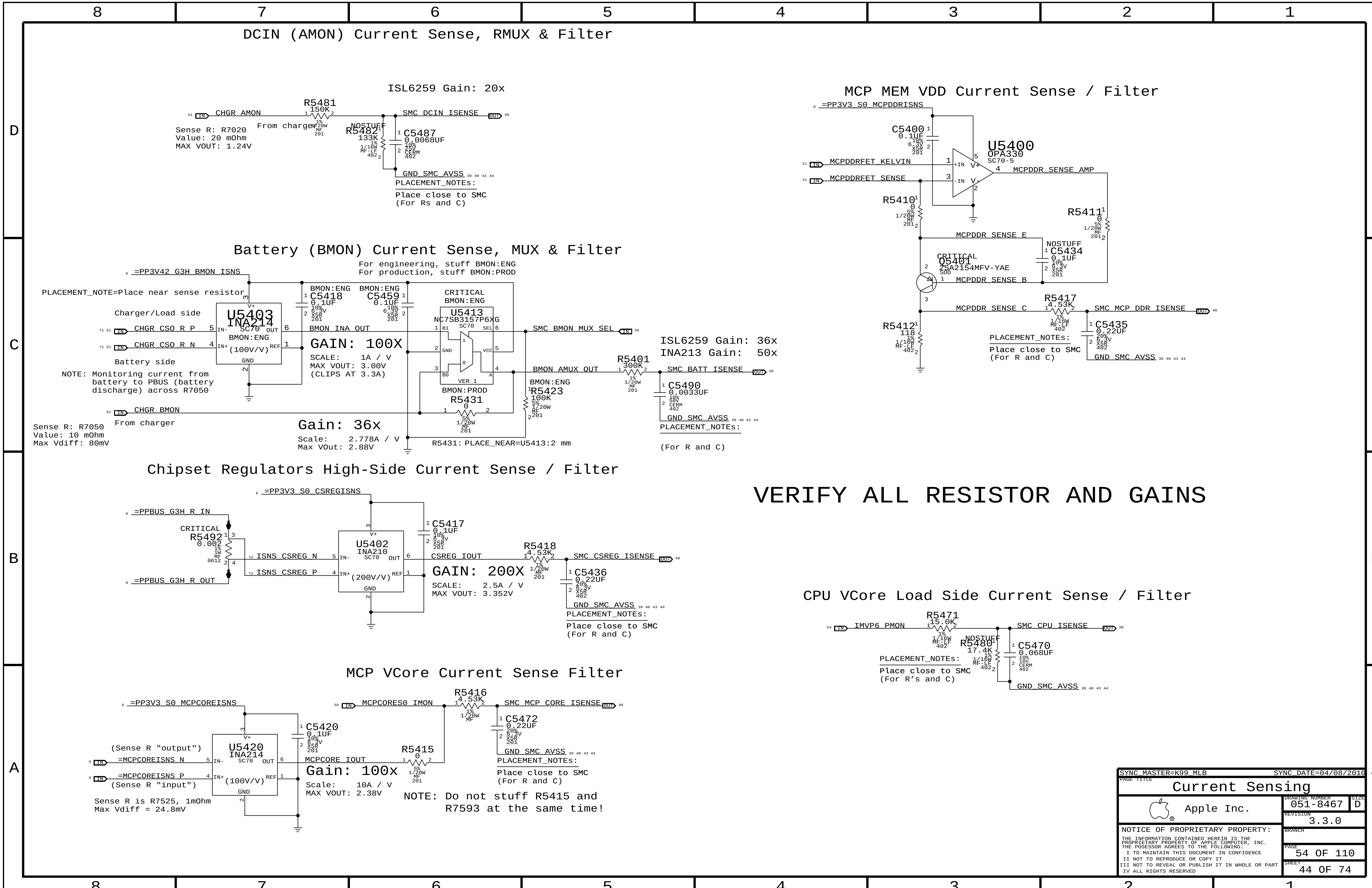

DCIN (AMON) Current Sense, RMUX & Filter
ISL6259 Gain: 20x
CHGR_AMON
R5481 150K 1% 1/20W MF 201
From charger
Sense R: R7020
Value: 20 mOhm
MAX VOUT: 1.24V
NOSTUFF
R5482 133K 1% 1/16W MF-LF 402
C5487 0.0068UF 10% 25V CERM 402
SMC_DCIN_ISENSE
GND_SMC_AVSS
PLACEMENT_NOTES:
Place close to SMC
(For Rs and C)
Battery (BMON) Current Sense, MUX & Filter
For engineering, stuff BMON:ENG
For production, stuff BMON:PROD
=PP3V42_G3H_BMON_ISNS
PLACEMENT_NOTE=Place near sense resistor
Charger/Load side
U5403 INA214 SC70
BMON:ENG
(100V/V)
CHGR_CSO_R_P
CHGR_CSO_R_N
Battery side
NOTE: Monitoring current from battery to PBUS (battery discharge) across R7050
BMON:ENG C5418 0.1UF 10% 6.3V X5R 201
BMON:ENG C5459 0.1UF 10% 6.3V X5R 201
BMON_INA_OUT
GAIN: 100X
SCALE: 1A / V
MAX VOUT: 3.00V
(CLIPS AT 3.3A)
CRITICAL BMON:ENG
U5413 NC7SB3157P6XG SC70
VER 1
BMON:PROD
SMC_BMON_MUX_SEL
BMON_AMUX_OUT
ISL6259 Gain: 36x
INA213 Gain: 50x
R5401 300K 1% 1/20W MF 201
SMC_BATT_ISENSE
C5490 0.0033UF 10% 50V CERM 402
GND_SMC_AVSS
PLACEMENT_NOTES:
(For R and C)
BMON:ENG R5423 100K 5% 1/20W MF 201
R5431 0 5% 1/20W MF 201
CHGR_BMON
From charger
Sense R: R7050
Value: 10 mOhm
Max Vdiff: 80mV
Gain: 36x
Scale: 2.778A / V
Max Vout: 2.88V
R5431: PLACE_NEAR=U5413:2 mm
Chipset Regulators High-Side Current Sense / Filter
=PP3V3_S0_CSREGISNS
=PPBUS_G3H_R_IN
CRITICAL
R5492 0.002 1% 1W MF 0612
=PPBUS_G3H_R_OUT
ISNS_CSREG_N
ISNS_CSREG_P
U5402 INA210 SC70
(200V/V)
C5417 0.1UF 10% 6.3V X5R 201
CSREG_IOUT
GAIN: 200X
SCALE: 2.5A / V
MAX VOUT: 3.352V
R5418 4.53K 1% 1/20W MF 201
SMC_CSREG_ISENSE
C5436 0.22UF 20% 6.3V X5R 402
GND_SMC_AVSS
PLACEMENT_NOTES:
Place close to SMC
(For R and C)
MCP VCore Current Sense Filter
=PP3V3_S0_MCPCOREISNS
(Sense R "output")
=MCPCOREISNS_N
=MCPCOREISNS_P
(Sense R "input")
U5420 INA214 SC70
(100V/V)
C5420 0.1UF 10% 6.3V X5R 201
MCPCORE_IOUT
Gain: 100x
Scale: 10A / V
MAX VOUT: 2.38V
Sense R is R7525, 1mOhm
Max Vdiff = 24.8mV
MCPCORES0_IMON
R5416 4.53K 1% 1/20W MF
SMC_MCP_CORE_ISENSE
C5472 0.22UF 20% 6.3V X5R 201
GND_SMC_AVSS
PLACEMENT_NOTES:
Place close to SMC
(For R and C)
R5415 0 5% 1/20W MF 201
NOTE: Do not stuff R5415 and R7593 at the same time!
MCP MEM VDD Current Sense / Filter
=PP3V3_S0_MCPDDRISNS
C5400 0.1UF 10% 6.3V X5R 201
U5400 OPA330 SC70-5
MCPDDRFET_KELVIN
MCPDDRFET_SENSE
MCPDDR_SENSE_AMP
R5410 0 5% 1/20W MF 201
R5411 0 5% 1/20W MF 201
MCPDDR_SENSE_E
NOSTUFF C5434 0.1UF 10% 6.3V X5R 201
CRITICAL Q5401 2SA2154MFV-YAE SOD
MCPDDR_SENSE_B
MCPDDR_SENSE_C
R5417 4.53K 1% 1/16W MF-LF 402
SMC_MCP_DDR_ISENSE
C5435 0.22UF 20% 6.3V X5R 402
R5412 118 1% 1/16W MF-LF 402
PLACEMENT_NOTES:
Place close to SMC
(For R and C)
GND_SMC_AVSS
VERIFY ALL RESISTOR AND GAINS
CPU VCore Load Side Current Sense / Filter
IMVP6_PMON
R5471 15.0K 1% 1/16W MF-LF 402
SMC_CPU_ISENSE
NOSTUFF R5480 17.4K 1% 1/16W MF-LF 402
C5470 0.068UF 10% 10V CERM 402
PLACEMENT_NOTES:
Place close to SMC
(For R's and C)
GND_SMC_AVSS
SYNC_MASTER=K99_MLB
SYNC_DATE=04/08/2010
Current Sensing
Apple Inc.
051-8467 D
3.3.0
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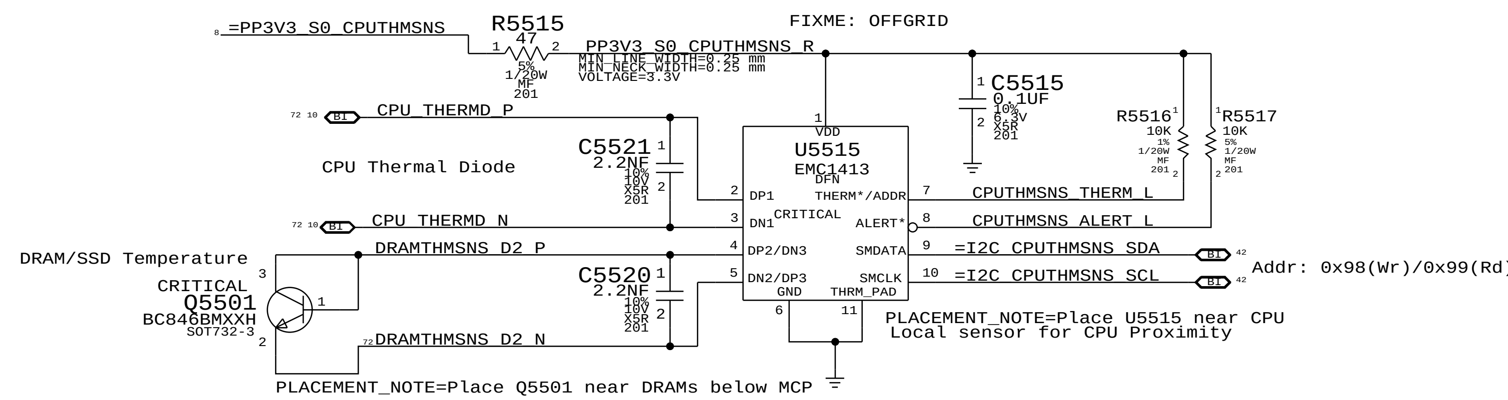
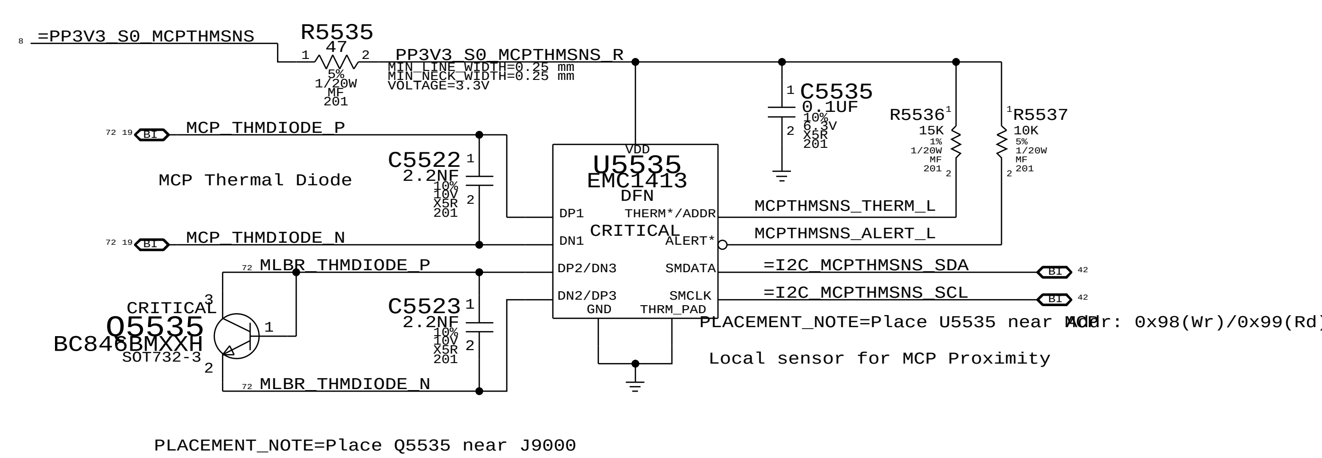
54 OF 110
44 OF 74

CPU T-Diode Thermal Sensor



MCP T-Diode Thermal Sensor



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE
Thermal Sensors

Apple Inc.

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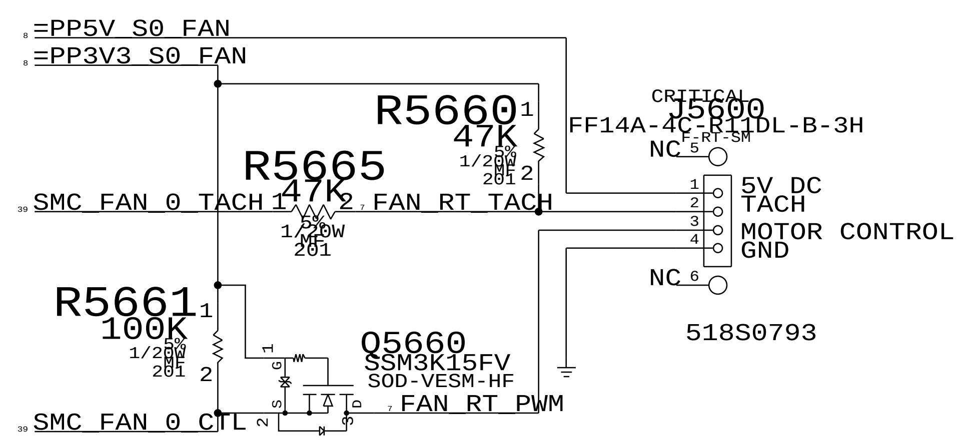
FAN CONNECTOR



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: Fan

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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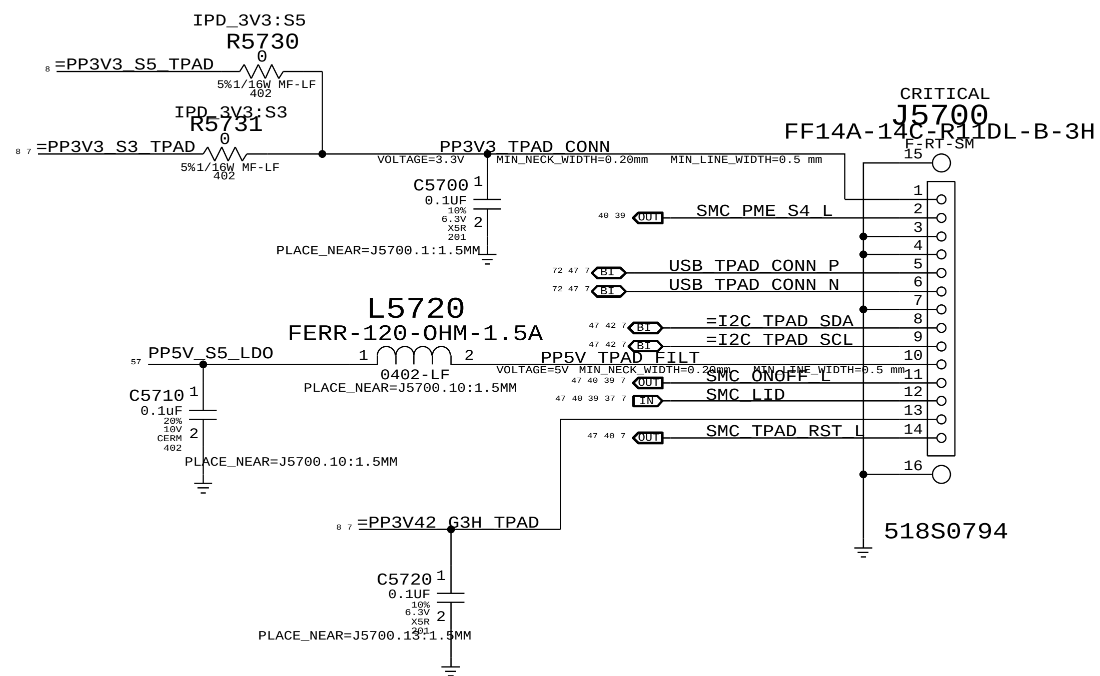
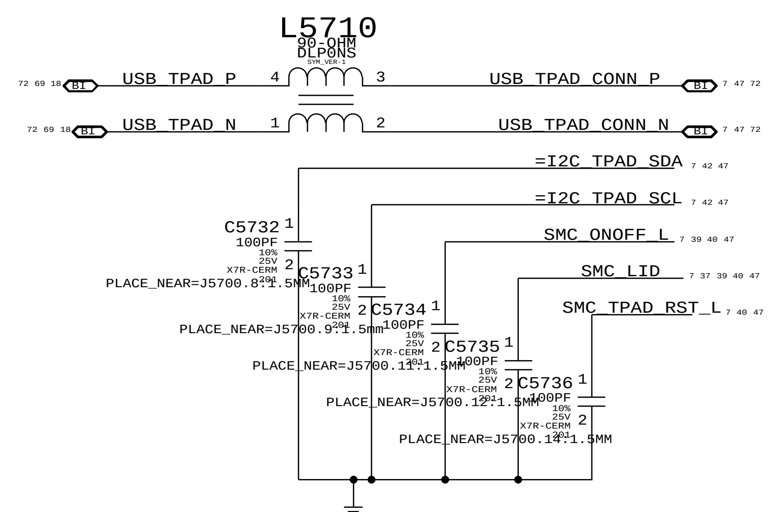
PAGE: 56 OF 110

SHEET: 46 OF 74

IPD Flex Connector





SYNC_MASTER=K99_MLB	SYNC_DATE=04/08/2010
PAGE TITLE: WELLSPRING 1	
Apple Inc.	DRAWING NUMBER 051-8467 SIZE D
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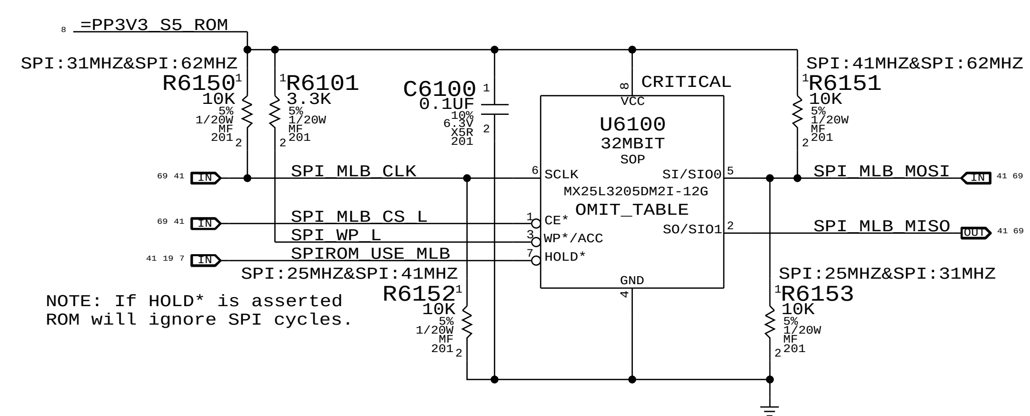
	SHEET 47 OF 74



MCP89 SPI Frequency Select

Frequency	SPI_MOSI	SPI_CLK
25.0 MHz	0	0
31.2 MHz	0	1
41.7 MHz	1	0
62.5 MHz	1	1

NOTE: 42 & 62 MHz use FAST_READ command.

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

SPI ROM

Apple Inc.

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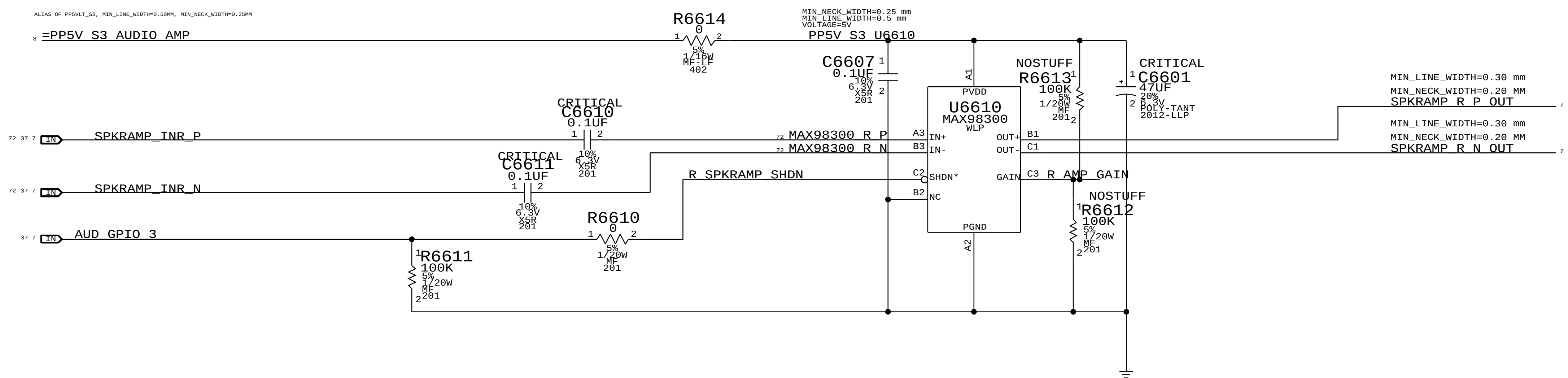
SPEAKER AMPLIFIERS

APN:353S2888

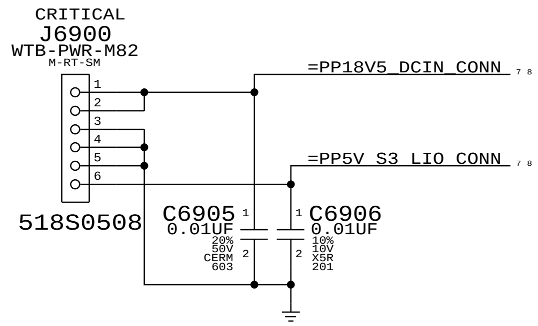
SPEAKER LOWPASS 80 HZ < FC < 132 HZ

GAIN 6DB



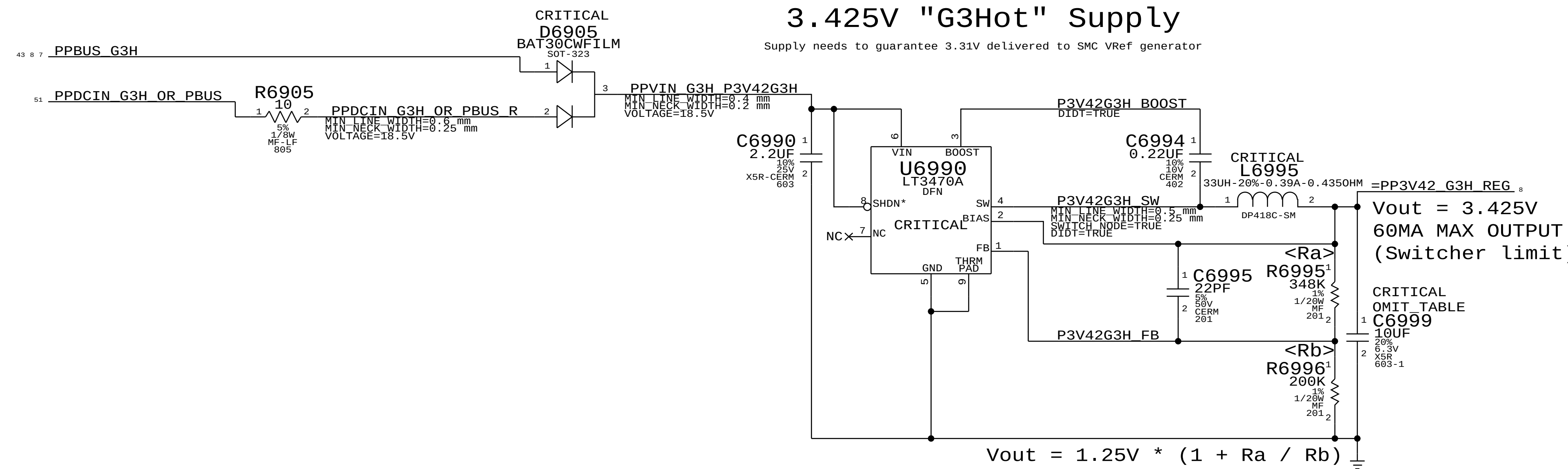
SYNC_MASTER=K99_MLB	SYNC_DATE=04/08/2010
PAGE TITLE	AUDIO: SPEAKER AMP
Apple Inc.	DRAWING NUMBER 051-8467 SIZE D
	REVISION 3.3.0
	BRANCH
	PAGE 66 OF 110
	SHEET 49 OF 74

MLB to LIO Power Cable Connector



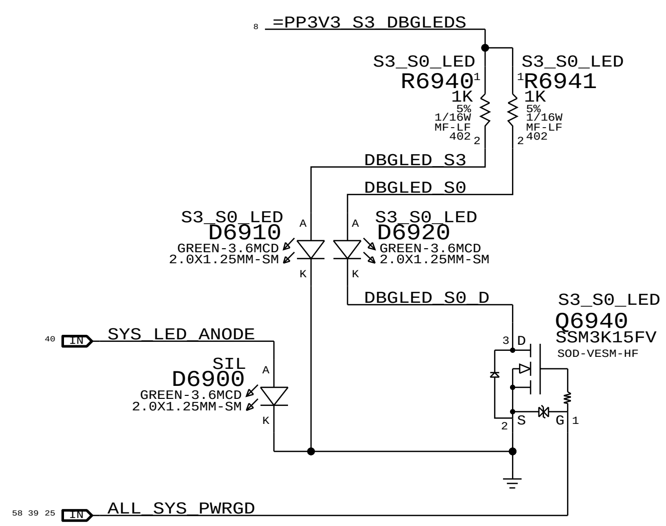
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

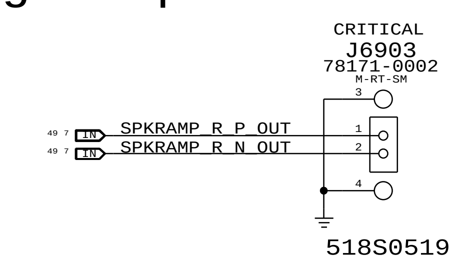


Debug LEDs

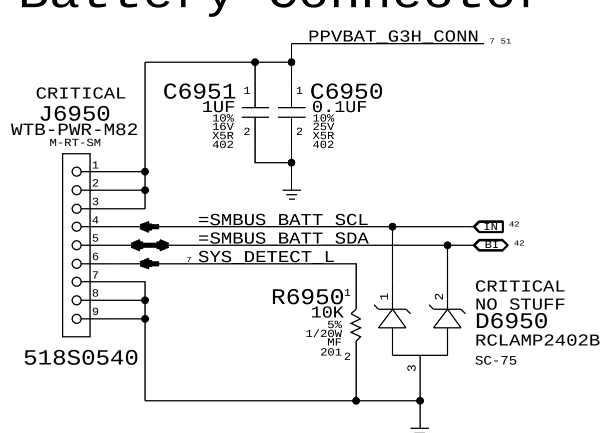
(For development only)



Right Speaker Connector



K16-Specific Battery Connector



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

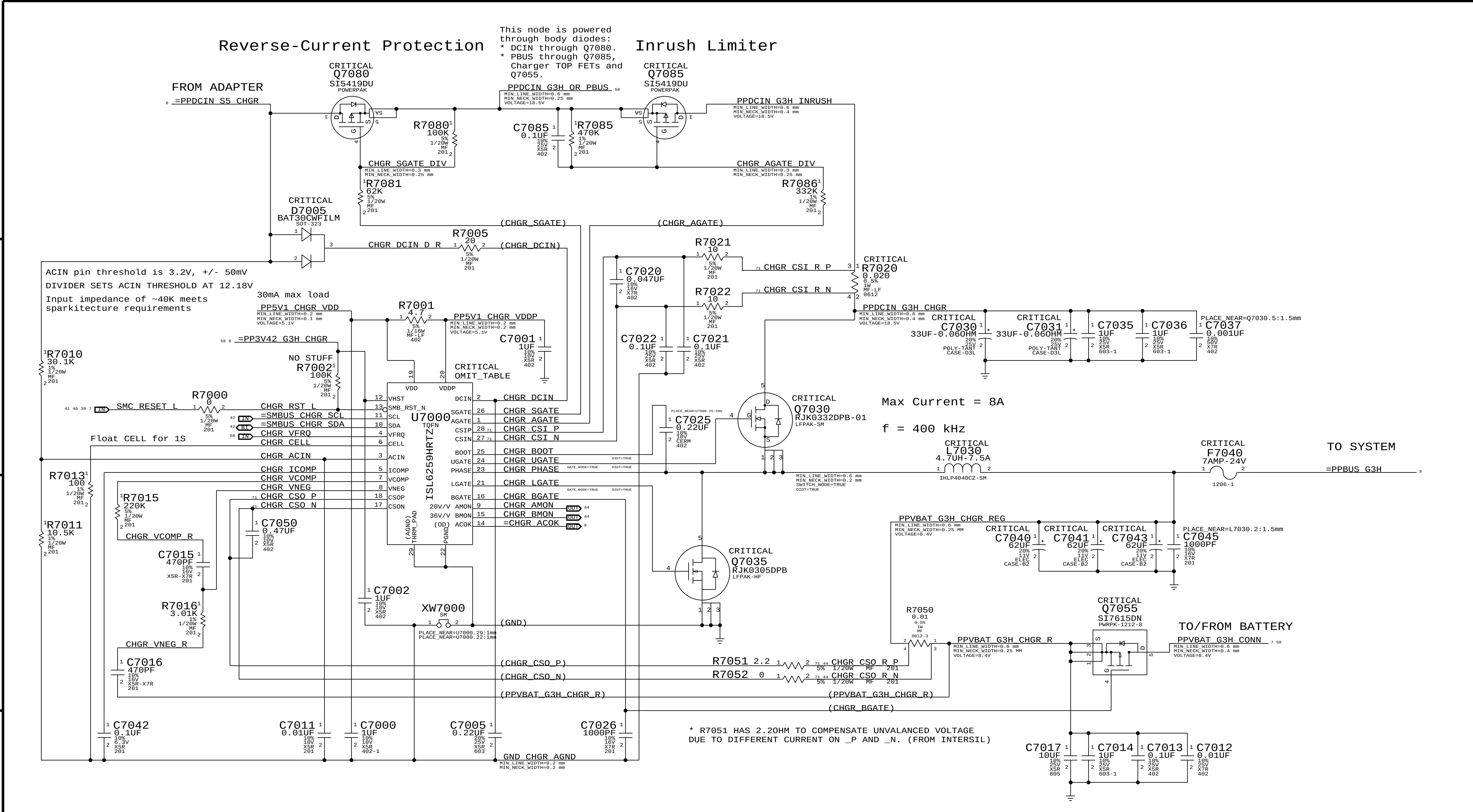
PAGE TITLE: DC-In & Battery Connectors

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

BRANCH



SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/16/2010)

PAGE TITLE

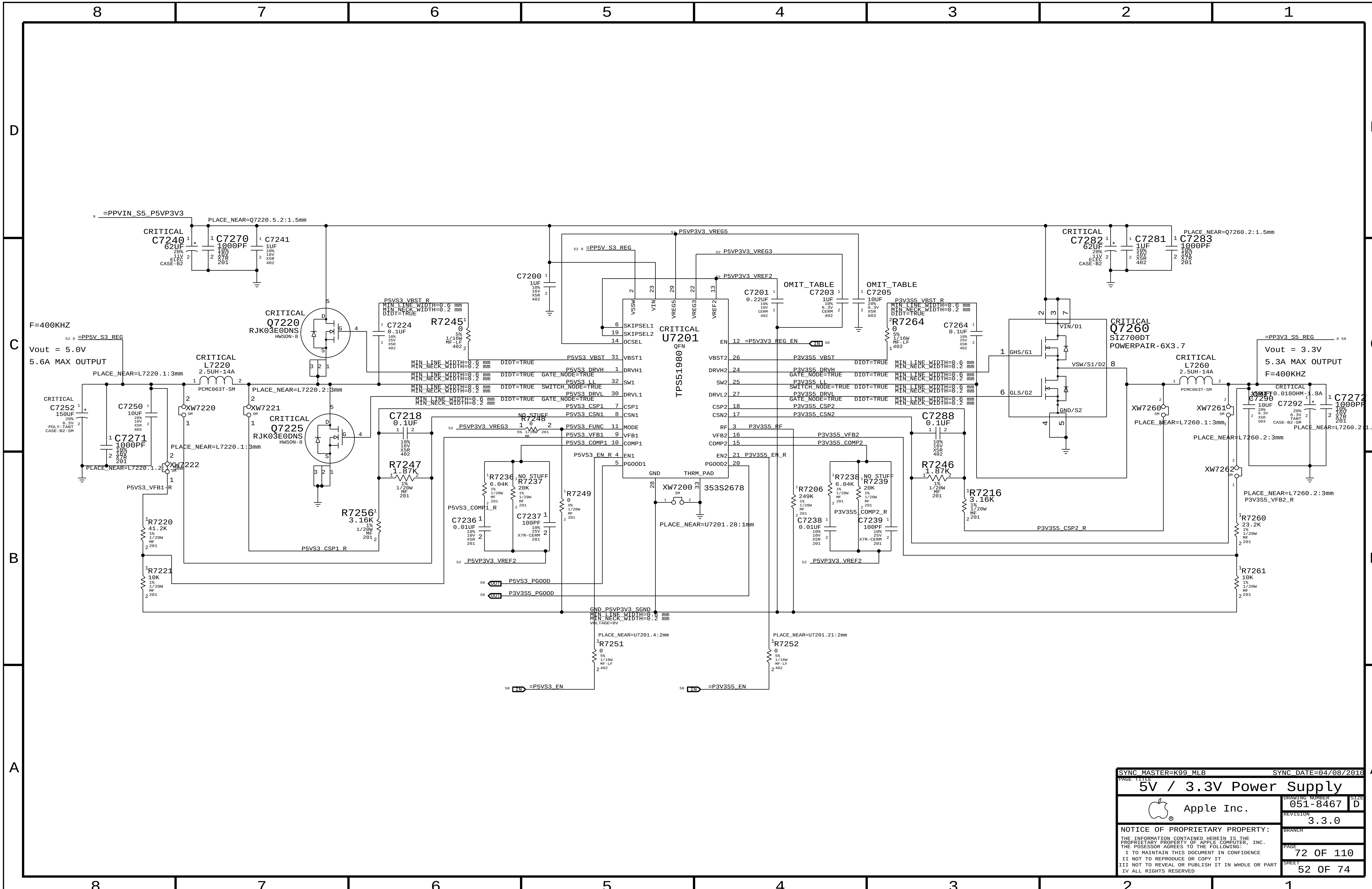
PBus Supply & Battery Charger

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

BRANCH



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

5V / 3.3V Power Supply

Apple Inc.

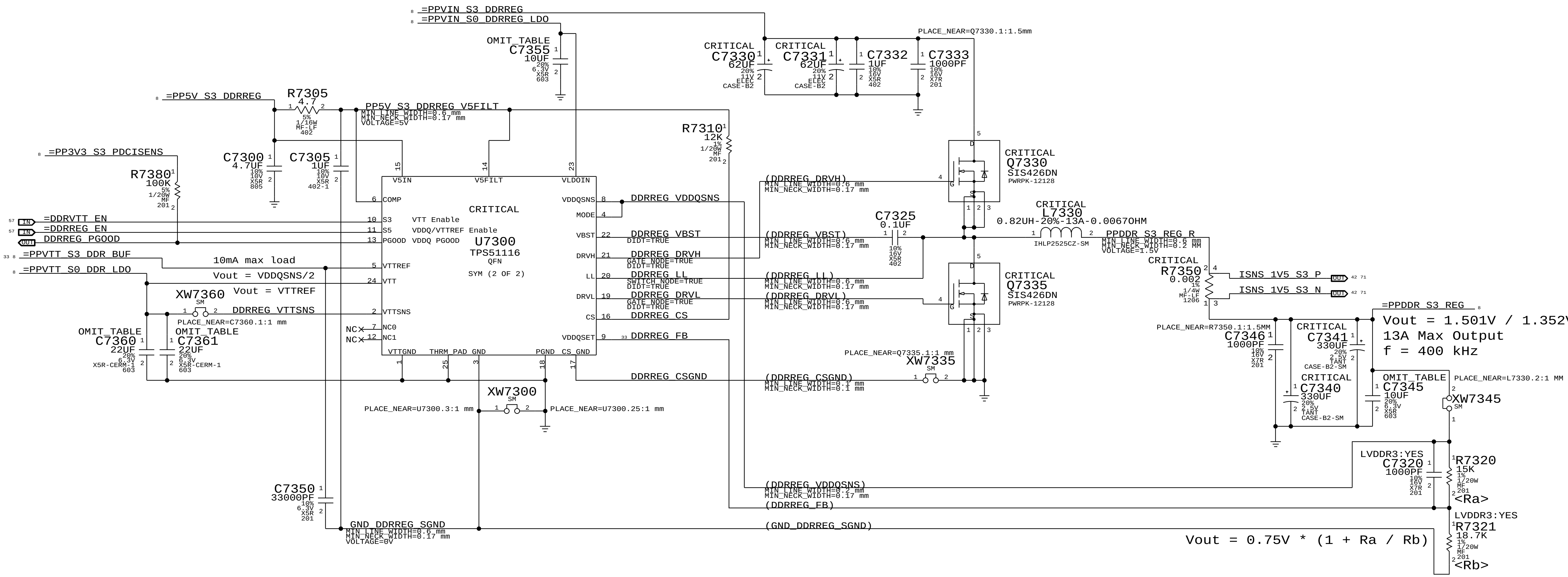
DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

BRANCH

PAGE 72 OF 110

SHEET 52 OF 74



Use LVDDR3:YES for fixed 1.35V operation or LVDDR3:NO for fixed 1.5V operation.

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0331	1	RES,15K,1%,1/16W,MF-LF,0402	R7321		LVDDR3:NO

SYNC_MASTER=K16_MLB SYNC_DATE=06/01/2010

PAGE TITLE

1.5V/1.35V LVDDR3 Supply

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

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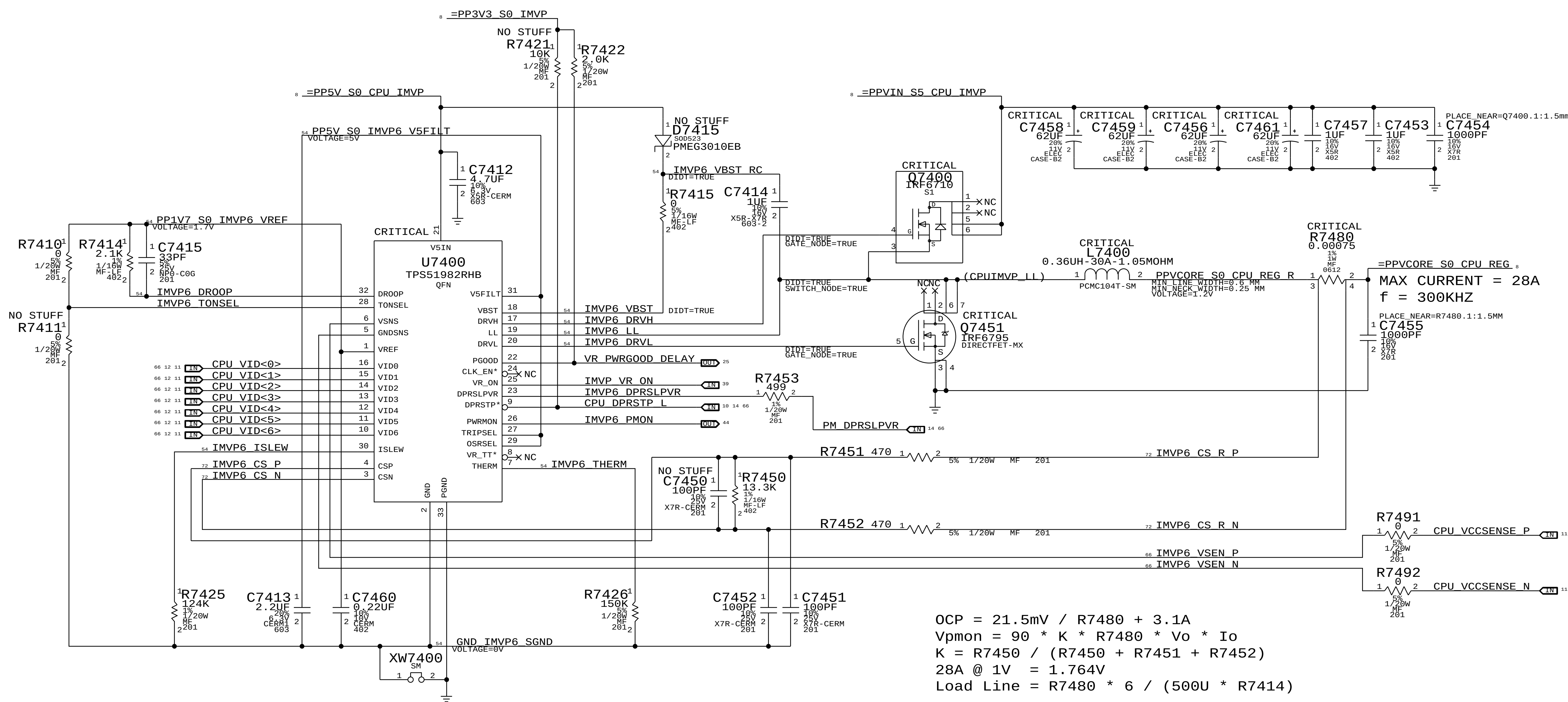

OCP = 21.5mV / R7480 + 3.1A
Vpmon = 90 * K * R7480 * Vo * Io
K = R7450 / (R7450 + R7451 + R7452)
28A @ 1V = 1.764V
Load Line = R7480 * 6 / (500U * R7414)

	MIN_LINE_WIDTH	MIN_NECK_WIDTH
GND_IMVP6_SGND	0.50 MM	0.20 MM
IMVP6_DROOP	0.25 MM	0.20 MM
IMVP6_THERM	0.25 MM	0.20 MM
IMVP6_ISLEW	0.25 MM	0.20 MM
PP1V7_S0_IMVP6_VREF	0.25 MM	0.20 MM
PP5V_S0_IMVP6_V5FILT	0.25 MM	0.20 MM
IMVP6_LL	1.5 MM	0.20 MM
IMVP6_VBST	0.25 MM	0.20 MM
IMVP6_DRVH	1.5 MM	0.20 MM
IMVP6_DRVL	1.5 MM	0.20 MM
IMVP6_VBST_RC	1.5 MM	0.20 MM

SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/16/2010)

PAGE TITLE

IMVP6 CPU VCore Regulator

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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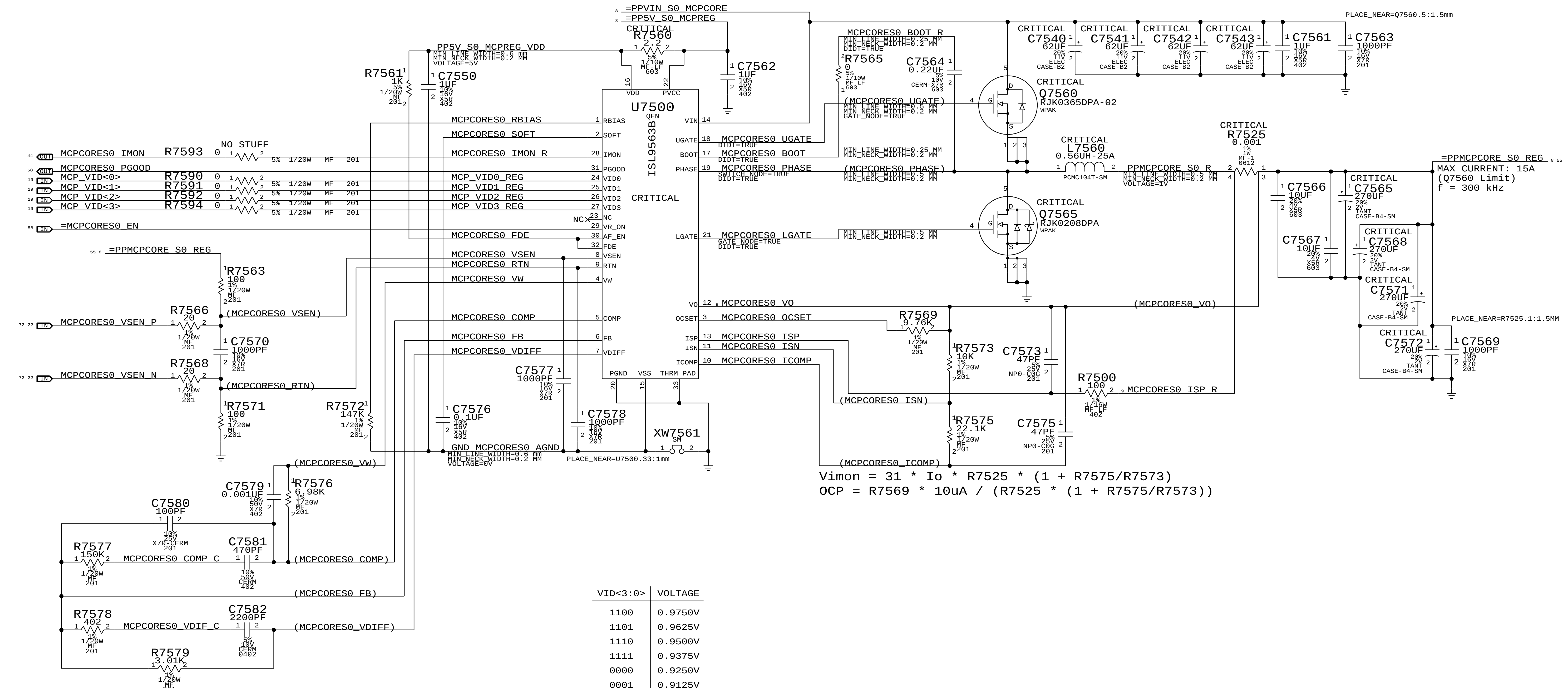

Vimon = 31 * Io * R7525 * (1 + R7575/R7573)

OCP = R7569 * 10uA / (R7525 * (1 + R7575/R7573))

VID<3:0>	VOLTAGE
1100	0.9750V
1101	0.9625V
1110	0.9500V
1111	0.9375V
0000	0.9250V
0001	0.9125V
0010	0.9000V
0011	0.8875V
0100	0.8750V
0101	0.8625V
0110	0.8500V
0111	0.8375V
1000	0.8250V
1001	0.8125V
1010	0.8000V
1011	0.7875V

K6 NOTES : XOR AND INVERTER IS REMOVED, CANNOT SYNC THIS PAGE FROM T27

SYNC_MASTER=(K99_MLB) SYNC_DATE=(02/11/2010)

PAGE TITLE

MCP VCore Regulator

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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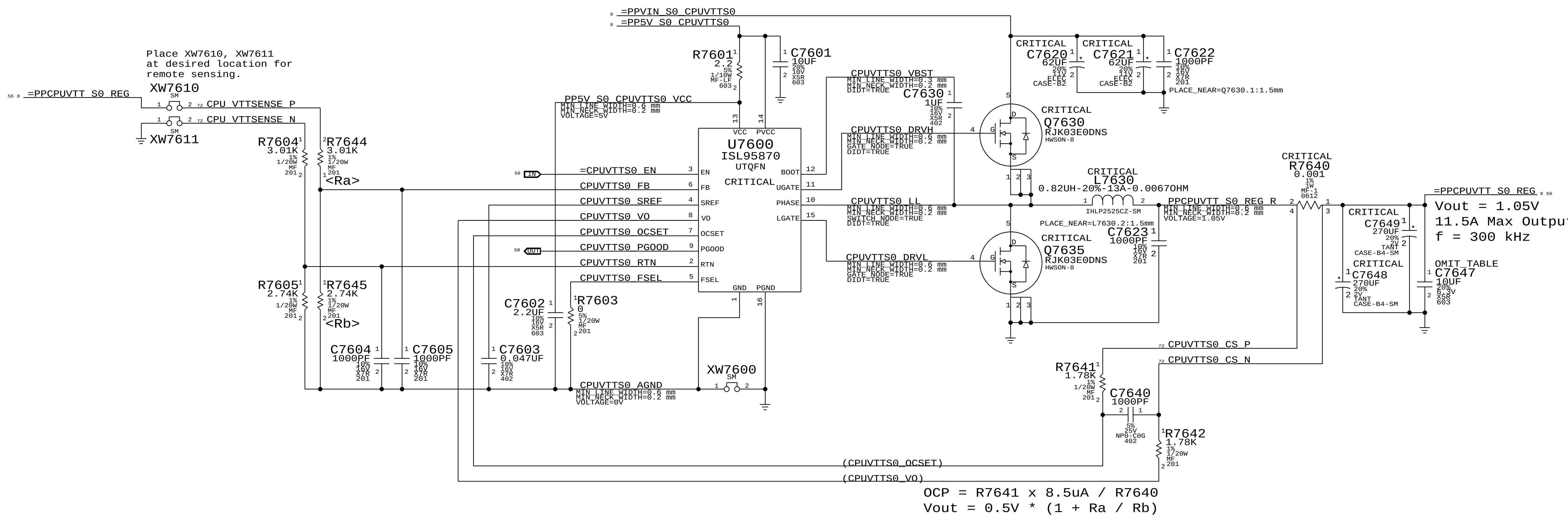

SYNC_MASTER=(K99_MLB) SYNC_DATE=(03/01/2010)

PAGE TITLE

CPUVTT (1.05V) Power Supply

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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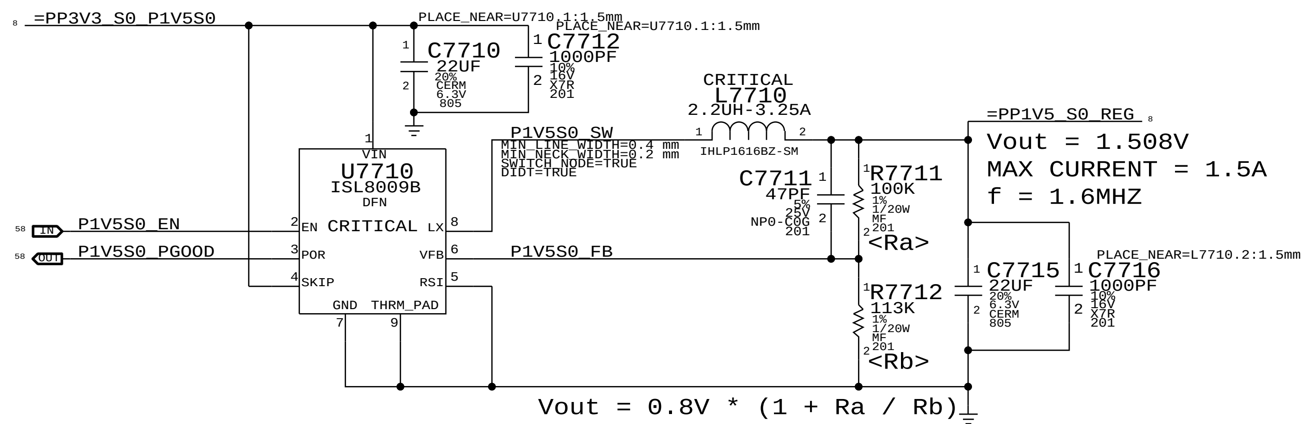
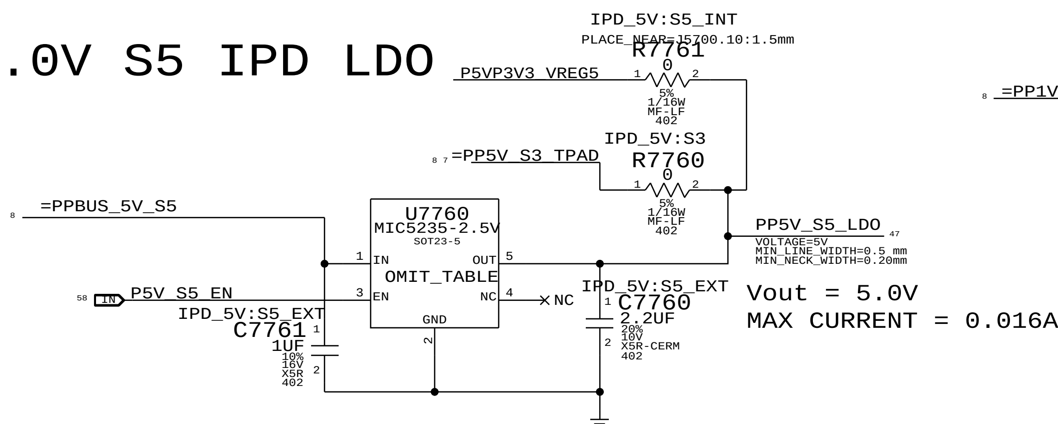
1.5V S0 Regulator

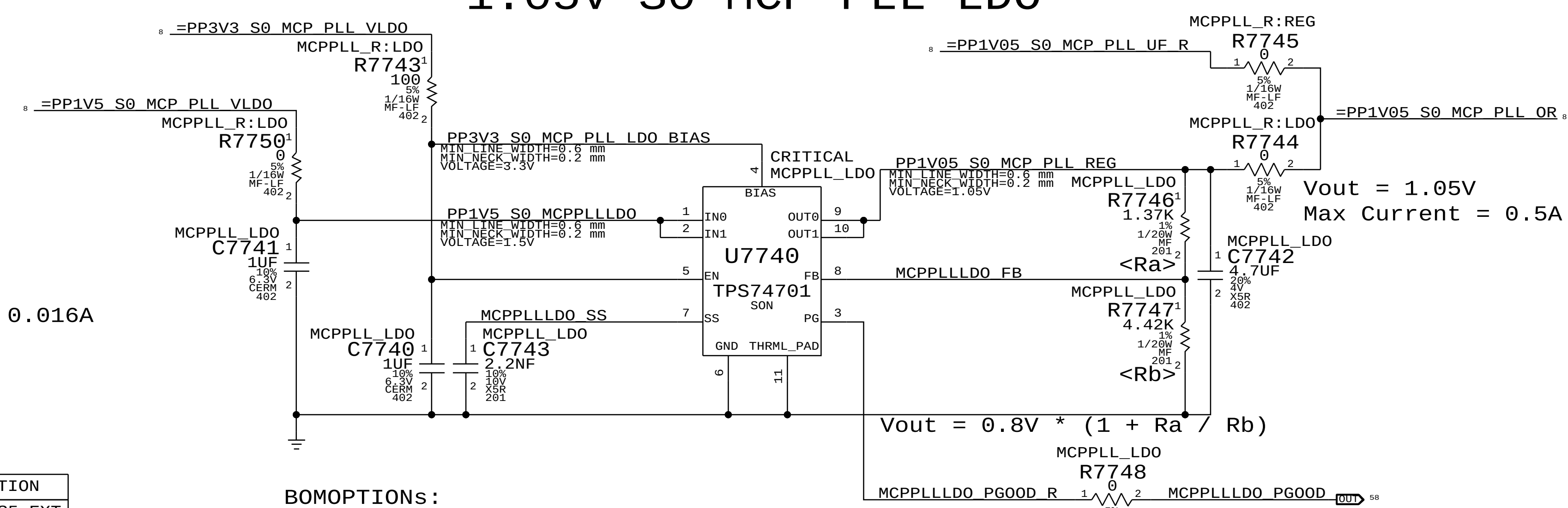


5.0V S5 IPD LDO



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S3034	1	IC,LDO,MIC5235,5V,1%,150MA,SOT23-5	U7760		IPD_5V:S5_EXT

1.05V S0 MCP PLL LDO

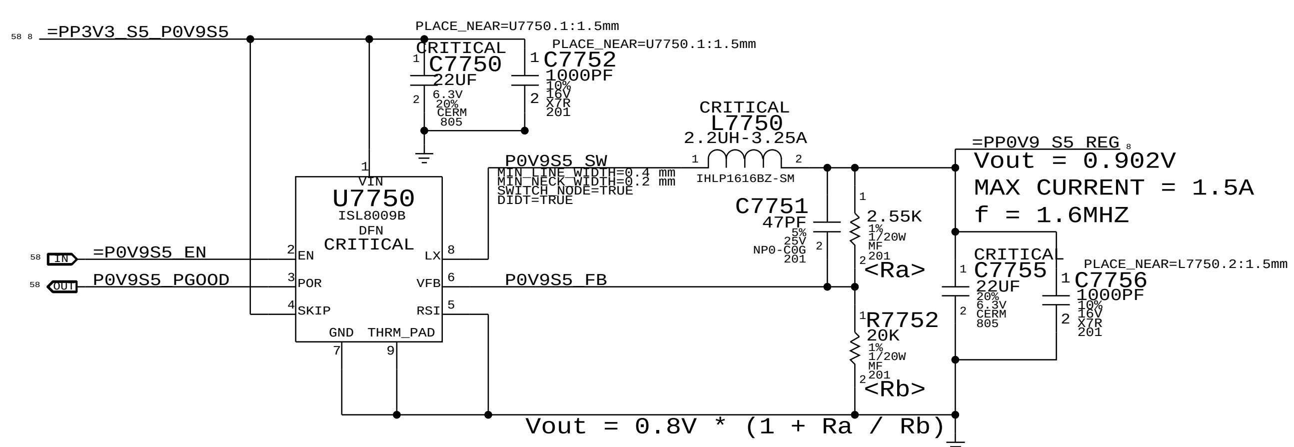


BOMOPTIONs:

MCPPLL_R:LDO - 1.05V S0 USED FOR MCP PLL LDO POWER.
MCPPLL_LDO - STUFFS U7740 AND RELATED CIRCUITRY.
TO USE U7740, MCPPLL_R:LDO AND MCPPLL_LDO MUST BE ACTIVE.
TO USE 1.05V S0, MCPPLL_R:REG MUST BE ACTIVE, MCPPLL_LDO CAN BE ACTIVE, MCPPLL_R:LDO MUST BE INACTIVE.

MCP 0.9V S5 (AUXC) Switcher

R7751



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: Misc Power Supplies

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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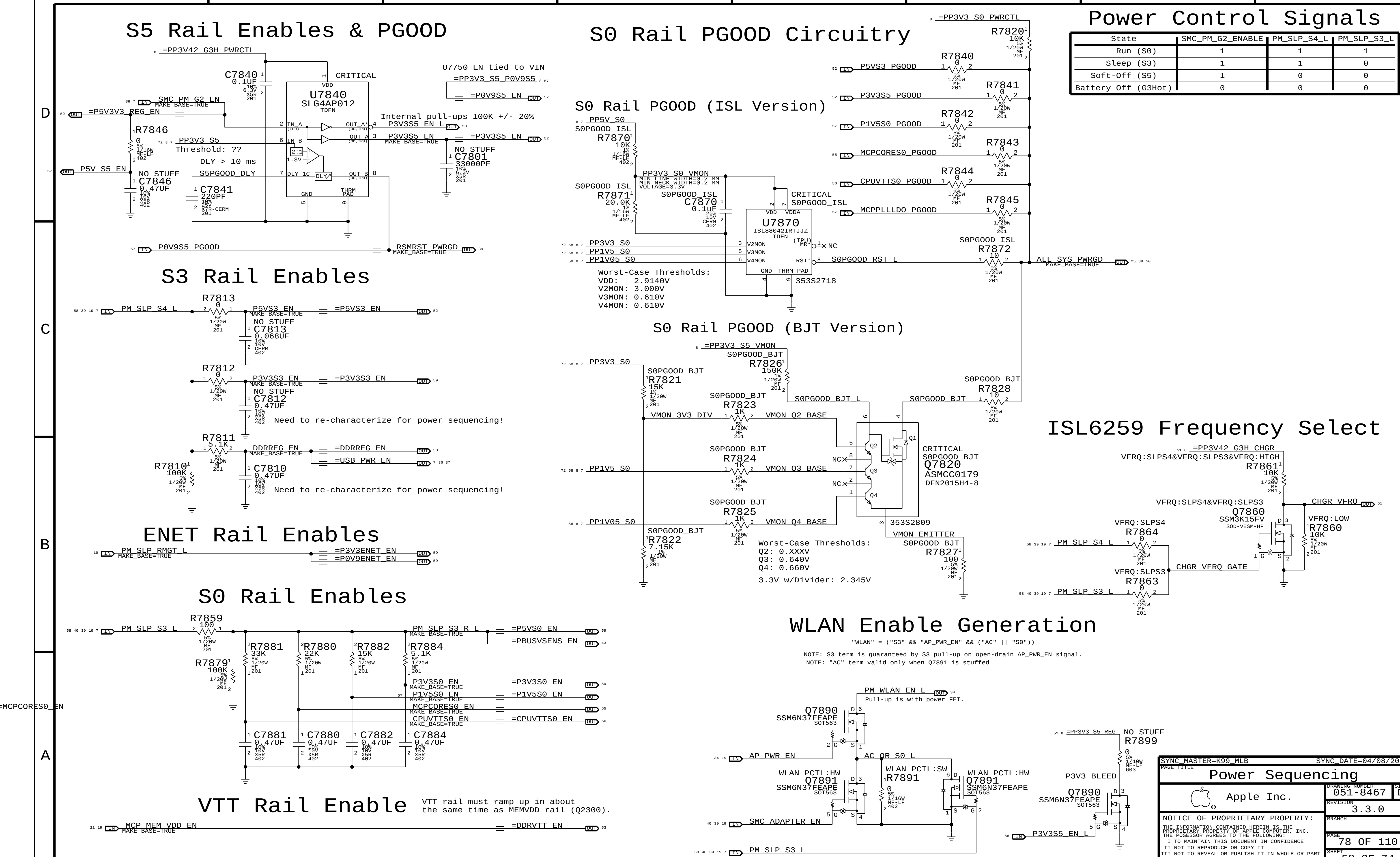
Power Control Signals

State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3Hot)	0	0	0



SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

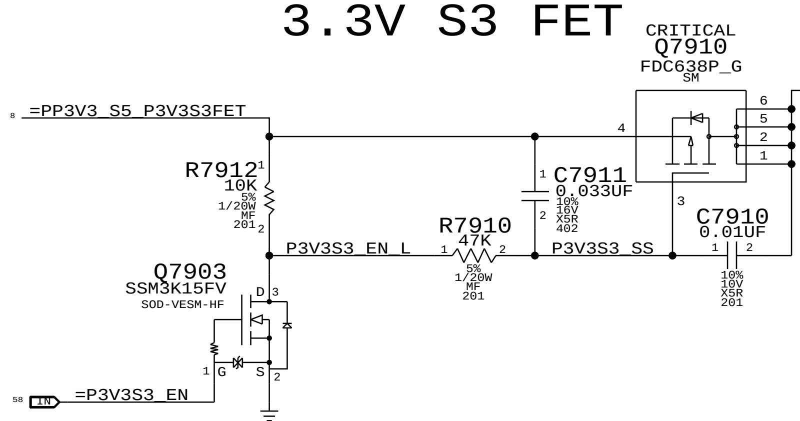
PAGE TITLE: Power Sequencing

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

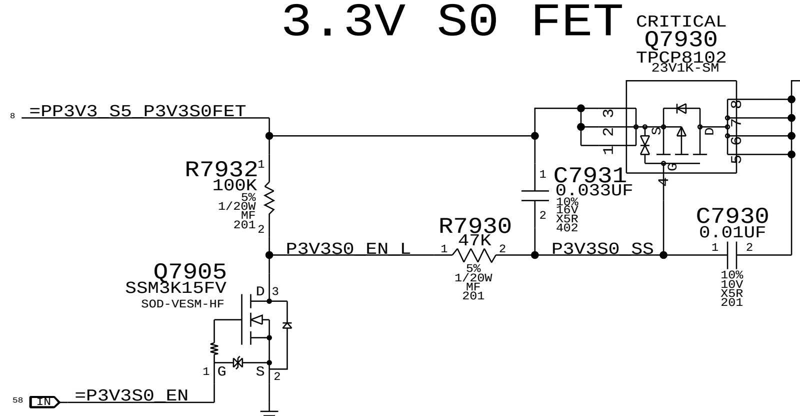
REVISION: 3.3.0

3.3V S3 FET



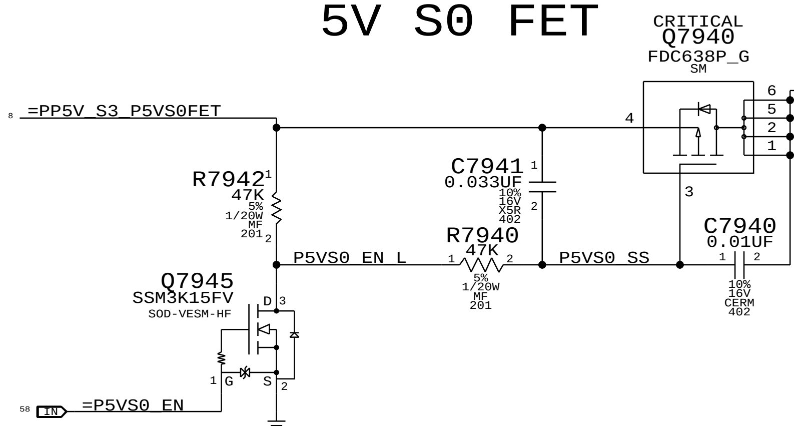
Q7910	
MOSFET	FDC638P
Type	P-Channel
Rds(on)	65 mOhm @2.5V
ID(max)	2.0 A @85C
Loading	1.274 A (EDP)

3.3V S0 FET



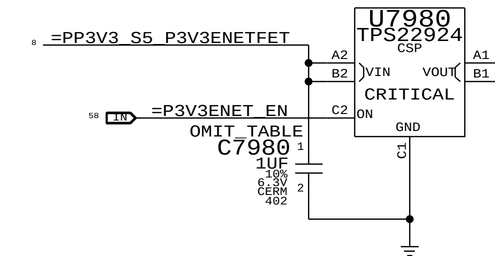
Q7930	
MOSFET	TPCP8102
Type	P-Channel
Rds(on)	14 mOhm @4.5V
ID(max)	7.2 A @85C
Loading	3.294 A (EDP)

5V S0 FET



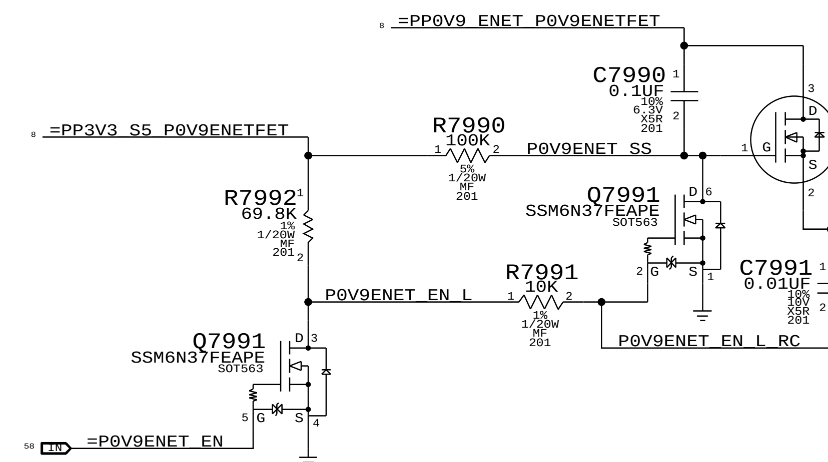
Q7940	
Part	FDC638P
Type	P-Channel
Rds(on)	65 mOhm @2.5V
ID(max)	2.0 A @85C
Loading	0.100 A (EDP)

3.3V ENET Switch



U7980	
Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max
I(max)	2 A
Loading	0.4 A (EDP)

0.9V ENET FET



Q7990	
MOSFET	SI2312BDS
Type	N-Channel
Rds(on)	37 mOhm @2.5V
ID(max)	3.25 A @85C
Loading	0.140 A (EDP)

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: Power FETs

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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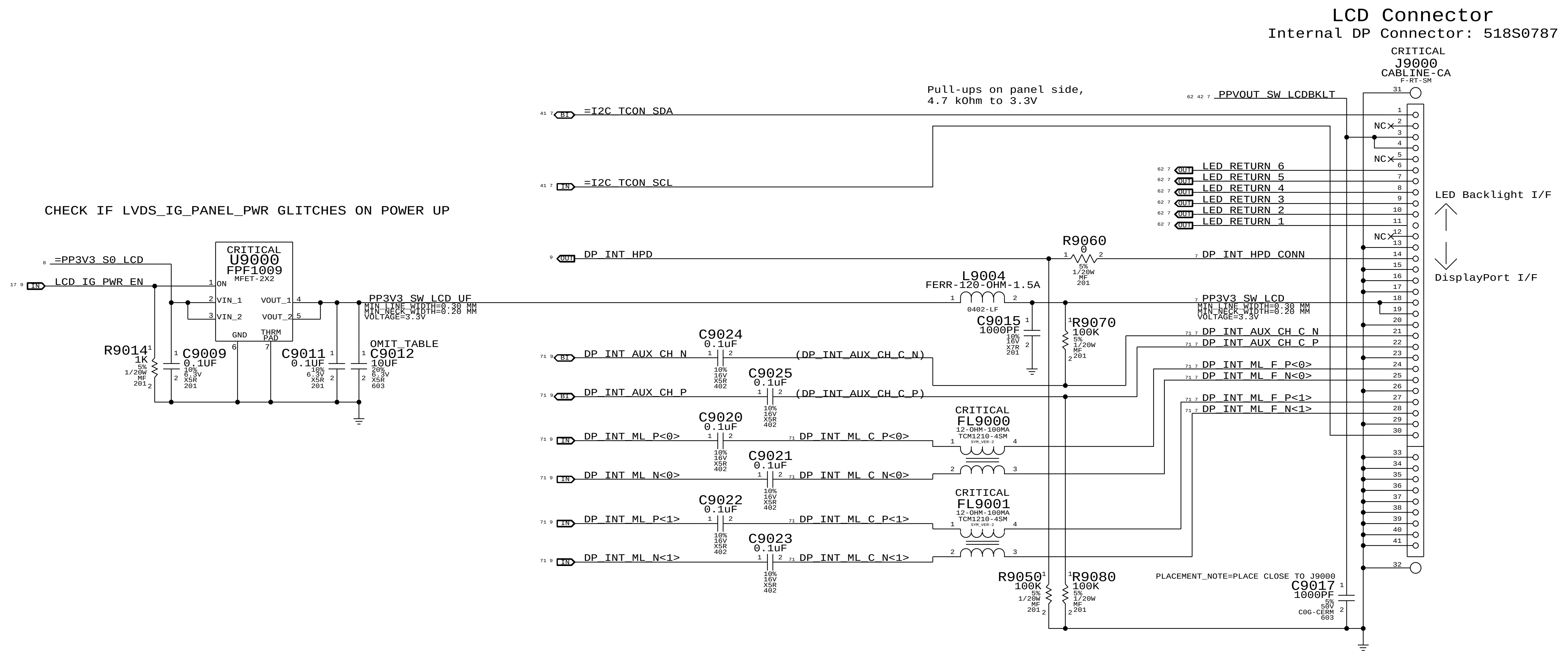
PAGE: 79 OF 110

SHEET: 59 OF 74



SYNC_MASTER=K99_MLB SYNC_DATE=07/23/2010

PAGE TITLE

Internal DisplayPort Connector

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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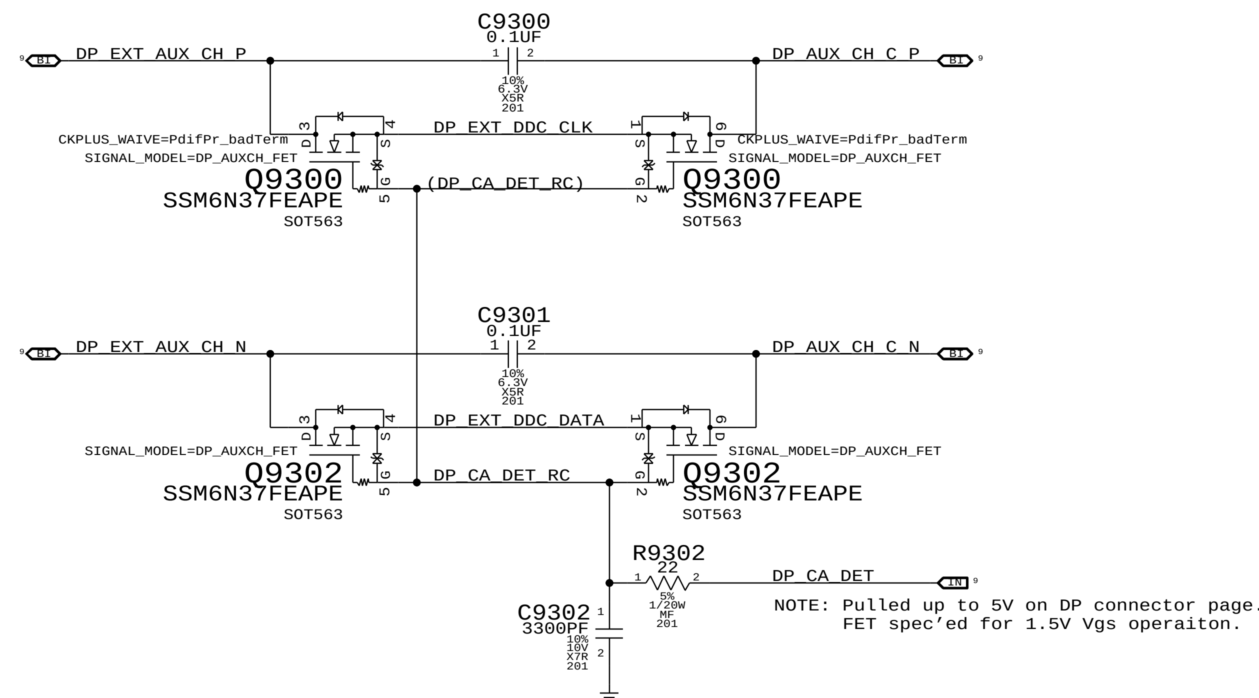

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

External DisplayPort Support

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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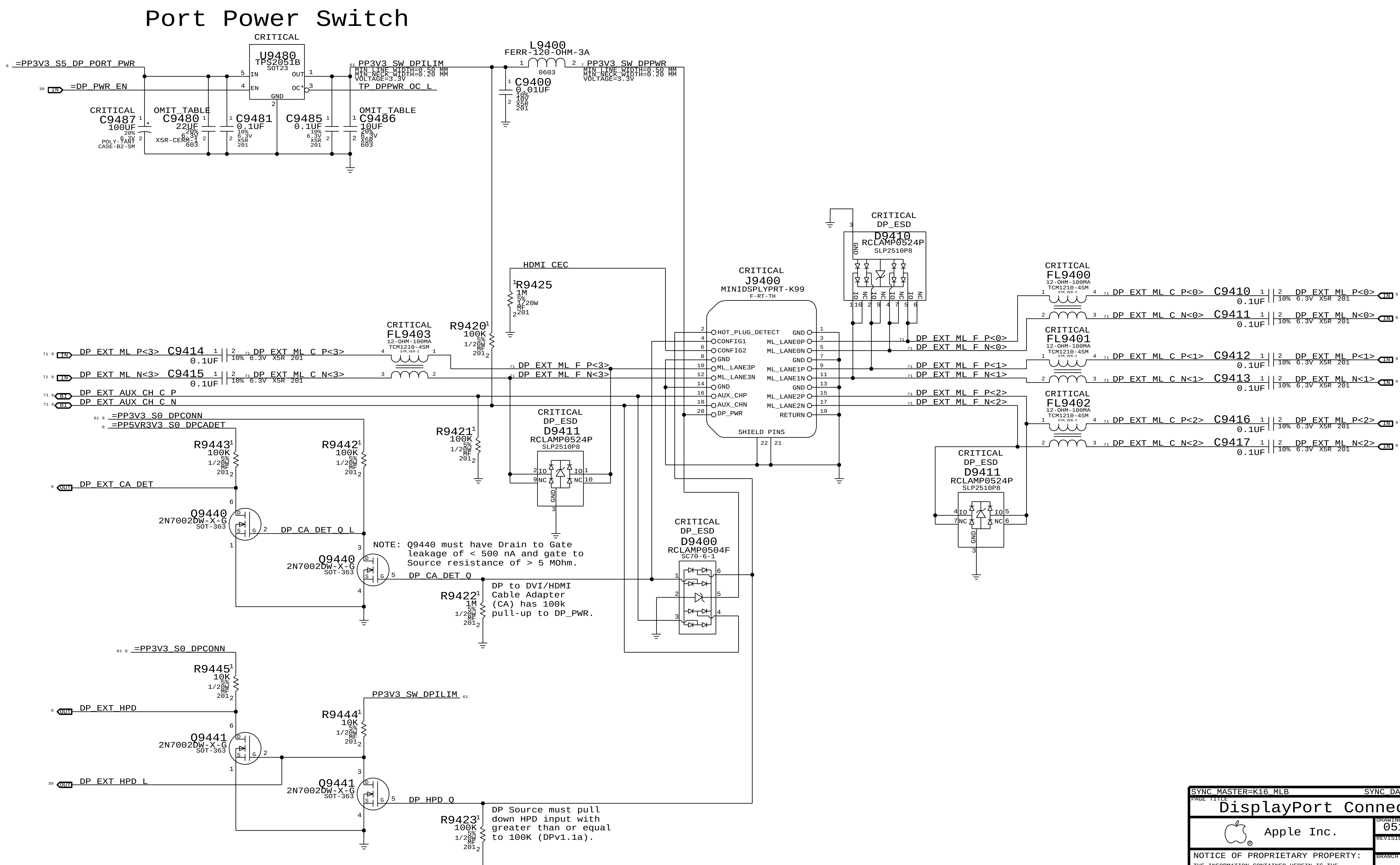

SYNC_MASTER=K16_MLB SYNC_DATE=06/01/2010

PAGE TITLE

DisplayPort Connector

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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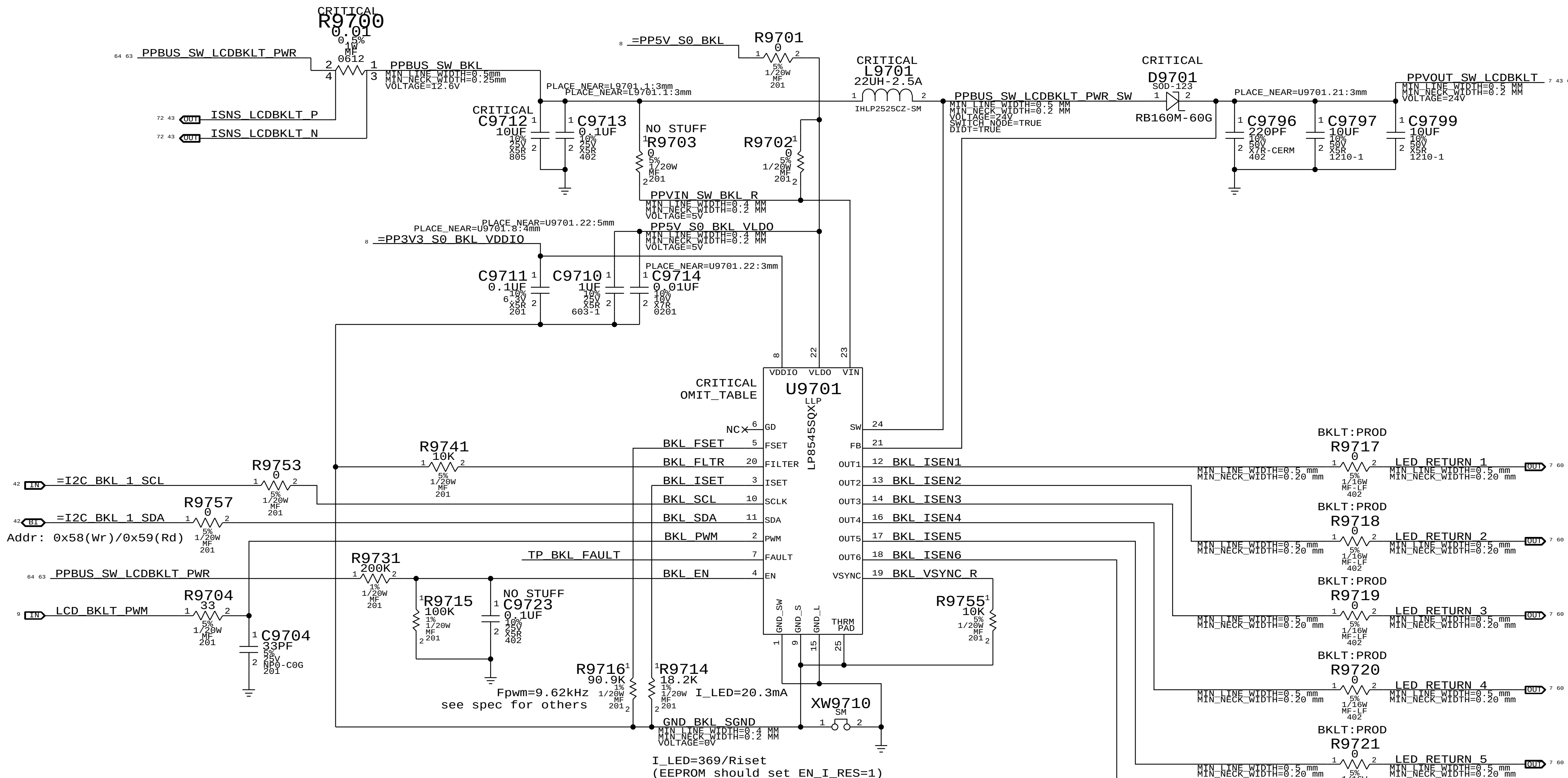
*L9701, D9701, C9796, C9797, C9799, C9712 AND C9713 SHOULD ALL BE PLACED NEAR EACHOTHER.
*PPBUS_SW_LCDBKLT_PWR_SW SHOULD BE KEPT AS SHORT AS POSSIBLE.
*LCD_BKLT_PWM SHOULD BE AWAY FROM BOOST CIRCUIT



FOR LP8543:
STUFF R9741
NO STUFF R9740, C9740, C9741, R9754

PART NUMBER	QTY		REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES,THIN FLIM,1/16W,10.2 OHM,0.1,0402,SM	R9717,R9718,R9719		BKLT:ENG
103S0198	3	RES,THIN FLIM,1/16W,10.2 OHM,0.1,0402,SM	R9720,R9721,R9722		BKLT:ENG
353S2896	1	IC,LP8545,LED BKLT CTRLR,PRODUCTIO,LLP24	U9701	CRITICAL	PROJ:K16
353S2967	1	IC,LP8545,LED BKLT CTRLR,LLP24,K99 VER	U9701	CRITICAL	PROJ:K99

10.2 ohm resistors for current measurement on LED strings.

SYNC_MASTER=(K99_MLB) SYNC_DATE=(03/01/2010)

PAGE TITLE
LCD Backlight Driver

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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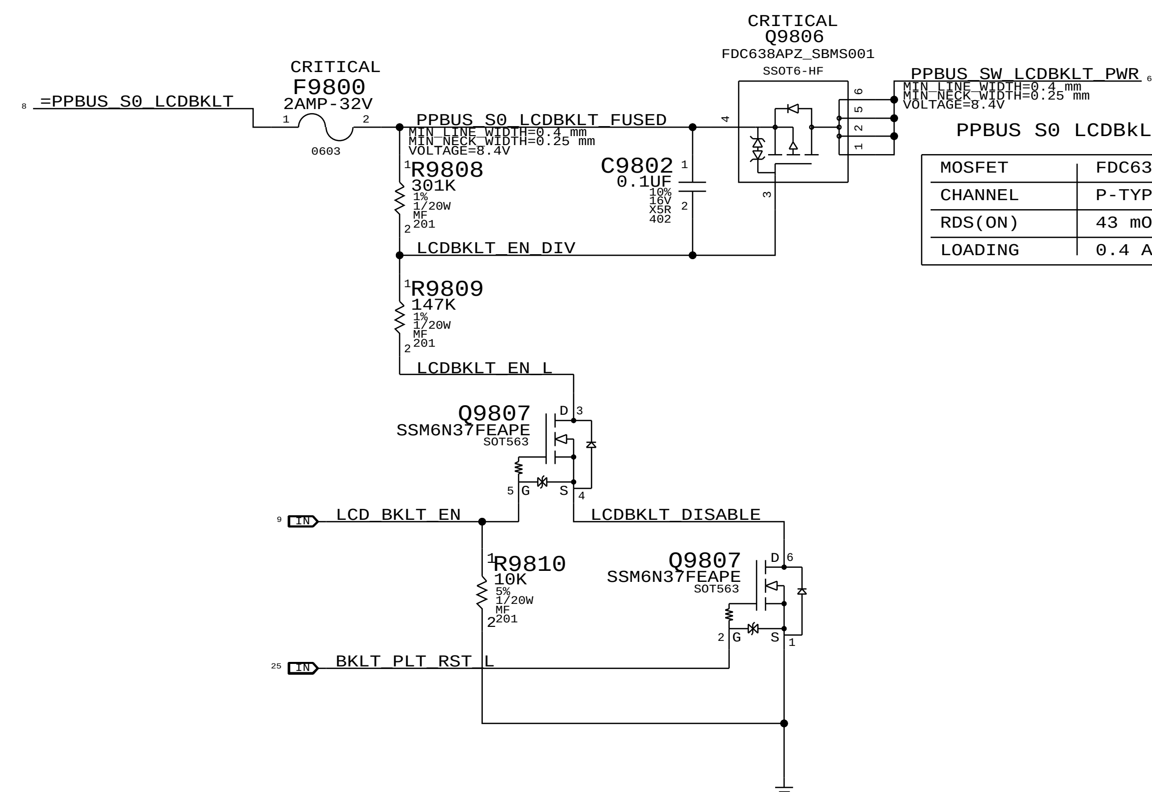

PPBUS S0 LCDBKLT FET

MOSFET	FDC638APZ
CHANNEL	P-TYPE
RDS(ON)	43 mOhm @4.5V
LOADING	0.4 A (EDP)

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE

LCD Backlight Support

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

REVISION 3.3.0

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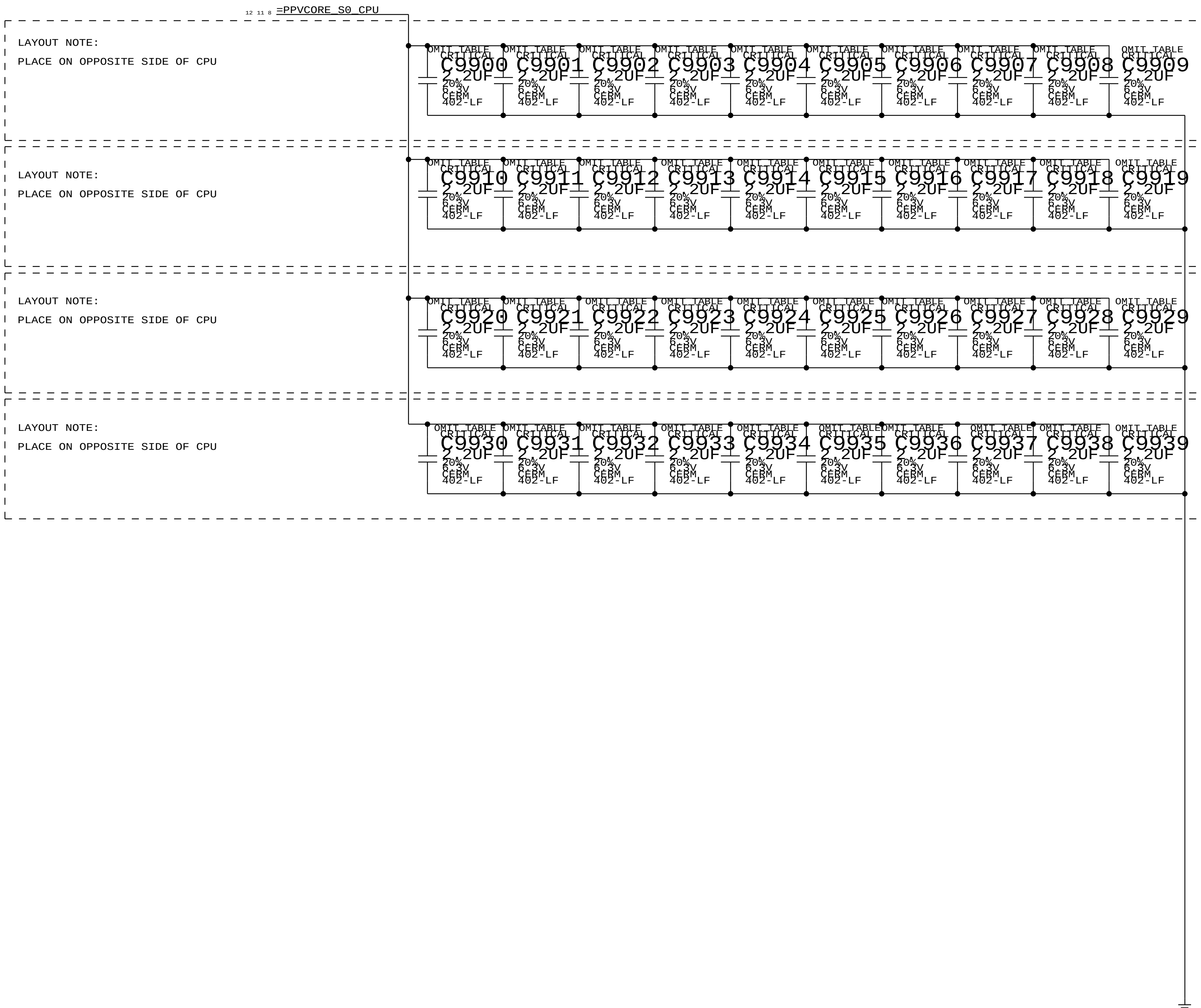
NOTICE OF PROPRIETARY PROPERTY:

ADDITIONAL CPU VCORE HF DECOUPLING

40x 1uF 0402



SYNC_MASTER=K99_MLB SYNC_DATE=05/18/2010

PAGE TITLE

Additional CPU/GPU Decoupling

Apple Inc.

DRAWING NUMBER 051-8467 | SIZE D

REVISION 3.3.0

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FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FSB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	*	=2x_DIELECTRIC	?
FSB_DSTB	*	=3x_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?
FSB_ADSTB	*	=2x_DIELECTRIC	?
FSB_1X	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_DSTB	TOP,BOTTOM	=5x_DIELECTRIC	?
FSB_ADDR	TOP,BOTTOM	=3x_DIELECTRIC	?
FSB_ADSTB	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_1X	TOP,BOTTOM	=3x_DIELECTRIC	?

All 4x/2x/1x FSB signals with impedance requirements are 50-ohm single-ended.

FSB 4X signals / groups shown in signal table on right.
Signals within each 4x group should be matched within 5 ps of strobe.
DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 135 ps.
Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.

FSB 2X signals / groups shown in signal table on right.
Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 270 ps.
Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1X signals shown in signal table on right.

Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1
SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	25 MIL	?
CPU_GTLREF	*	25 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?

SR DG recommends at least 25 mils, >50 mils preferred

Most CPU signals with impedance requirements are 55-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1
SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_FSB_COMP	*	8 MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.1.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_FSB_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

Group	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	Pages
FSB 4X Signal Groups	FSB_DATA_GROUP0	FSB_55S	FSB_DATA	FSB_D_L<15..0>	7 10 14
	FSB_DATA_GROUP0	FSB_55S	FSB_DATA	FSB_DINV_L<0>	7 10 14
	FSB_DSTB0	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_P<0>	7 10 14
	FSB_DSTB0	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_N<0>	7 10 14
	FSB_DATA_GROUP1	FSB_55S	FSB_DATA	FSB_D_L<31..16>	7 10 14
	FSB_DATA_GROUP1	FSB_55S	FSB_DATA	FSB_DINV_L<1>	7 10 14
	FSB_DSTB1	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_P<1>	7 10 14
	FSB_DSTB1	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_N<1>	7 10 14
	FSB_DATA_GROUP2	FSB_55S	FSB_DATA	FSB_D_L<47..32>	7 10 14
	FSB_DATA_GROUP2	FSB_55S	FSB_DATA	FSB_DINV_L<2>	7 10 14
	FSB_DSTB2	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_P<2>	7 10 14
	FSB_DSTB2	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_N<2>	7 10 14
	FSB_DATA_GROUP3	FSB_55S	FSB_DATA	FSB_D_L<63..48>	7 10 14
	FSB_DATA_GROUP3	FSB_55S	FSB_DATA	FSB_DINV_L<3>	7 10 14
	FSB_DSTB3	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_P<3>	7 10 14
	FSB_DSTB3	FSB_DSTB_55S	FSB_DSTB	FSB_DSTB_L_N<3>	7 10 14
FSB 2X Signals	FSB_ADDR_GROUP0	FSB_55S	FSB_ADDR	FSB_A_L<16..3>	7 10 14
	FSB_ADDR_GROUP0	FSB_55S	FSB_ADDR	FSB_REQ_L<4..0>	7 10 14
	FSB_ADSTB0	FSB_55S	FSB_ADSTB	FSB_ADSTB_L<0>	7 10 14
	FSB_ADDR_GROUP1	FSB_55S	FSB_ADDR	FSB_A_L<35..17>	7 10 14
	FSB_ADSTB1	FSB_55S	FSB_ADSTB	FSB_ADSTB_L<1>	7 10 14
FSB 1X Signals	FSB_1X	FSB_55S	FSB_1X	FSB_ADS_L	7 10 14
	FSB_BREQ0_L	FSB_55S	FSB_1X	FSB_BREQ0_L	10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_BNR_L	10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_BPRI_L	10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_DBSY_L	10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_DEFER_L	10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_DRDY_L	10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_HIT_L	7 10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_HITM_L	7 10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_LOCK_L	7 10 14
	FSB_CPURST_L	FSB_55S	FSB_1X	FSB_CPURST_L	10 13 14
	FSB_1X	FSB_55S	FSB_1X	FSB_RS_L<2..0>	10 14
	FSB_1X	FSB_55S	FSB_1X	FSB_TRDY_L	10 14
	CPU_ASYNC	CPU_55S	CPU_AGTL	CPU_A20M_L	10 14
	CPU_BSEL	CPU_55S	CPU_AGTL	CPU_BSEL<2..0>	9 10
	CPU_FERR_L	CPU_55S	CPU_8MIL	CPU_FERR_L	10 14
	CPU_ASYNC	CPU_55S	CPU_AGTL	CPU_IGNNE_L	10 14
	CPU_INIT_L	CPU_55S	CPU_AGTL	CPU_INIT_L	10 14
	CPU_ASYNC_R	CPU_55S	CPU_AGTL	CPU_INTR	10 14
	CPU_ASYNC_R	CPU_55S	CPU_AGTL	CPU_NMI	10 14
	CPU_PROCHOT_L	CPU_55S	CPU_AGTL	CPU_PROCHOT_L	10 14 40
	CPU_PWRGD	CPU_55S	CPU_AGTL	CPU_PWRGD	10 13 14
	CPU_ASYNC	CPU_55S	CPU_AGTL	CPU_SMI_L	10 14
	CPU_ASYNC	CPU_55S	CPU_AGTL	CPU_STPCLK_L	10 14
	PM_THRMTRIP_L	CPU_55S	CPU_8MIL	PM_THRMTRIP_L	10 14 40
	FSB_CPUSLP_L	CPU_55S	CPU_AGTL	FSB_CPUSLP_L	10 14
	CPU_FROM_SB	CPU_55S	CPU_AGTL	CPU_DPSLP_L	10 14
	CPU_DPRSTP_L	CPU_55S	CPU_AGTL	CPU_DPRSTP_L	10 14 54
	CPU_ASYNC	CPU_55S	CPU_AGTL	FSB_DPWR_L	10 14
	FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB_CLK_CPU_P	10 14
	FSB_CLK_CPU	CLK_FSB_100D	CLK_FSB	FSB_CLK_CPU_N	10 14
	FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB_CLK_ITP_P	13 14
	FSB_CLK_ITP	CLK_FSB_100D	CLK_FSB	FSB_CLK_ITP_N	13 14
	FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB_CLK_MCP_P	14
	FSB_CLK_MCP	CLK_FSB_100D	CLK_FSB	FSB_CLK_MCP_N	14
	CPU_IERR_L	CPU_55S		CPU_IERR_L	10
	PM_DPRSLPVR	CPU_55S	CPU_AGTL	PM_DPRSLPVR	14 54
	(See above)	CPU_55S	CPU_AGTL	IMVP_DPRSLPVR	
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_VDD	14
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_GND	14
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_VCC	14
	MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_GND	14
	CPU_GTLREF	CPU_50S	CPU_GTLREF	CPU_GTLREF	10 33
	CPU_COMP	CPU_50S	CPU_COMP	CPU_COMP<3>	10
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<2>	10
	CPU_COMP	CPU_50S	CPU_COMP	CPU_COMP<1>	10
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<0>	10
	XDP_TDI	CPU_55S	CPU_ITP	XDP_TDI	10 13
	XDP_TDO	CPU_55S	CPU_ITP	XDP_TDO	10 13
	XDP_TMS	CPU_55S	CPU_ITP	XDP_TMS	10 13
	XDP_TCK	CPU_55S	CPU_ITP	XDP_TCK	10 13
	XDP_TRST_L	CPU_55S	CPU_ITP	XDP_TRST_L	10 13
	XDP_BPM_L	CPU_55S	CPU_ITP	XDP_BPM_L<4..0>	10 13
	XDP_BPM_L5	CPU_55S	CPU_ITP	XDP_BPM_L<5>	10 13
	(FSB_CPURST_L)	CPU_55S	CPU_ITP	XDP_CPURST_L	13
		CPU_55S	CPU_8MIL	CPU_VID<6..0>	11 12 54
		CPU_55S	CPU_8MIL	IMVP6_VID<6..0>	12
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	11 54
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	11 54
	(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_P	54
	(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_N	54

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: CPU/FSB Constraints

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

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PAGE: 100 OF 110

SHEET: 66 OF 74

8	7	6	5	4	3	2	1

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	
MEM_CLK2MEM	*	=4:1_SPACING	?	NV DG says 3x inner, 4x outer
MEM_CTRL2CTRL	*	=2:1_SPACING	?	NV DG says 2x inner, 4x outer
MEM_CTRL2MEM	*	=2.5:1_SPACING	?	NV DG says 2x inner, 4x outer
MEM_CMD2CMD	*	=1.5:1_SPACING	?	NV DG says 2x inner, 4x outer
MEM_CMD2MEM	*	=3:1_SPACING	?	NV DG says 2x inner, 4x outer
MEM_DATA2DATA	*	=1.5:1_SPACING	?	NV DG says 2x inner, 4x outer
MEM_DATA2MEM	*	=3:1_SPACING	?	NV DG says 2x inner, 4x outer
MEM_DQS2MEM	*	=3:1_SPACING	?	NV DG says 4x inner, 5x outer
MEM_2OTHER	*	25 MIL	?	

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_DATA	*	*	MEM_2OTHER
MEM_DQS	*	*	MEM_2OTHER

Need to support MEM_*-style wildcards!

DDR3:
DQ signals should be matched within 5 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, inter-pair matching shoulw be within 360 ps
No DQS to clock matching requirement.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.
CMD/CTRL signals should be matched within 150 ps.
All memory signals maximum length is 1.030 ps.

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.3
SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MEM_COMP	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_MEM_COMP	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.2.2

MEM_A/B_CKE EC SET NAME IS CHANGED ON K6, CANNOT SYNC THIS PAGE FROM T27

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
MEM_A_CLK	MEM_70D	MEM_CLK	MEM_A_CLK_P<5..0>	9 15 26 27 32
MEM_A_CLK	MEM_70D	MEM_CLK	MEM_A_CLK_N<5..0>	9 15 26 27 32
MEM_A_CKE	MEM_50S	MEM_CTRL	MEM_A_CKE<3..0>	15 21 26 27 32
MEM_A_CNTL	MEM_50S	MEM_CTRL	MEM_A_CS_L<3..0>	15 26 27 32
MEM_A_CNTL	MEM_50S	MEM_CTRL	MEM_A_ODT<3..0>	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM_A_A<15..0>	9 15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM_A_BA<2..0>	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM_A_RAS_L	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM_A_CAS_L	15 26 27 32
MEM_A_CMD	MEM_50S	MEM_CMD	MEM_A_WE_L	15 26 27 32
MEM_A_DQ_BYTE0	MEM_55S	MEM_DATA	MEM_A_DQ<7..0>	15 26
MEM_A_DQ_BYTE1	MEM_55S	MEM_DATA	MEM_A_DQ<15..8>	15 26
MEM_A_DQ_BYTE2	MEM_55S	MEM_DATA	MEM_A_DQ<23..16>	15 26
MEM_A_DQ_BYTE3	MEM_55S	MEM_DATA	MEM_A_DQ<31..24>	15 26
MEM_A_DQ_BYTE4	MEM_55S	MEM_DATA	MEM_A_DQ<39..32>	15 27
MEM_A_DQ_BYTE5	MEM_55S	MEM_DATA	MEM_A_DQ<47..40>	15 27
MEM_A_DQ_BYTE6	MEM_55S	MEM_DATA	MEM_A_DQ<55..48>	15 27
MEM_A_DQ_BYTE7	MEM_55S	MEM_DATA	MEM_A_DQ<63..56>	15 27
MEM_A_DQ_BYTE0	MEM_55S	MEM_DATA	MEM_A_DM<0>	15 26
MEM_A_DQ_BYTE1	MEM_55S	MEM_DATA	MEM_A_DM<1>	15 26
MEM_A_DQ_BYTE2	MEM_55S	MEM_DATA	MEM_A_DM<2>	15 26
MEM_A_DQ_BYTE3	MEM_55S	MEM_DATA	MEM_A_DM<3>	15 26
MEM_A_DQ_BYTE4	MEM_55S	MEM_DATA	MEM_A_DM<4>	15 27
MEM_A_DQ_BYTE5	MEM_55S	MEM_DATA	MEM_A_DM<5>	15 27
MEM_A_DQ_BYTE6	MEM_55S	MEM_DATA	MEM_A_DM<6>	15 27
MEM_A_DQ_BYTE7	MEM_55S	MEM_DATA	MEM_A_DM<7>	15 27
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM_A_DQS_P<0>	15 26
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM_A_DQS_N<0>	15 26
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM_A_DQS_P<1>	15 26
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM_A_DQS_N<1>	15 26
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM_A_DQS_P<2>	15 26
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM_A_DQS_N<2>	15 26
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM_A_DQS_P<3>	15 26
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM_A_DQS_N<3>	15 26
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM_A_DQS_P<4>	15 27
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM_A_DQS_N<4>	15 27
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM_A_DQS_P<5>	15 27
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM_A_DQS_N<5>	15 27
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM_A_DQS_P<6>	15 27
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM_A_DQS_N<6>	15 27
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM_A_DQS_P<7>	15 27
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM_A_DQS_N<7>	15 27
MEM_B_CLK	MEM_70D	MEM_CLK	MEM_B_CLK_P<5..0>	9 15 28 29 32
MEM_B_CLK	MEM_70D	MEM_CLK	MEM_B_CLK_N<5..0>	9 15 28 29 32
MEM_B_CKE	MEM_50S	MEM_CTRL	MEM_B_CKE<3..0>	15 21 28 29 32
MEM_B_CNTL	MEM_50S	MEM_CTRL	MEM_B_CS_L<3..0>	15 28 29 32
MEM_B_CNTL	MEM_50S	MEM_CTRL	MEM_B_ODT<3..0>	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM_B_A<15..0>	9 15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM_B_BA<2..0>	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM_B_RAS_L	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM_B_CAS_L	15 28 29 32
MEM_B_CMD	MEM_50S	MEM_CMD	MEM_B_WE_L	15 28 29 32
MEM_B_DQ_BYTE0	MEM_55S	MEM_DATA	MEM_B_DQ<7..0>	15 28
MEM_B_DQ_BYTE1	MEM_55S	MEM_DATA	MEM_B_DQ<15..8>	15 28
MEM_B_DQ_BYTE2	MEM_55S	MEM_DATA	MEM_B_DQ<23..16>	15 28
MEM_B_DQ_BYTE3	MEM_55S	MEM_DATA	MEM_B_DQ<31..24>	15 28
MEM_B_DQ_BYTE4	MEM_55S	MEM_DATA	MEM_B_DQ<39..32>	15 29
MEM_B_DQ_BYTE5	MEM_55S	MEM_DATA	MEM_B_DQ<47..40>	15 29
MEM_B_DQ_BYTE6	MEM_55S	MEM_DATA	MEM_B_DQ<55..48>	15 29
MEM_B_DQ_BYTE7	MEM_55S	MEM_DATA	MEM_B_DQ<63..56>	15 29
MEM_B_DQ_BYTE0	MEM_55S	MEM_DATA	MEM_B_DM<0>	15 28
MEM_B_DQ_BYTE1	MEM_55S	MEM_DATA	MEM_B_DM<1>	15 28
MEM_B_DQ_BYTE2	MEM_55S	MEM_DATA	MEM_B_DM<2>	15 28
MEM_B_DQ_BYTE3	MEM_55S	MEM_DATA	MEM_B_DM<3>	15 28
MEM_B_DQ_BYTE4	MEM_55S	MEM_DATA	MEM_B_DM<4>	15 29
MEM_B_DQ_BYTE5	MEM_55S	MEM_DATA	MEM_B_DM<5>	15 29
MEM_B_DQ_BYTE6	MEM_55S	MEM_DATA	MEM_B_DM<6>	15 29
MEM_B_DQ_BYTE7	MEM_55S	MEM_DATA	MEM_B_DM<7>	15 29
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM_B_DQS_P<0>	15 28
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM_B_DQS_N<0>	15 28
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM_B_DQS_P<1>	15 28
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM_B_DQS_N<1>	15 28
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM_B_DQS_P<2>	15 28
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM_B_DQS_N<2>	15 28
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM_B_DQS_P<3>	15 28
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM_B_DQS_N<3>	15 28
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM_B_DQS_P<4>	15 29
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM_B_DQS_N<4>	15 29
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM_B_DQS_P<5>	15 29
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM_B_DQS_N<5>	15 29
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM_B_DQS_P<6>	15 29
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM_B_DQS_N<6>	15 29
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM_B_DQS_P<7>	15 29
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM_B_DQS_N<7>	15 29
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_VDD	15
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_GND	15

SYNC_MASTER=K99_MLB SYNC_DATE=04/08/2010

PAGE TITLE: Memory Constraints

Apple Inc.

DRAWING NUMBER: 051-8467 SIZE: D

REVISION: 3.3.0

BRANCH

D

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?
MCP_PEX_COMP	*	8 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.3

NEED PCIe Gen1/Gen2 notes!

Analog Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CRT_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CRT	*	20 MIL	?
CRT_2CRT	*	15 MIL	?
CRT_2CLK	*	50 MIL	?
CRT_2SWITCHER	*	250 MIL	?
CRT_SYNC	*	=4x_DIELECTRIC	?
MCP_DAC_COMP	*	=2x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CRT	CRT	*	CRT_2CRT

C

CRT signal single-ended impedence varies by location:
- 37.5-ohm from MCP to first termination resistor.
- 50-ohm from first to second termination resistor.
- 75-ohm from output of three-pole filter to connector (if possible).

R/G/B signals should be matched as close as possible and < 10 inches.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.1.

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	20 MIL	20 MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be matched within 100 mils.
NOTE: NV DG recommends 90 ohm differential for LVDS, but cable/display assume 100 ohm.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 100 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max trace length: LVDS 10 inches, DP 8.5 inches.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.4.2

B

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=3x_DIELECTRIC	?
SATA_TERMP	*	8 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=4x_DIELECTRIC	?

SATA intra-pair matching should be 1 ps.
Max trace length: 12 inches for SATA Gen1/Gen2, TBD for SATA Gen3.
SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.6

A

MCP89 Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
	PCIE_90D	PCIE	PEG_R2D_P<15..0>	
	PCIE_90D	PCIE	PEG_R2D_N<15..0>	
PEG_R2D	PCIE_90D	PCIE	PEG_R2D_C_P<15..0>	
	PCIE_90D	PCIE	PEG_R2D_C_N<15..0>	
PEG_D2R	PCIE_90D	PCIE	PEG_D2R_P<15..0>	
	PCIE_90D	PCIE	PEG_D2R_N<15..0>	
	PCIE_90D	PCIE	PEG_D2R_C_P<15..0>	
	PCIE_90D	PCIE	PEG_D2R_C_N<15..0>	
	PCIE_90D	PCIE	PCIE_AP_R2D_P	7 34
	PCIE_90D	PCIE	PCIE_AP_R2D_N	7 34
PCIE_AP_R2D	PCIE_90D	PCIE	PCIE_AP_R2D_C_P	16 34
	PCIE_90D	PCIE	PCIE_AP_R2D_C_N	16 34
PCIE_AP_D2R	PCIE_90D	PCIE	PCIE_AP_D2R_P	7 16 34
	PCIE_90D	PCIE	PCIE_AP_D2R_N	7 16 34
	PCIE_90D	PCIE	PCIE_ENET_R2D_P	
	PCIE_90D	PCIE	PCIE_ENET_R2D_N	
PCIE_ENET_R2D	PCIE_90D	PCIE	PCIE_ENET_R2D_C_P	
	PCIE_90D	PCIE	PCIE_ENET_R2D_C_N	
PCIE_ENET_D2R	PCIE_90D	PCIE	PCIE_ENET_D2R_P	
	PCIE_90D	PCIE	PCIE_ENET_D2R_N	
	PCIE_90D	PCIE	PCIE_ENET_D2R_C_P	
	PCIE_90D	PCIE	PCIE_ENET_D2R_C_N	
	PCIE_90D	PCIE	PCIE_FW_R2D_P	
	PCIE_90D	PCIE	PCIE_FW_R2D_N	
PCIE_FW_R2D	PCIE_90D	PCIE	PCIE_FW_R2D_C_P	
	PCIE_90D	PCIE	PCIE_FW_R2D_C_N	
PCIE_FW_D2R	PCIE_90D	PCIE	PCIE_FW_D2R_P	
	PCIE_90D	PCIE	PCIE_FW_D2R_N	
	PCIE_90D	PCIE	PCIE_FW_D2R_C_P	
	PCIE_90D	PCIE	PCIE_FW_D2R_C_N	
MCP_PE0_REFCLK	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_P	9 16
	CLK_PCIE_100D	CLK_PCIE	PEG_CLK100M_N	9 16
MCP_PE1_REFCLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_AP_P	7 16 34
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_AP_N	7 16 34
MCP_PE2_REFCLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_ENET_P	
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_ENET_N	
MCP_PE3_REFCLK	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_P	
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_N	
MCP_PEX_CLK_COMP		MCP_PEX_COMP	MCP_PEX0_TERMP	16
CRT_RED	CRT_50S	CRT	CRT_IG_R_C_PR	
CRT_GREEN	CRT_50S	CRT	CRT_IG_G_Y_Y	
CRT_BLUE	CRT_50S	CRT	CRT_IG_B_COMP_PB	
CRT_SYNC	CRT_50S	CRT_SYNC	CRT_IG_HSYNC	
CRT_SYNC	CRT_50S	CRT_SYNC	CRT_IG_VSYNC	
MCP_DAC_RSET		MCP_DAC_COMP	MCP_TV_DAC_RSET	
MCP_DAC_VREF		MCP_DAC_COMP	MCP_TV_DAC_VREF	
DP_INT_ML	DP_90D	DISPLAYPORT	DP_IG_ML1_P<1..0>	9 17
DP_INT_ML	DP_90D	DISPLAYPORT	DP_IG_ML1_N<1..0>	9 17
DP_INT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH1_P	9 17
DP_INT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH1_N	9 17
DP_EXT_ML	DP_90D	DISPLAYPORT	DP_IG_ML0_P<3..0>	9 17
DP_EXT_ML	DP_90D	DISPLAYPORT	DP_IG_ML0_N<3..0>	9 17
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH0_P	9 17
DP_EXT_AUX_CH	DP_90D	DISPLAYPORT	DP_IG_AUX_CH0_N	9 17
MCP_TMDS0_RSET	MCP_DV_COMP		MCP_TMDS0_RSET	17 24
MCP_TMDS0_VPROBE			MCP_TMDS0_VPROBE	17 24
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_P	
LVDS_IG_A_CLK	LVDS_100D	LVDS	LVDS_IG_A_CLK_N	
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<2..0>	
LVDS_IG_A_DATA	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<2..0>	
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS_IG_A_DATA_P<3>	
LVDS_IG_A_DATA3	LVDS_100D	LVDS	LVDS_IG_A_DATA_N<3>	
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS_IG_B_CLK_P	
LVDS_IG_B_CLK	LVDS_100D	LVDS	LVDS_IG_B_CLK_N	
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<2..0>	
LVDS_IG_B_DATA	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<2..0>	
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS_IG_B_DATA_P<3>	
LVDS_IG_B_DATA3	LVDS_100D	LVDS	LVDS_IG_B_DATA_N<3>	
MCP_IFPAB_RSET	MCP_DV_COMP		MCP_IFPAB_RSET	17 24
MCP_IFPAB_VPROBE			MCP_IFPAB_VPROBE	17 24
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_C_P	18 35
	SATA_90D	SATA	SATA_HDD_R2D_C_N	18 35
	SATA_90D	SATA	SATA_HDD_R2D_P	7 35
	SATA_90D	SATA	SATA_HDD_R2D_N	7 35
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_P	18 35
	SATA_90D	SATA	SATA_HDD_D2R_N	18 35
	SATA_90D	SATA	SATA_HDD_D2R_C_P	7 35
	SATA_90D	SATA	SATA_HDD_D2R_C_N	7 35
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_P	9 18
	SATA_90D	SATA	SATA_ODD_R2D_C_N	9 18
	SATA_90D	SATA	SATA_ODD_R2D_P	
	SATA_90D	SATA	SATA_ODD_R2D_N	
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_P	9 18
	SATA_90D	SATA	SATA_ODD_D2R_N	9 18
	SATA_90D	SATA	SATA_ODD_D2R_C_P	
	SATA_90D	SATA	SATA_ODD_D2R_C_N	
MCP_SATA_TERMP		SATA_TERMP	MCP_SATA_TERMP	18

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PAGE TITLE
MCP Constraints 1

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5x_DIELECTRIC	?
CLK_LPC	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.7

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.8

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.9

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?
MCP_HDA_COMP	*	8 MIL	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.10

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.11

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=1.5x_DIELECTRIC	?

SOURCE: MCP89 Interface DG (DG-04625-001_v0.9), Section 2.12

MCP89 Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
LPC_AD	LPC_55S	LPC	LPC_AD<3..0>	7 19 39 41
LPC_FRAME_L	LPC_55S	LPC	LPC_FRAME_L	7 19 39 41
LPC_RESET_L	LPC_55S	LPC	LPC_RESET_L	19 25
MCP_LPC_CLK0	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	19 25
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	25 39
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	7 25 41
USB_EXTA	USB_90D	USB	USB_EXTA_P	18 36
	USB_90D	USB	USB_EXTA_N	18 36
	USB_90D	USB	USB_EXTA_MUXED_P	36 72
	USB_90D	USB	USB_EXTA_MUXED_N	36 72
USB_MINI	USB_90D	USB	USB_MINI_P	9 18
	USB_90D	USB	USB_MINI_N	9 18
USB_EXTD	USB_90D	USB	USB_EXTD_P	7 18 37
	USB_90D	USB	USB_EXTD_N	7 18 37
USB_CAMERA	USB_90D	USB	USB_CAMERA_P	7 18 37
	USB_90D	USB	USB_CAMERA_N	7 18 37
USB_BT	USB_90D	USB	USB_BT_P	7 18 34
	USB_90D	USB	USB_BT_N	7 18 34
USB_TPAD	USB_90D	USB	USB_TPAD_P	18 47 72
	USB_90D	USB	USB_TPAD_N	18 47 72
USB_IR	USB_90D	USB	USB_IR_P	
	USB_90D	USB	USB_IR_N	
USB_EXTB	USB_90D	USB	USB_EXTB_P	
	USB_90D	USB	USB_EXTB_N	
USB_T57	USB_90D	USB	USB_T57_P	
	USB_90D	USB	USB_T57_N	
USB_EXTC	USB_90D	USB	USB_EXTC_P	9 18
	USB_90D	USB	USB_EXTC_N	9 18
USB_SDCARD	USB_90D	USB	USB_SDCARD_P	18 38
	USB_90D	USB	USB_SDCARD_N	18 38
USB_WM	USB_90D	USB	USB_WM_P	
	USB_90D	USB	USB_WM_N	
MCP_USB_RBIAS	MCP_USB_RBIAS		MCP_USB_RBIAS_GND	18
SMBUS_MCP_0_CLK	SMB_55S	SMB	SMBUS_MCP_0_CLK	19 42
SMBUS_MCP_0_DATA	SMB_55S	SMB	SMBUS_MCP_0_DATA	19 42
(SMBUS_SMC_MGMT_SCL)	SMB_55S	SMB	SMBUS_MCP_1_CLK	19 42
(SMBUS_SMC_MGMT_SDA)	SMB_55S	SMB	SMBUS_MCP_1_DATA	19 42
HDA_BIT_CLK	HDA_55S	HDA	HDA_BIT_CLK	7 19 37
	HDA_55S	HDA	HDA_BIT_CLK_R	19
HDA_SYNC	HDA_55S	HDA	HDA_SYNC	7 19 37
	HDA_55S	HDA	HDA_SYNC_R	19
HDA_RST_L	HDA_55S	HDA	HDA_RST_R_L	19
	HDA_55S	HDA	HDA_RST_L	7 19 37
HDA_SDIN0	HDA_55S	HDA	HDA_SDIN0	7 19 37
	HDA_55S	HDA	HDA_SDIN_CODEC	
HDA_SDOUT	HDA_55S	HDA	HDA_SDOUT	7 19 37
	HDA_55S	HDA	HDA_SDOUT_R	19
MCP_HDA_PULLDN_COMP		MCP_HDA_COMP	MCP_HDA_PULLDN_COMP	19
MCP_SUS_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK_R	19 25
	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK	25 39
SPI_CLK	SPI_55S	SPI	SPI_CLK_R	19 41
	SPI_55S	SPI	SPI_CLK	41
SPI_MOSI	SPI_55S	SPI	SPI_MOSI_R	19 41
	SPI_55S	SPI	SPI_MOSI	41
SPI_MISO	SPI_55S	SPI	SPI_MISO	19 41
SPI_CS0	SPI_55S	SPI	SPI_CS0_R_L	19 41
	SPI_55S	SPI	SPI_CS0_L	41
	SPI_55S	SPI	SPI_MLB_CLK	41 48
	SPI_55S	SPI	SPI_MLB_MOSI	41 48
	SPI_55S	SPI	SPI_MLB_MISO	41 48
	SPI_55S	SPI	SPI_MLB_CS_L	41 48
	SPI_55S	SPI	SPI_ALT_CLK	7 41
	SPI_55S	SPI	SPI_ALT_MOSI	7 41
	SPI_55S	SPI	SPI_ALT_MISO	7 41
	SPI_55S	SPI	SPI_ALT_CS_L	7 41

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PAGE TITLE
MCP Constraints 2

Apple Inc.

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MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MII_COMP	*	=STANDARD	7.5 MIL	7.5 MIL	=STANDARD	=STANDARD	=STANDARD
ENET_MII_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	12 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

88E1116R (Ethernet PHY) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_MDI_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	25 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

SD Card Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_INTERFACE	*	=3X_DIELECTRIC	?

RGMII Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
MCP_MII_COMP	MCP_MII_COMP		MCP_MII_COMP_VDD	18
MCP_MII_COMP	MCP_MII_COMP		MCP_MII_COMP_GND	18
MCP_CLK25M_BUF0	ENET_MII_55S	MCP_BUF0_CLK	MCP_CLK25M_BUF0_R	
	ENET_MII_55S	MCP_BUF0_CLK	RTL8211_CLK25M_CKXTAL1	
ENET_INTR_L	ENET_MII_55S	ENET_MII	ENET_INTR_L	
ENET_MDIO	ENET_MII_55S	ENET_MII	ENET_MDIO	9 18
ENET_MDC	ENET_MII_55S	ENET_MII	ENET_MDC	
ENET_PWRDWN_L	ENET_MII_55S	ENET_MII	ENET_PWRDWN_L	
ENET_RXCLK	ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK_R	
	ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK	9 18
	ENET_MII_55S	ENET_MII	ENET_RXD_R<3..0>	
ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET_RXD<0>	9 18
ENET_RXD_STRAP	ENET_MII_55S	ENET_MII	ENET_RXD<3..1>	9 18
ENET_RXD	ENET_MII_55S	ENET_MII	ENET_RX_CTRL	9 18
ENET_TXCLK	ENET_MII_55S	ENET_MII	ENET_CLK125M_TXCLK	
ENET_TXD	ENET_MII_55S	ENET_MII	ENET_TXD<0>	
ENET_TXD	ENET_MII_55S	ENET_MII	ENET_TXD<3..1>	
ENET_TXD	ENET_MII_55S	ENET_MII	ENET_TX_CTRL	
	ENET_MII_55S	ENET_MII	ENET_RESET_L	

Ethernet Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET
ENET_MDI	ENET_MDI_100D	ENET_MDI	ENET_MDI_P<3..0>
	ENET_MDI_100D	ENET_MDI	ENET_MDI_N<3..0>

SD Card Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
SD_DATA	SD_55S	SD_INTERFACE	SD_D<4..0>	7 38
	SD_55S	SD_INTERFACE	SDCONN_DATA<4..0>	
	SD_55S	SD_INTERFACE	BCM57765_CR_DATA<4>	
SD_DATA_R	SD_55S	SD_INTERFACE	SD_D<7..5>	7 38
	SD_55S	SD_INTERFACE	SDCONN_DATA<7..5>	
	SD_55S	SD_INTERFACE	BCM57765_CR_DATA<7..5>	
SD_CLK	SD_55S	SD_INTERFACE	SD_CLK	7 38
	SD_55S	SD_INTERFACE	SD_CLK_R	38
	SD_55S	SD_INTERFACE	SDCONN_CLK	
SD_CMD	SD_55S	SD_INTERFACE	SD_CMD	7 38
	SD_55S	SD_INTERFACE	SDCONN_CMD	
	SD_55S	SD_INTERFACE	BCM57765_CR_CMD	

NOTE: SD_D<7..5> are different to support BCM5764M/BCM57765 co-layout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SMBUS_SMC_A_S3_SCL	SMB_55S	SMB	SMBUS_SMC_A_S3_SCL	42
SMBUS_SMC_A_S3_SDA	SMB_55S	SMB	SMBUS_SMC_A_S3_SDA	42
SMBUS_SMC_B_S0_SCL	SMB_55S	SMB	SMBUS_SMC_B_S0_SCL	42
SMBUS_SMC_B_S0_SDA	SMB_55S	SMB	SMBUS_SMC_B_S0_SDA	42
SMBUS_SMC_0_S0_SCL	SMB_55S	SMB	SMBUS_SMC_0_S0_SCL	42
SMBUS_SMC_0_S0_SDA	SMB_55S	SMB	SMBUS_SMC_0_S0_SDA	42
SMBUS_SMC_BSA_SCL	SMB_55S	SMB	SMBUS_SMC_BSA_SCL	7 42
SMBUS_SMC_BSA_SDA	SMB_55S	SMB	SMBUS_SMC_BSA_SDA	7 42
SMBUS_SMC_MGMT_SCL	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	42
SMBUS_SMC_MGMT_SDA	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	42

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
CHGR_CSI	1TO1_DIFFPAIR		CHGR_CSI_P	51
	1TO1_DIFFPAIR		CHGR_CSI_N	51
	1TO1_DIFFPAIR		CHGR_CSI_R_P	51
	1TO1_DIFFPAIR		CHGR_CSI_R_N	51
CHGR_CSO	1TO1_DIFFPAIR		CHGR_CSO_P	51
	1TO1_DIFFPAIR		CHGR_CSO_N	51
	1TO1_DIFFPAIR		CHGR_CSO_R_P	44 51
	1TO1_DIFFPAIR		CHGR_CSO_R_N	44 51

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PAGE TITLE
SMC Constraints

Apple Inc.

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REVISION 3.3.0

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=1:1_SPACING	?
THERM	*	=1:1_SPACING	?
AUDIO	*	=1:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
MEM_POWER	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_POWER	*	PWR_P2MM
MEM_CMD	MEM_POWER	*	PWR_P2MM
MEM_CTRL	MEM_POWER	*	PWR_P2MM
MEM_DATA	MEM_POWER	*	PWR_P2MM
MEM_DQS	MEM_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_FSB	GND	*	GND_P2MM
CPU_COMP	GND	*	GND_P2MM
CPU_GTLREF	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

SD CARD READER LAYOUT RELAXATIONS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_55S OVERRIDE	* OVERRIDE	OVERRIDE	=STANDARD OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE

MCP Fanout Constraint Relaxations

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	5.8 MM OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MEM_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MII_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_USB_RBIAS OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.25 MM OVERRIDE	250 MIL OVERRIDE	OVERRIDE	OVERRIDE

Misc Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
(USB_EXTA)	USB_90D	USB	USB_EXTA_MUXED_P	36 69
(USB_EXTA)	USB_90D	USB	USB_EXTA_MUXED_N	36 69
(USB_EXTA)	USB_90D	USB	USB_LT1_P	36
(USB_EXTA)	USB_90D	USB	USB_LT1_N	36
(USB_TPAD)	USB_90D	USB	USB_TPAD_P	18 47 69
(USB_TPAD)	USB_90D	USB	USB_TPAD_N	18 47 69
(USB_TPAD)	USB_90D	USB	USB_TPAD_CONN_P	7 47
(USB_TPAD)	USB_90D	USB	USB_TPAD_CONN_N	7 47
SMBUS_SMC_MGMT_SDA	SMB_55S	SMB	I2C_SMC_SMS_SDA_R	
SMBUS_SMC_MGMT_SCL	SMB_55S	SMB	I2C_SMC_SMS_SCL_R	
	SMB_55S	SMB	I2C_TCON_SCL	42
	SMB_55S	SMB	I2C_TCON_SDA	42
	SMB_55S	SMB	I2C_TCON_SCL_CONN	60
	SMB_55S	SMB	I2C_TCON_SDA_CONN	60

Graphics Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	DP_90D	DISPLAYPORT	DP_INT_ML_P<1..0>	9 60
	DP_90D	DISPLAYPORT	DP_INT_ML_N<1..0>	9 60
	DP_90D	DISPLAYPORT	DP_INT_ML_C_P<1..0>	60
	DP_90D	DISPLAYPORT	DP_INT_ML_C_N<1..0>	60
	DP_90D	DISPLAYPORT	DP_INT_ML_F_P<1..0>	7 60
	DP_90D	DISPLAYPORT	DP_INT_ML_F_N<1..0>	7 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_C_P	7 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_C_N	7 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_P	9 60
	DP_90D	DISPLAYPORT	DP_INT_AUX_CH_N	9 60
(DP_EXT_ML)	DP_90D	DISPLAYPORT	DP_EXT_ML_P<3..0>	9 62
	DP_90D	DISPLAYPORT	DP_EXT_ML_N<3..0>	9 62
	DP_90D	DISPLAYPORT	DP_EXT_ML_C_P<3..0>	62
	DP_90D	DISPLAYPORT	DP_EXT_ML_C_N<3..0>	62
	DP_90D	DISPLAYPORT	DP_EXT_ML_F_P<3..0>	62
	DP_90D	DISPLAYPORT	DP_EXT_ML_F_N<3..0>	62
(DP_EXT_AUX_CH)	DP_90D	DISPLAYPORT	DP_EXT_AUX_CH_C_P	9 62
	DP_90D	DISPLAYPORT	DP_EXT_AUX_CH_C_N	9 62

Power Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
CPUTHMSNS_D2	THERM_1TO1_55S	THERM	DRAMTHMSNS_D2_P	45
	THERM_1TO1_55S	THERM	DRAMTHMSNS_D2_N	45
CPU_THERMD	THERM_1TO1_55S	THERM	CPU_THERMD_P	10 45
	THERM_1TO1_55S	THERM	CPU_THERMD_N	10 45
MCPTHMSNS_D2	THERM_1TO1_55S	THERM	MLBR_THMDIODE_P	45
	THERM_1TO1_55S	THERM	MLBR_THMDIODE_N	45
MCP_THMDIODE	THERM_1TO1_55S	THERM	MCP_THMDIODE_P	19 45
	THERM_1TO1_55S	THERM	MCP_THMDIODE_N	19 45
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_1V5_S3_P	43 53
	SENSE_1TO1_55S	SENSE	ISNS_1V5_S3_N	43 53
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_AIRPORT_P	34 43
	SENSE_1TO1_55S	SENSE	ISNS_AIRPORT_N	34 43
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_CSREG_P	44
	SENSE_1TO1_55S	SENSE	ISNS_CSREG_N	44
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_HDD_P	35 43
	SENSE_1TO1_55S	SENSE	ISNS_HDD_N	35 43
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	ISNS_LCDBKLT_P	43 63
	SENSE_1TO1_55S	SENSE	ISNS_LCDBKLT_N	43 63
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPUVTTS0_CS_P	56
	SENSE_1TO1_55S	SENSE	CPUVTTS0_CS_N	56
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	IMVP6_CS_P	54
	SENSE_1TO1_55S	SENSE	IMVP6_CS_N	54
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	IMVP6_CS_R_P	54
	SENSE_1TO1_55S	SENSE	IMVP6_CS_R_N	54
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_VTTSENSE_P	56
	SENSE_1TO1_55S	SENSE	CPU_VTTSENSE_N	56
SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	MCPCORES0_VSEN_P	22 55
	SENSE_1TO1_55S	SENSE	MCPCORES0_VSEN_N	22 55
		MEM_POWER	PP1V5R1V35_S3	7 8
		SB_POWER	PP3V3_S5	7 8 58
		SB_POWER	PP3V3_S0	7 8 58
		SB_POWER	PP1V5_S0	7 8 58
		GND	GND	

Audio Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SPKRAMP_INR	DIFFPAIR	AUDIO	SPKRAMP_INR_P	7 37 49
	DIFFPAIR	AUDIO	SPKRAMP_INR_N	7 37 49
MAX98300_R	DIFFPAIR	AUDIO	MAX98300_R_P	49
	DIFFPAIR	AUDIO	MAX98300_R_N	49

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PAGE TITLE

K16/K99 Specific Constraints

Apple Inc.

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BRANCH

K99 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA	MM	15.5.1

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.100 MM	0.076 MM	30 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
27P4_OHM_SE	ISL3,ISL10	Y	0.250 MM	0.250 MM			
27P4_OHM_SE	ISL4,ISL9	Y	0.250 MM	0.250 MM			

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
40_OHM_SE	TOP,BOTTOM	Y	0.170 MM	0.170 MM			
40_OHM_SE	ISL3,ISL4,ISL9,ISL10	Y	0.140 MM	0.140 MM			

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.110 MM			
50_OHM_SE	ISL3,ISL4,ISL9,ISL10	Y	0.090 MM	0.090 MM			

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	ISL3,ISL4,ISL9,ISL10	Y	0.076 MM	0.076 MM			

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	TOP,BOTTOM	Y	0.175 MM	0.175 MM		0.130 MM	0.130 MM
70_OHM_DIFF	ISL3,ISL10	Y	0.135 MM	0.135 MM		0.130 MM	0.130 MM
70_OHM_DIFF	ISL4,ISL9	Y	0.155 MM	0.155 MM		0.130 MM	0.130 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	TOP,BOTTOM	Y	0.140 MM	0.140 MM		0.160 MM	0.160 MM
80_OHM_DIFF	ISL3,ISL10	Y	0.109 MM	0.109 MM		0.160 MM	0.160 MM
80_OHM_DIFF	ISL4,ISL9	Y	0.125 MM	0.125 MM		0.160 MM	0.160 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
75_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
75_OHM_DIFF	TOP,BOTTOM	Y	0.160 MM	0.160 MM		0.160 MM	0.160 MM
75_OHM_DIFF	ISL3,ISL10	Y	0.120 MM	0.120 MM		0.140 MM	0.140 MM
75_OHM_DIFF	ISL4,ISL9	Y	0.140 MM	0.140 MM		0.140 MM	0.140 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.115 MM		0.210 MM	0.210 MM
90_OHM_DIFF	ISL3,ISL10	Y	0.089 MM	0.089 MM		0.210 MM	0.210 MM
90_OHM_DIFF	ISL4,ISL9	Y	0.105 MM	0.105 MM		0.210 MM	0.210 MM

.210 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
95_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
95_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.115 MM			0.210 MM
95_OHM_DIFF	ISL3,ISL10	Y	0.089 MM	0.089 MM		0.210 MM	0.210 MM
95_OHM_DIFF	ISL4,ISL9	Y	0.105 MM	0.105 MM		0.210 MM	0.210 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	TOP,BOTTOM	Y	0.091 MM	0.091 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL3,ISL10	Y	0.075 MM	0.075 MM		0.300 MM	0.300 MM
100_OHM_DIFF	ISL4,ISL9	Y	0.085 MM	0.085 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	0.1 MM	?
BGA_P2MM	*	0.2 MM	?
BGA_P3MM	*	0.3 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1.5:1_SPACING	*	0.15 MM	?
1.8:1_SPACING	*	0.18 MM	?
2:1_SPACING	*	0.2 MM	?
2.28:1_SPACING	*	0.228 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
PP1V5_MEM	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
NB_STATIC	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
1.5X_DIELECTRIC	*	0.105 MM	?
5X_DIELECTRIC	*	0.350 MM	?

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PAGE TITLE

K99 RULE DEFINITIONS

Apple Inc.

DRAWING NUMBER 051-8467 SIZE D

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BRANCH

1UF 0402 CAPACITOR VENDOR TABLES FOR ACOUSTICS

SAMSUNG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0629	2	CAP, 1UF, 6.3V, 10%, 0402	C7203,C7980	CRITICAL	SS_CAP_1UF

MURATA

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0628	2	CAP, 1UF, 6.3V, 10%, 0402	C7203,C7980	CRITICAL	MU_CAP_1UF

TAIYO YUDEN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0630	2	CAP, 1UF, 6.3V, 10%, 0402	C7203,C7980	CRITICAL	TY_CAP_1UF

2.2UF 0402 CAPACITOR VENDOR TABLES FOR ACOUSTICS

SAMSUNG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1240,C1241,C1242,C1243,C1244,C1245,C1246,C1247,C1248,C1249	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1250,C1251,C1252,C1253,C1254,C1255,C1256,C1257,C1258,C1259	CRITICAL	SS_CAP_2_2UF
138S0632	8	CAP, 2.2UF, 6.3V, 20%, 0402	C1260,C1261,C1262,C1263,C1264,C1265,C1266,C1267	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9900,C9901,C9902,C9903,C9904,C9905,C9906,C9907,C9908,C9909	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9910,C9911,C9912,C9913,C9914,C9915,C9916,C9917,C9918,C9919	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9920,C9921,C9922,C9923,C9924,C9925,C9926,C9927,C9928,C9929	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9930,C9931,C9932,C9933,C9934,C9935,C9936,C9937,C9938,C9939	CRITICAL	SS_CAP_2_2UF
138S0632	12	CAP, 2.2UF, 6.3V, 20%, 0402	C1283,C1284,C1285,C1286,C1287,C1288,C1291,C1292,C1293,C1294,C1295,C1296	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3500,C3501,C3504,C3505,C3510,C3511,C3512,C3514,C3515,C3516	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3520,C3521,C3524,C3525,C3530,C3531,C3534,C3535,C3540,C3541	CRITICAL	SS_CAP_2_2UF
138S0632	8	CAP, 2.2UF, 6.3V, 20%, 0402	C3542,C3544,C3545,C3546,C3550,C3551,C3554,C3555	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3600,C3601,C3604,C3605,C3610,C3611,C3612,C3614,C3615,C3616	CRITICAL	SS_CAP_2_2UF
138S0632	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3620,C3621,C3624,C3625,C3630,C3631,C3634,C3635,C3640,C3641	CRITICAL	SS_CAP_2_2UF
138S0632	9	CAP, 2.2UF, 6.3V, 20%, 0402	C3642,C3644,C3645,C3646,C3650,C3651,C3654,C3655,C5[illegible]10	CRITICAL	SS_CAP_2_2UF

MURATA

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1240,C1241,C1242,C1243,C1244,C1245,C1246,C1247,C1248,C1249	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1250,C1251,C1252,C1253,C1254,C1255,C1256,C1257,C1258,C1259	CRITICAL	MU_CAP_2_2UF
138S0633	8	CAP, 2.2UF, 6.3V, 20%, 0402	C1260,C1261,C1262,C1263,C1264,C1265,C1266,C1267	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9900,C9901,C9902,C9903,C9904,C9905,C9906,C9907,C9908,C9909	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9910,C9911,C9912,C9913,C9914,C9915,C9916,C9917,C9918,C9919	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9920,C9921,C9922,C9923,C9924,C9925,C9926,C9927,C9928,C9929	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9930,C9931,C9932,C9933,C9934,C9935,C9936,C9937,C9938,C9939	CRITICAL	MU_CAP_2_2UF
138S0633	12	CAP, 2.2UF, 6.3V, 20%, 0402	C1283,C1284,C1285,C1286,C1287,C1288,C1291,C1292,C1293,C1294,C1295,C1296	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3500,C3501,C3504,C3505,C3510,C3511,C3512,C3514,C3515,C3516	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3520,C3521,C3524,C3525,C3530,C3531,C3534,C3535,C3540,C3541	CRITICAL	MU_CAP_2_2UF
138S0633	8	CAP, 2.2UF, 6.3V, 20%, 0402	C3542,C3544,C3545,C3546,C3550,C3551,C3554,C3555	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3600,C3601,C3604,C3605,C3610,C3611,C3612,C3614,C3615,C3616	CRITICAL	MU_CAP_2_2UF
138S0633	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3620,C3621,C3624,C3625,C3630,C3631,C3634,C3635,C3640,C3641	CRITICAL	MU_CAP_2_2UF
138S0633	9	CAP, 2.2UF, 6.3V, 20%, 0402	C3642,C3644,C3645,C3646,C3650,C3651,C3654,C3655,C5[illegible]10	CRITICAL	MU_CAP_2_2UF

TAIYO YUDEN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1240,C1241,C1242,C1243,C1244,C1245,C1246,C1247,C1248,C1249	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C1250,C1251,C1252,C1253,C1254,C1255,C1256,C1257,C1258,C1259	CRITICAL	TY_CAP_2_2UF
138S0634	8	CAP, 2.2UF, 6.3V, 20%, 0402	C1260,C1261,C1262,C1263,C1264,C1265,C1266,C1267	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9900,C9901,C9902,C9903,C9904,C9905,C9906,C9907,C9908,C9909	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9910,C9911,C9912,C9913,C9914,C9915,C9916,C9917,C9918,C9919	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9920,C9921,C9922,C9923,C9924,C9925,C9926,C9927,C9928,C9929	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C9930,C9931,C9932,C9933,C9934,C9935,C9936,C9937,C9938,C9939	CRITICAL	TY_CAP_2_2UF
138S0634	12	CAP, 2.2UF, 6.3V, 20%, 0402	C1283,C1284,C1285,C1286,C1287,C1288,C1291,C1292,C1293,C1294,C1295,C1296	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3500,C3501,C3504,C3505,C3510,C3511,C3512,C3514,C3515,C3516	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3520,C3521,C3524,C3525,C3530,C3531,C3534,C3535,C3540,C3541	CRITICAL	TY_CAP_2_2UF
138S0634	8	CAP, 2.2UF, 6.3V, 20%, 0402	C3542,C3544,C3545,C3546,C3550,C3551,C3554,C3555	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3600,C3601,C3604,C3605,C3610,C3611,C3612,C3614,C3615,C3616	CRITICAL	TY_CAP_2_2UF
138S0634	10	CAP, 2.2UF, 6.3V, 20%, 0402	C3620,C3621,C3624,C3625,C3630,C3631,C3634,C3635,C3640,C3641	CRITICAL	TY_CAP_2_2UF
138S0634	9	CAP, 2.2UF, 6.3V, 20%, 0402	C3642,C3644,C3645,C3646,C3650,C3651,C3654,C3655,C5[illegible]10	CRITICAL	TY_CAP_2_2UF

10UF 0603 CAPACITOR VENDOR TABLES FOR ACOUSTICS

SAMSUNG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0626	1	CAP, 10UF, 6.3V, 20%, 0603	C1280	CRITICAL	SS_CAP_10UF
138S0626	8	CAP, 10UF, 6.3V, 20%, 0603	C4690,C4695,C5025,C7205,C7290,C7345,C7355,C7647	CRITICAL	SS_CAP_10UF
138S0626	8	CAP, 10UF, 6.3V, 20%, 0603	C9012,C9486,C2500,C2520,C2560,C2567,C2600,C6999	CRITICAL	SS_CAP_10UF

MURATA

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0625	1	CAP, 10UF, 6.3V, 20%, 0603	C1280	CRITICAL	MU_CAP_10UF
138S0625	8	CAP, 10UF, 6.3V, 20%, 0603	C4690,C4695,C5025,C7205,C7290,C7345,C7355,C7647	CRITICAL	MU_CAP_10UF
138S0625	8	CAP, 10UF, 6.3V, 20%, 0603	C9012,C9486,C2500,C2520,C2560,C2567,C2600,C6999	CRITICAL	MU_CAP_10UF

TAIYO YUDEN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0627	1	CAP, 10UF, 6.3V, 20%, 0603	C1280	CRITICAL	TY_CAP_10UF
138S0627	8	CAP, 10UF, 6.3V, 20%, 0603	C4690,C4695,C5025,C7205,C7290,C7345,C7355,C7647	CRITICAL	TY_CAP_10UF
138S0627	8	CAP, 10UF, 6.3V, 20%, 0603	C9012,C9486,C2500,C2520,C2560,C2567,C2600,C6999	CRITICAL	TY_CAP_10UF

22UF 0603 CAPACITOR VENDOR TABLES FOR ACOUSTICS

SAMSUNG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0635	4	CAP, 22UF, 6.3V, 20%, 0603	C1210,C1214,C1217,C1218	CRITICAL	SS_CAP_22UF
138S0635	3	CAP, 22UF, 6.3V, 20%, 0603	C1223,C1226,C1227	CRITICAL	SS_CAP_22UF
138S0635	5	CAP, 22UF, 6.3V, 20%, 0603	C1230,C4902,C7360,C7361,C9480	CRITICAL	SS_CAP_22UF

MURATA

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0676	4	CAP, 22UF, 6.3V, 20%, 0603	C1210,C1214,C1217,C1218	CRITICAL	MU_CAP_22UF
138S0676	3	CAP, 22UF, 6.3V, 20%, 0603	C1223,C1226,C1227	CRITICAL	MU_CAP_22UF
138S0676	5	CAP, 22UF, 6.3V, 20%, 0603	C1230,C4902,C7360,C7361,C9480	CRITICAL	MU_CAP_22UF

TAIYO YUDEN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0688	4	CAP, 22UF, 6.3V, 20%, 0603	C1210,C1214,C1217,C1218	CRITICAL	TY_CAP_22UF
138S0688	3	CAP, 22UF, 6.3V, 20%, 0603	C1223,C1226,C1227	CRITICAL	TY_CAP_22UF
138S0688	5	CAP, 22UF, 6.3V, 20%, 0603	C1230,C4902,C7360,C7361,C9480	CRITICAL	TY_CAP_22UF

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